Indira Gandhi
National Open University
School of Health Sciences

BNS-040
Basics of First Aid

Block

4

FIRST AID IN COMMON AND ENVIRONMENTAL EMERGENCIES

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BLOCK INTRODUCTION

In this last Block of this Theory course , we will be discussing about various emergency situations in continuation with the previous block which was on Accidents and Injuries. In this Block, we will discuss the Common and Environmental Emergencies. These situations can be common medical emergencies, extreme heat and cold conditions, bites and stings, altitude illness and allergies and shock. These situations can occur anywhere and in varied age groups. As first aid provider, you must know about these common emergencies occurring day to day in various settings. In this block, we will discuss about various important emergencies. You must develop knowledge about these emergencies in the light of knowledge developed by virtue of previous blocks as to how to deal with specific emergency situations.

This block consists of 5 units as follows:

Unit 1 focuses on common emergencies;

Unit 2 discusses extreme heat and cold conditions;

Unit 3 deals with bites and stings;

Unit 4 explains altitude illness and

Unit 5 describes allergies and shock.

So, let us start reading this block and understand how to provide first aid in these emergencies.

UNIT 1 COMMON EMERGENCIES

Structure

1.0 INTRODUCTION

This is the fourth and last theory block of this course. In the first Unit of this Block we will be discussing regarding the common emergencies which one encounters in day to day life. Someone may faint or have fits. Some of your own family members may complain of chest pain or have episodes of stroke, asthma and other medical diseases. You may also encounter problems of Drowning and Poisoning and so on. Hence, it is important as a first aid provider to know how to recognize and how to act in these situations.

These emergencies are common because they occur frequently and affect a majority of individuals around you, someone known to you or otherwise. It is important for you to understand first aid actions that need to be undertaken in these common emergencies. So , let's begin this Unit by understanding the concept of common emergencies and learning about the different common emergencies that can be witnessed by you.

1.1 OBJECTIVES

After completion of this unit, you shall be able to:

- define Common Emergencies;
- enumerate the causes of Common Emergencies;
- recognize the types of Common Emergencies;
- explain the assessment of victim affected by Common Emergencies;
- describe the first aid to be provided in various Common Emergencies; and
- list the Do's and Don'ts in various Common Emergencies.

1.2 COMMON EMERGENCIES

Common emergencies is a term used to denote the emergencies which occur commonly. These emergencies are associated with medical conditions, illness or sickness and are experienced many a times. These emergencies occur everywhere and many persons/people report these problems on day to day basis. That's why they are called Common Emergencies.

There are many common emergencies like:

- Bleeding
- Breathing difficulty (asthma)
- Fits and Fainting
- Chest Pain
- Stroke
- Drowning
- Poisoning
- High and Low Blood Sugar
- Sprains and Strains
- Burns

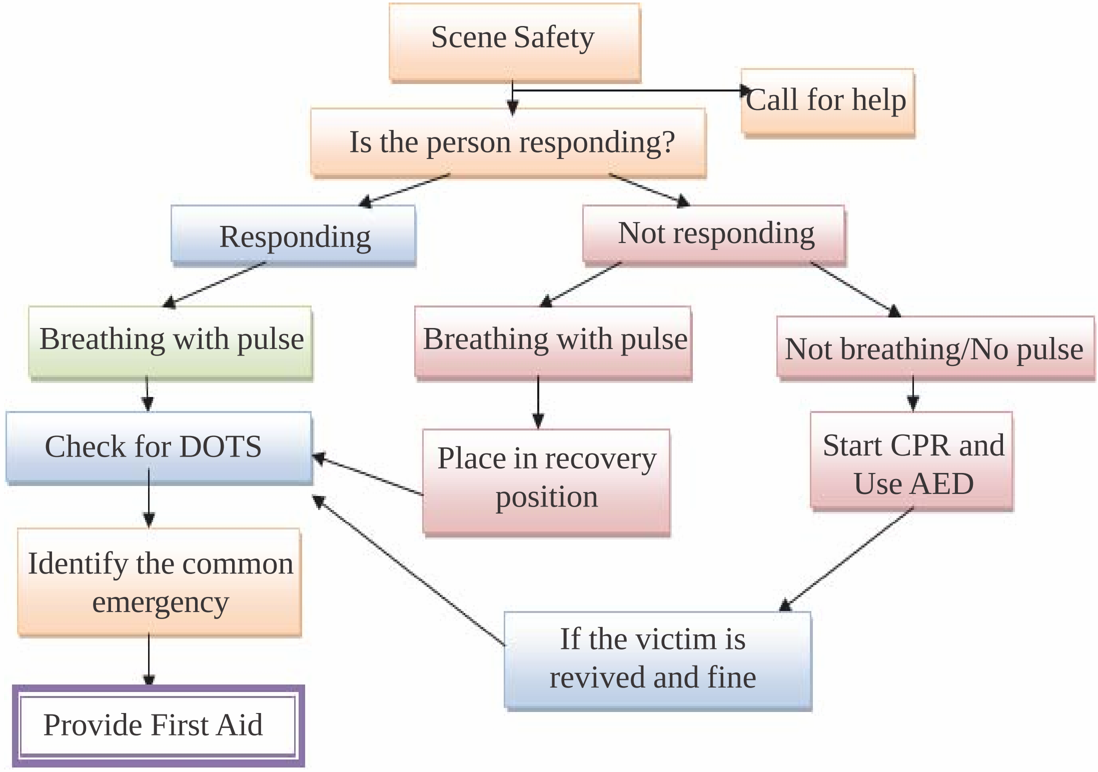
Some of these emergencies have been discussed in their specific units and some will be discussed in the upcoming sections.

Assessment of the Victim and General First Aid

The assessment and general first aid is as follows:

General Assessment

You have already learnt the assessment to be done while providing first aid in Unit 1 of Block 2 of the Theory Course. The general assessment is done as per following flowchart.



Specific Assessment

After general assessment, perform specific assessment as:

1) Check for DOTS and look for deformity, swelling, tenderness and any other sign and symptoms.

2) Check for the sign and symptoms of common emergencies (given in upcoming sections). Check how severe is the condition.

3) Ask History from the victim or onlookers present.

General First Aid

- First and Foremost, assess and recognize the type of emergency.
- Maintain Scene Safety.
- Comfort and reassure the victim .
- Provide Specific First aid as per the problem detected.

In the next section, we will discuss the first aid in various common emergencies. So, lets continue.

1.3 FITS/SEIZURES/CONVULSION

In this section, we will discuss the definition, causes, recognition and first aid in Fits.

1.3.1 Definition, Causes, Types and Recognizing Fits /Seizures/ Convulsions

Definition

A fit or seizure or convulsion is an episode of vigorous and fast shaking of muscles of body along with loss of consciousness caused by sudden release of electrical activity in the brain.

A full-scale seizure associated with the medical problem of epilepsy involves violent jerking of the limbs, facial twitching, foaming at the mouth and clenched teeth. The seizure episode may last for a few minutes (up to 3-4 minutes) and the victim may need several hours to recover (Fig. 1.1).

Fig. 1.1: Fits

Causes

As also discussed in Unit 2 of Block 1 of this Theory course regarding "Understanding Human Body" you know the structure of Human Brain and associated Nerves. The major role of this system is to send signals around the human body in form of electrical signals. These signals are produced by the Brain and if due to any injury or problem or disease, this electrical activity becomes abnormal or increases, it causes uncontrollable muscular shaking and loss of consciousness (Fig. 1.2).

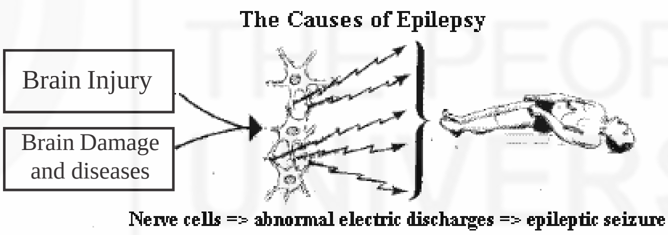


Fig. 1.2 : Cause of Epilepsy

Some of the associated causes/risk factors are:

- High Fever over 102º F in children below 5 years
- Head Injury
- Brain diseases like Epilepsy, Brain tumour, Stroke (discussed later on in this unit)
- Poisons and drug abuse or overdose
- Diabetes.

Types

There are two types of fits:

- Convulsive fits which occur in infants and young children due to fever or illness or infection and is an ALARMING SIGN.

- Epileptic fits which occur due to injury or disorder (epilepsy disease) or infections of the brain.

Recognition

The various sign and symptoms which help in the recognition of fits are:

1) If it is a seizure due to fever (Fig.1.3):
   - Jerking or twitching of the body
   - Stiffness in arms or legs
   - Difficulty breathing
   - Child regains consciousness after the episode but is sleepy and tired.

Fig. 1.3: Seizures in Children

2) If the seizure is because of other causes (Fig.1.4):
   - Person suddenly falls on the ground
   - Headache and fever
   - Stiffness and Rigidity in the body
   - Excessive shaking of the body
   - Frothy Mouth
   - Eye balls rolled up
   - Deep, Noisy breathing
   - Unconsciousness for sometime
   - If awakened , the person does not remember what happened and will be confused.

Fig. 1.4: Movements in Epileptic Fit

1.3.2 First Aid in Fits/Seizures/Convulsions

The first aid steps in case of Fits are as follows:

1) Protect the victim from injury during the fit

- Check the immediate/surrounding area for any hazards and remove them if possible.
- Move furniture away from the person and take special care with electrical appliances or cooking utensils to avoid a burn or scald.
- If the victim is close to a wall or hard furniture, pad the area with clothing or a pillow to avoid further injury.
- If there is danger due to fall, fire, accident or drowning , remove the victim from the scene.
- Place a soft padding or flat pillow under the head to avoid injury, in case available or improvisation from towel or blanket or folded jacket or cloth can be done.
- In case of child with fits due to fever, place pillows or soft padding around the child so that violent movement doesn't cause injury (Fig. 1.5).

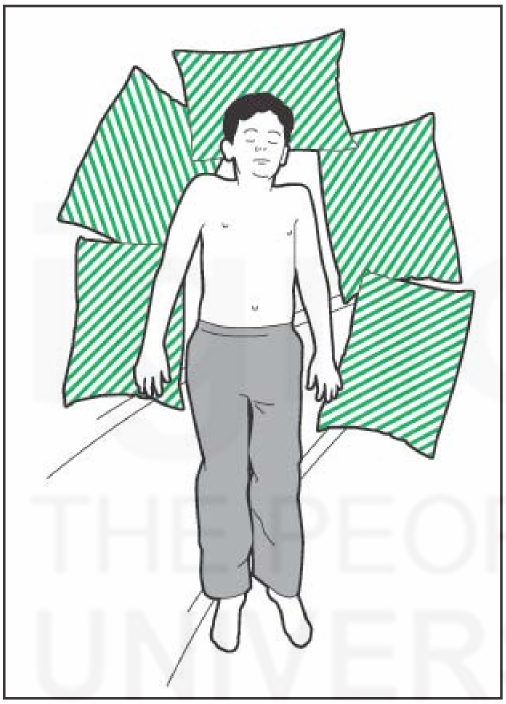


Fig. 1.5: Placing padding /pillows around the victim to maintain safety

2) Manage the seizure or convulsion

- Stay with the victim until the seizure ends.
- If in a public place, keep bystanders clear and reassure them that the seizure will end soon (Fig. 1.6).

Fig. 1.6: Be with vicitm and keep bystanders clear

- If the victim is a child and has fever, cool the child by removing wet nappy or pants to expose and cool the body. Make sure they get some fresh air by opening a door or window. Also, you can bring down his/her temperature by sponging with wet towel using lukewarm water for babies and children (Fig. 1.7).

Fig. 1.7: Bring down the body temperature by sponging

3) After the seizure

- As soon as the seizure ends, quickly roll the unconscious victim onto their side which opens the airway and any secretions can fall out (Fig.1.8).

Fig. 1.8: Turning the victim onto their side (recovery position)

- Cover the victim lightly with a coat or blanket (Fig. 1.9).

Fig. 1.9: Covering the victim

- Check that normal breathing has resumed.
- Allow the victim to sleep until fully recovered, but check for a response every few minutes.

4) Reassure and Assess
   - Talk to victim calmly and reassure.
   - Check for signs of injury and infection and apply necessary first aid.
   - Collect the history from the victim. Try to find out if this is his first episode and he is a known case or not.

5) Transport

   Transfer the victim to medical facility if
   - The condition doesn't improve, worsens/deteriorates after the seizure or
   - If the seizure does not stop after 5 minutes or
   - If the victim does not wake up within 10 minutes after the seizure has stopped or is not breathing well or
   - Repeated seizures occur or
   - Victim has suffered injury or
   - Victim is pregnant or
   - Seizure has occurred in water or
   - It is first seizure or
   - The victim doesn't respond or gain consciousness after the episode.

Do's and Don'ts

Do's

- Note the time when the episode of fit begins and ends. Share this information with the emergency medical team in Ambulance or with medical personnel's in the Hospital or medical facility.
- Check for RCAB after the episode of seizure (Response, Circulation, Airway and Breathing).
- After the episode of fit, advise victim not to drive. Try to arrange for someone to be with the victim until he/she safely reaches home.
- Advise the victim to contact his/her doctor to report the seizure.
- Check for any prescribed medication of the victim.
- If the victim is known to have epilepsy or tells you that he has a medical problem and is on medicine, there is no need for medical aid or an ambulance unless the seizure lasts more than 5 minutes or a second seizure follows the first one.
- If it is the first known seizure, medical help is vital to avoid any future complications.
- Avoid giving anything to drink until the victim is fully conscious or alert.

- Stay with the victim till they fully recover from the episode of seizure.
- Loosen any tight constrictive clothing. Remove eye glasses or spectacles, if any.
- Wipe secretions present on the mouth.

Don't's

- Don't hold the person down or try to stop his or her movements because this may result in a broken bone or soft tissue injury.
- Don't put anything in the person's mouth and also don't use cloth/spoon/ fingers etc. This can injure teeth or the jaw or damage the tissues of the mouth. A person having a seizure cannot swallow his or her tongue.
- Don't try to give mouth-to-mouth breaths (like CPR). People usually start breathing again on their own after a seizure. But be alert for any symptoms that require resuscitation as discussed in Unit 3 of Theory Block 2.
- Don't hold the tongue of the victim. It will only hurt your fingers as it can come in between teeth or jaw of the victim.
- Don't cool the child by immersing in water or pouring cool or luke warm water . Only sponge with towel wet and squeezed in luke-warm water .
- Don't give anything to eat or drink till fully alert.
- Don't make the victim smell shoes. This is of no help.
- Don't roll into recovery position if you suspect a spinal/back/neck injury.

1.4 FAINTING

In this section, we shall discuss the definition, causes, recognition and first aid in Fainting.

1.4.1 Definition, Causes and Recognizing Fainting

Definition

Fainting is a brief loss of consciousness that is caused by a temporary reduction of blood flow to the brain. The loss of responsiveness is usually short when fainting occurs and lasts for few seconds to 1 or 2 minutes. The other name for fainting is syncope (Fig. 1.10).

Fig. 1.10: Fainting or Syncope

Fainting is also called as "passing out" or "blacking out". It is a most common problem in elderly but can also occur in other age groups.

Causes

The main reason for fainting is lack of oxygen in the brain which occurs rapidly, has small duration and person recovers fast from the episode. The various causes are:

- Standing for long time especially in hot weather or a hot shower
- Sitting for long periods and then suddenly standing due to which the blood collects in leg veins and heart is not able to pump it to brain immediately when standing
- Sight of needles or blood, injections and pain which makes a person anxious or nervous
- Fear and Pain
- Use of certain drugs and alcohol
- Taking in too little food and fluids or their loss e.g. in dehydrea/diarrhoea
- Low blood pressure
- Lack of sleep
- Over exhaustion
- Low Blood Sugar
- Massive Bleeding in accidents or trauma
- Brain diseases like stroke (discussed in next section).

Recognition

The sign and symptoms of Fainting are as follows:

- A brief loss of consciousness causing the victim to fall to the floor
- A slow pulse
- Pale, cold skin and sweating
- Dizziness
- Weakness
- Seeing spots in the eyes
- Visual blurring
- Nausea
- Pale skin, Flushing of the face
- Sweating.

1.4.2 First Aid in Fainting

The following steps must be followed when providing first aid in fainting:

- When the person has the fainting sensation, make the person sit or lie down.
- Note the time of start of episode.
- If sitting, keep the person's head between knees so that head and heart are at same level (Fig. 1.11).

Fig. 1.11: Keeping head between knees

- If lying,
  - Make the person to lie flat at least for 15 minutes.
  - Check the victim's response, airway to ensure it is clear and also check his/her vital signs.
  - Raise his/her legs above heart level (Fig. 1.12).

Fig. 1.12: Raising legs above heart level

- Loosen clothing (neck ties, collars, belts etc.)
- If the person vomits, roll the entire body as a unit to the side. Support the neck and back while rolling. You can also cover him up with coat or blanket.
- Wait till the victim gains consciousness, it takes few minutes only.
- If consciousness is not regained within one minute immediately call the ambulance or transfer to medical facility and keep on monitoring/assessing the victim.

Do's and Don'ts

Do's

- Try to catch the victim before he/she falls.
- Loosen any tight constrictive clothing.
- Try to keep the victim in well ventilated area.
- Always stay with the victim.
- Note the time when the episode starts.
- Provide support if the victim tries to get up to sit or stand.

Don'ts

- Don't keep the victim in standing or sitting or upright position if fainted. You should make them keep head between knees when siting.
- Don't allow the victim to sit up or stand up quickly when gaining consciousness.

- Don’t try to move the unconscious person to a sitting position.
- Don’t give an unconscious person any food or drink.
- Don’t slap face of the victim or throw cold water for waking the victim up.
- Don’t place pillow under the head of an unconscious person.
- Don’t roll if you suspect a spinal/back injury.

1.5 STROKE

In this section, we will discuss the definition, causes, recognition and first aid in Stroke.

1.5.1 Definition, Causes and Recognizing Stroke

Definition

A stroke is a condition in which part of the brain is affected by disturbance to the normal blood supply. It is also called as “Brain Attack”. In this, the blood flow is cut off and brain cells don’t receive oxygen and glucose causing them to die. This can lead to permanent brain damage and death.

Causes

This can result from a clot in a blood vessel of brain that stops blood from passing through it to reach the brain tissue. Sometimes a stroke is caused by burst of blood vessels causing internal bleeding in the skull with pressure on brain tissue. At first, the victim will have a severe headache, but it can lead to paralysis of one side of the body and even the loss of the ability to speak (Fig. 1.13).

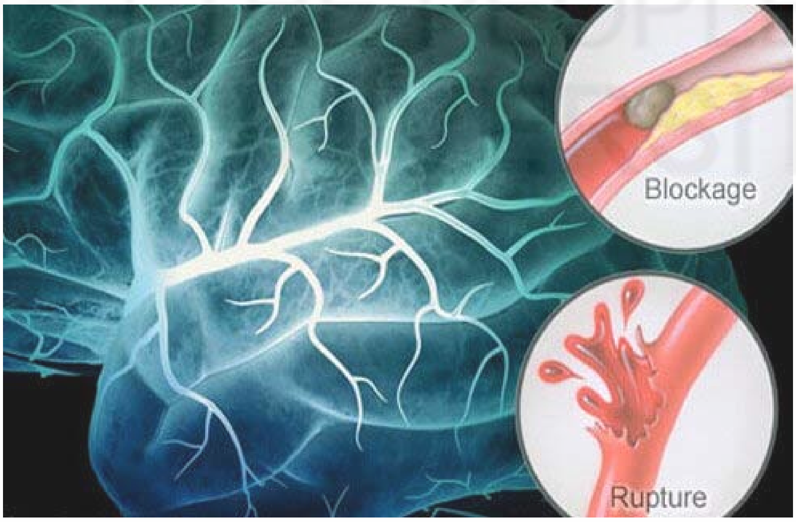


Fig. 1.13: Cause of Stroke

Recognition

The sign and symptoms which show the episode of stroke are:

- Sudden severe head ache
- Tingling, weakness or numbness in one side of the body
- Loss of muscle tone of the face, with dribbling from one side

- Loss of speech, difficulty to talk or understanding what others say
- Loss of vision in one or both eyes
- Loss of bladder or bowel control
- Loss of balance of body
- Deteriorating conscious state or unconsciousness or fainting
- Loss of memory –long-term or short-term.

Note:

In order to simplify, Always Remember to check or use **F.A.S.T. acronym** (Fig. 1.14) to help remember warning signs.

F for Face. Does the face droop on one side when the person tries to smile?

A for Arms. Is one arm lower when the person tries to raise both arms?

S for Speech. Can the person repeat a simple sentence? Is speech slurred or hard to understand?

T for Time. During a stroke every minute counts. If you observe any of these signs, call emergency number or medical facility immediately.

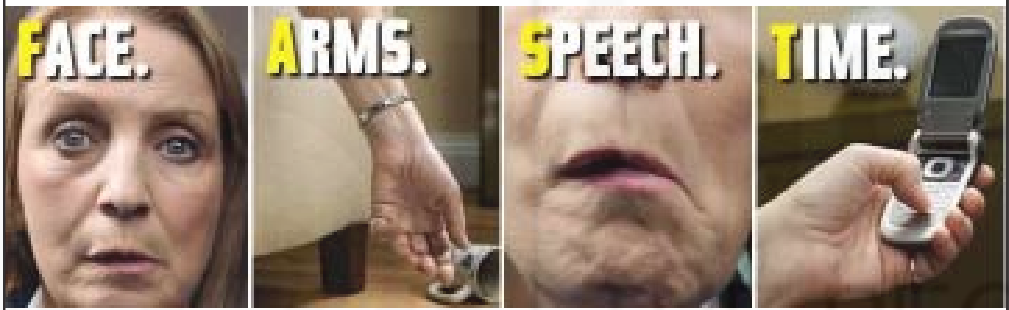


Fig. 1.14: F.A.S.T. for Stroke assessment

1.5.2 First Aid in Stroke

1) Assess the level of consciousness and call for help
    - Check and assess the condition of the victim.
    - Call for an ambulance immediately because it is important for the victim to be assessed as soon as possible as treatment must be started within 1-2 hours of episode of stroke depending on the cause.
2) Care for victim
    - If unconscious and breathing normally, place the victim by rolling onto their side (Recovery position).
    - If conscious, put the victim into the position of greatest comfort. You can make the person lie down with their head and shoulders raised and supported (use pillows).
    - Cover the victim with blanket or coat to reduce heat loss.
3) Observe
    - While waiting for the ambulance to arrive, observe closely for any change in condition.

- Observe for signs of Shock (discussed in Unit 5 of this Block).

4) Transport

- Shift to hospital immediately.

Do's and Don'ts

Do's

- Try to catch the victim before he/she falls.
- Note the time when the episode starts.
- Always stay with the victim.
- Be prepared to resuscitate if required.
- If fainting occurs, care like a fainting victim as discussed in Section 1.4.2 of this Unit.
- Prompt medical treatment must be taken since this is a MEDICAL EMERGENCY, so call an ambulance immediately or transport to medical facility quickly.
- Loosen tight clothing.
- Wipe secretions present on the mouth.

Don'ts

- Don't give any food or drink.
- Don't give any medicine.
- Don't mishandle the victim.

1.6 CHEST PAIN

In this section, we will discuss the definition, causes, recognition and first aid in Chest Pain.

1.6.1 Definition, Causes and Recognizing Chest Pain

Definition

People suffer chest pain because of number of reasons. Any chest pain that is more than10-15 minutes should be considered as serious (Fig. 1.15). There are various reasons for Chest Pain.

Fig. 1.15: Chest Pain

Causes

The causes of Chest Pain can be any of the following:

- Conditions of Heart like Angina and Myocardial Infarction (Heart Attack)
- Conditions like Chest Injuries, Swelling in the ribs, Muscle Strain
- Gastrointestinal problems like Indigestion.

> **Note:**
>
> **Angina is pain in chest and occurs when part of heart muscle doesnot receive enough oxygenated blood due to narrowing of coronary arteries which is common in elderly during over-exertion, exercise, mental strain, worry, anxiety, excitement etc. In Myocardial Infarction, part of heart muscle dies due to lack of blood supply. It is serious and can cause death of the victim.**

Recognition

The following sign and symptoms help in the recognition of Chest Pain according to its causes:

1) Angina : Pain in the chest that radiates to jaw, arm and neck with time duration of 5 to 8 minutes. The pain increases with the exertion and reduces or stops if the victim rests. (Fig. 1.16).

2) Myocardial Infarction (Fig. 1.17).
   - Pain or an uncomfortable pressure in the middle of the chest that does not stop AFTER 15 minutes of rest
   - Pain that spreads up the neck and into the jaw and teeth, shoulder or down one or both arms
   - Pain feels like a crushing weight is resting on the patient's chest
   - Shortness of breath
   - Pale , cool or bluish skin
   - Fainting, Sweating
   - Nausea, Vomiting
   - A feeling of great anxiety, weakness or fatigue
   - Sudden collapse.

3) Chest Problems
   - Sudden, sharp chest pain often with shortness of breath
   - Cough, anxiety
   - Fast heartbeat
   - Fainting
   - Sweating
   - Fever in some cases

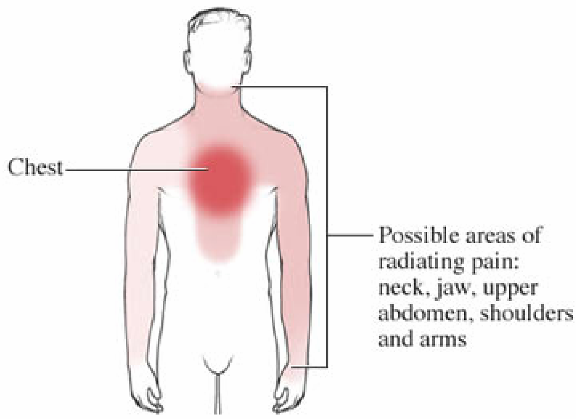


Fig. 1.16 : Angina Pain

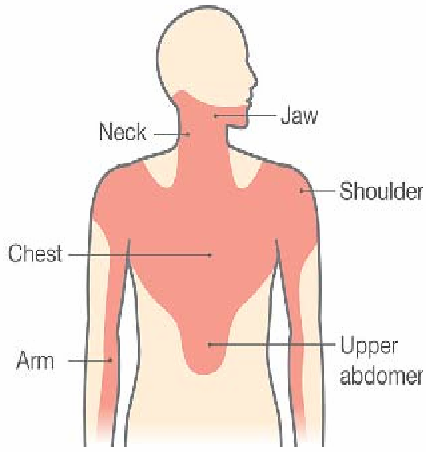


Fig. 1.17: Pain due to Heat Attack

4) Gastric problems

- Severe Pain
- Extreme Nausea and sometimes vomiting
- Cramping or spasms in stomach.

1.6.2 First Aid in Chest Pain

The First Aid in Chest Pain is as follows:

1) Make the victim to rest

- Help the conscious victim to rest in the position of greatest comfort, generally in a half-sitting position with support for the back and head. (Fig. 1.18)

Fig. 1.18 : Position for vicitm of Chest Pain

- If outdoors with no chair available, kneel behind the victim to provide support.

2) Assess the victim

- Assess the type of pain, response of the victim and be ready to start resuscitation (CPR) if collapse occurs.

Note:

Assess pain by using any of these acronyms for finding out the characteristic of the pain :

PQRST	**SOCRATES**
P- Provoke : what causes pain? **Q-** Quality: what is pain like? Sharp? Dull? **R-** Radiates – does pain spread anywhere else? **S-** Severity – how bad is the pain? **T-** Time – when did the pain start/ finish?	**S-** Site : where is pain? **O-** Onset: when did the pain begin. **C-** Character: it is sharp or dull? **R-** Radiation**:** where does it spread? **A-** Associated Symptoms with the pain **T-** Time: when does the pain start? **E-** Effecting factors which increase or decrease pain. **S-** Severity: how bad is the pain?

3) Care for the Victim

- Ask the victim if he has been prescribed any medicine. Assist him/her to take that medication.
- Try to keep the victim calm and reassure because any stress or activity can cause complications or even collapse.
- Loosen any tight clothing at the neck and waist to assist breathing.

4) Transport as soon as the ambulance arrives.

Do's and Don'ts

Do's

- Always call for an ambulance.
- Make the victim rest completely.
- You can give prescribed medication, if any for the victim who has known history of the heart disease and knows how to take his/her medicine.
- Loosen any constrictive clothing.
- Treat for shock , if present.
- It is advisable to not to transport a victim with chest pain in a car. Additional exertion could cause the heart to suffer more damage.

  In many cases, it is difficult to differentiate between Angina and heart attack. In case of doubt, First aid provider must always treat it as a case of Heart Attack and provide imediate rest.

Don'ts

- Don't allow any movement.
- Don't delay getting medical help.
- Don't give any food, fluids or stimulants such as alcohol, cigarettes, tea or coffee.
- Don't allow the victim to sit upright or walk.
- Don't leave the victim alone.
- Don't wait to see if the symptoms go away.
- Don't allow the person to deny the symptoms.
- Don't allow crowding; ensure sufficient ventilation around the person.
- Don't give medicine on your behalf. Only give the medicine that the victim has been prescribed and is available with him. Don't borrow medicine from any other person.

Check Your Progress 1

1) Define Common Emergency.

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2) How will you recognize febrile fits?

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3) Expand F.A.S.T used for assessment of victim with Stroke.

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4) List three Do's and Don'ts in Chest Pain.

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1.7 ASTHMA

In this section , we will discuss the definition, causes, recognition and first aid in Asthma..

1.7.1 Definition, Causes and Recognizing Asthma

Definition

Asthma is a common condition in which breathing becomes difficult because of swelling/inflammation of the air passages. The airways become narrowed by muscle spasm, swelling and increased mucus production, often causing a wheeze to be heard. Air is trapped in the lungs by the swollen airways and the victim has most difficulty breathing out (Fig. 1.19).

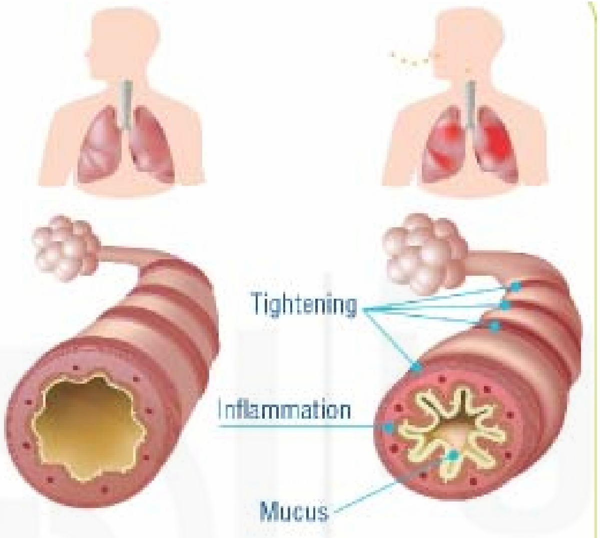

Fig. 1.19: Difference of Normal and Asthmatic lung

Asthma causes the lungs to become very sensitive and unable to function properly. Due to it, the airways become swollen, smaller and tighter causing difficulty to breathe properly. The lining of air tubes becomes swollen and excessive mucous is produced causing narrowing of airways. This hampers the process of respiration and the body receives lesser oxygen than what a normal human body does.

Causes

- Chest infection
- Allergy due to Pollens, Dust, Animal Fur, Smoke, Cold air or exercise
- Changes in temperature
- Stress and Exercise.

Recognition

The sign and symptoms which help to recognize Asthma are:

- Breathlessness and difficulty speaking more than a few words
- Wheezing and lot of coughing
- Poor skin colour, especially blueness of lips and fingertips
- Difficulty in breathing
- Tightness in the Chest
- In severe cases, gasping, confusion, collapse occurs.

1.7.2 First Aid in Asthma

The first aid care in Asthma is:

1) Help the victim to Rest

- Help the victim to sit upright (Fig. 1.20).

Fig. 1.20: Siting upright

- Loosen tight clothing.

2) Use Inhaler prescribed for asthma

- If the person has asthma medication, such as an inhaler, assist in using it. If the person doesn't have an inhaler, use one from a first aid kit or borrow from someone else.
- When using the Inhaler without spacer (Fig. 1.21)
  - Remove cap and shake inhaler well.
  - Have the person breathe out completely and put mouth tightly around mouthpiece.
  - Press inhaler once to deliver a puff.
  - Have the person breathe in slowly through the mouth and then hold breath for 4-10 seconds.
  - Give a total of four puffs, waiting about a minute between each puff and shaking inhaler each time when using.
- When using Inhaler with spacer (Fig. 1.22)
  - Remove cap and shake inhaler well.
  - Connect spacer to inhaler.
  - Have the person breathe out completely and put mouth tightly around mouthpiece of spacer.
  - Press inhaler once to deliver a puff.
  - Have the person breathe in slowly through the mouth taking 4 breaths waiting for about a minute between each puff.

Mouth piece

Fig. 1.21: Inhaler

Fig. 1.22: Inhaler with spacer

3) Continue Using Inhaler if Breathing is Still a Problem

- After four puffs, wait for four minutes. If the person still has trouble breathing, give another set of four puffs.
- If there's still little or no improvement, give four puffs every four minutes until ambulance arrives. If the person is having a severe attack, give up to six to eight puffs every five minutes.

4) Monitor the Person Until Help Arrives

- Keep on observing the victim.
- Be alert to resuscitate if required.

5) Transport immediately when the ambulance arrives.

Do's and Don'ts

Do's

- If the victim does not have any personal medication available, be prepared to borrow the inhaler from another person.
- Call the ambulance immediately.
- Try to use inhaler with spacer , if available.
- Loosen tight clothing.
- Ask the victim to breathe slowly and deeply.

Don'ts

- Don't Assume the person's asthma is improving if you no longer hear wheezing.

1.8 HIGH AND LOW BLOOD SUGAR

In this section , we will discuss the definition, causes, recognition and first aid in High and Low Sugar.

1.8.1 Definition, Causes and Recognizing High and Low Blood Sugar

Definition

High Blood Sugar

This is the condition where the body's blood sugar level is high and is also called as hyperglycemia (i.e. above 130mg/dl). It is common in diabetes patients who have very less or little insulin in the body which you have already studied in Unit 2 of Block 1 of this Theory Course. Hence, they are unable to maintain the blood sugar level in the body and therefore, suffer from frequent rise in blood sugar levels.

If it's not treated and gets worse, the person can gradually become unresponsive. So it's important to get them to see a doctor in case they need emergency treatment.

Low Blood Sugar

Low blood sugar is just the opposite of high blood sugar and is the condition in which the body's blood sugar level decreases below the normal levels (i.e. less than 70 mg/dl). It is also called as hypoglycaemia. It develops when the insulin-sugar balance is not maintained.

Also, when a person with diabetes misses a meal or does too much exercise, he becomes prone to suffer from Low Blood Sugar. It is more common in a person with newly diagnosed diabetes while they are becoming used to balancing sugar levels.

Causes

As already discussed the main reason for the high and low blood sugar is the insulin imbalance in body. In low concentration, insulin is not able to control the blood sugar level and hence, the blood sugar rises. If the concentration of blood sugar decreases due to any reason (e.g. medicines for diabetes), the person suffers from low blood sugar levels.

Hence, the various reasons for high and low blood sugar are listed as under:

1) High Blood Sugar
   - Low Insulin in the body due to low production
   - Not taking insulin or diabetic medicine correctly
   - High carbohydrate intake
   - Older people
   - Illness like dehydration, vomiting , infection , high fever which effects the sugar level especially in diabetics
   - Taking steroid medicines.
2) Low Blood Sugar
   - Increased energy demands from being cold or fatigued or stress or exertion
   - Taking too much medicines for Diabetes i.e. insulin (which improve Insulin Function and hence, causing the blood sugar imbalance)
   - Metabolic Problems
   - Lacking nutritious diet.

Recognition

If you think someone is having high or low blood sugar, then you need to check against the symptoms listed below which are as follows:

1) High Blood Sugar (Hyperglycemia)
   - Warm, dry skin
   - Rapid pulse and breathing
   - Fruity sweet breath
   - Feeling thirsty
   - Drowsiness, leading to unresponsiveness if not treated
   - Frequent passing urine.
2) Low Blood Sugar (Hypoglycaemia)
   - Weakness, fatigue or hunger
   - Confusion, irritation and nervousness
   - Sweating with cold, clammy skin

- Rapid pulse
- Trembling / seizures
- Decreasing levels of response/feeling sleepy/fainting.

1.8.2 First Aid in High and Low Blood Sugar

The first aid care is:

1) Assess the condition of the victim
   - Find out his problem. Ask for history of diabetes.
   - Try to recognize the problem with help of symptoms as given above in section 1.8.1.
2) Call for the ambulance immediately
3) Care for the victim
   – If you suspect High Blood Sugar
     - Make him/her sit down.
     - Assess his/her condition.
     - Ask him/her to drink plenty of water.

   – If you suspect low blood sugar
     - Give something sugary to eat like fruit juice, candy, sugar.
     - If they improve quickly, give them more sugary food or drink and let them rest.

Note:

In most of the cases, it is very difficult to recognize these two conditions as their sign and symptoms are almost same. If you recognize what the victim is having , well and good and if not, give the victim sugar anyway, as this will quickly relieve low blood sugar and is not going to do much harm in case of high blood sugar. So, always try to treat such victims with first aid of low blood sugar and take medical help immediately, if you are unable to recognize what the victim may be having.

4) While you wait for help to arrive, keep checking their breathing, pulse and level of response.
5) If the victim has becomes unconscious, assess him and be prepared to do CPR and use AED.
6) Thereafter, transport to medical facility.

Do's and Don'ts

Do's

- When providing sugar or sugary drink, tell the person to eat or drink it slowly.
- If the victim loses consciousness, be prepared to resuscitate the victim.
- In all conditions, take medical help.

Don'ts

- If unconscious, don't give the victim anything to eat or drink or put anything in mouth.

1.9 DROWNING

In this section, we will discuss regarding definition, causes, recognition and first aid in Drowning.

1.9.1 Definition, Causes and Recognizing Drowning

Definition and Causes

The condition Drowning is often an emergency encountered even in households or near large or small water storage areas e.g. swimming pools, lakes, storage tanks. In this situation, the person with no or little knowledge of swimming or instability due to various reasons when in the water gets completely immersed in the water, there by drowning. When a person is drowning, the air passages close to prevent water from entering the lungs. This prevents air from entering the lungs, thus reducing the oxygen supply to the lungs and hence, whole of the body. This leads to unconsciousness and even death if the person is not rescued and the airway and breathing is not restored .

Filling up of water in the lungs occurs only if the victim has been unconscious in the water for some time. More commonly, the water goes into the stomach. The death is not because of water filling up in the lungs but due to lack of oxygen in absence of breathing action in the drowning emergency. Another risk for the rescued person is that he or she may choke or vomit as water in the stomach forces the stomach contents upwards. Children and young adults are at the greatest risk of drowning.

Near drowning is the term given when the person who has witnessed drowning can be revived back to life (Fig. 1.23).

Fig. 1.23: Drowning

Recognition

The recognition of drowning can be done by the following signs and symptoms:

- Visible sight that the victim is drowning and calling for help.
- Victim will have:
  - Head low in the water, mouth at water level
  - Head tilted back with mouth open
  - Eyes unable to focus/closed

- Not using legs
- Trying to swim in a particular direction but not able to
- Trying to roll over on the back
- Difficulty or inability to wave for help.

- When taken out
  - Vomiting
  - Gasping or Coughing or difficulty breathing
  - Restlessness
  - Confusion/feeling sleepy
  - Pale appearance of the victim
  - Cold skin
  - Bluish skin – mostly around the victim's lips
  - Abdominal distention
  - Unconsciousness.

1.9.2 First Aid in Drowning

The first aid care is:

1) Rescuing a drowning person**:**
   - Call up the life guard and/or ambulance.
   - Shout to victim to get their attention.
   - If possible, reach to the person from the safety of solid ground using a pole rope or buoyancy aid to enable him to help himself out of the water.
2) Bring the victim to the shore/out of water.
3) Check for breathing by feeling air through the nose. See for the chest movement and listen for breath sounds. At the same time check the pulse.
4) If unconscious and not breathing, give CPR and use AED (Follow all precautions if using AED as leart in Unit 3 of Thoery Block 2).
5) If the person becomes conscious and starts breathing, turn him on the side and keep monitoring.
6) If the person vomits then also, log roll the person to the side and try to remove vomitus with help of fingers.
7) Remove wet clothing and provide dry clothes and cover with dry warm cloth to prevent hypothermia.
8) Transport the victim to medical facility.

Do's and Don'ts

Do's

- Victim must be taken to medical facility even if he/she regains consciousness.
- If the victim loses consciousness, be prepared to resuscitate the victim.

- Be cautious when bringing the victim out of water. You don't want to hurt his/her spine or neck. Avoid bending or turning his/her neck.
- Make sure to keep the victim still and calm.
- Keep the person as warm as possible, in order to prevent occurrence of hypothermia (discussed in Unit 2 of this Block).

Don'ts

- Don't rescue the victim if the water is deep and if you don't know swimming.
- Don't jump into ice water. Instead reach out to the victim with an extended hand or object.

1.10 POISONING

In this section, we shall be discussing regarding various types of poisoning and first aid in poisoning.

1.10.1 Definition, Causes, Types, Recognition and General First Aid for Poisoning

Definition

A poison is a substance that causes injury, illness or death if it enters the body. Poisons may enter the body in the form of liquids, solids, or gas and vapour fumes.

Poisons are substances that can cause temporary or permanent damage if too much is absorbed by the body. Poisons can be swallowed, inhaled, injected or absorbed through the skin.

Causes and Types of Poisons

The various types of Poisoning are as follows:

1) **Swallowed/Ingested poisons**

   Swallowed poisons include poisons which are swallowed through mouth. They are mostly the reasons for poisoning in children and mentally ill. The various causes of swallowed poisons are:

   A. Chemicals like acid, alkali, insecticides and pesticides etc.

   B. Drugs

   C. Contaminated food

   D. Alcohol

   E. Metals

   F. Household Cleaning Items.

2) **Inhaled Poisons**

   Inhaled poisons include poisons which are inhaled through nose. The various causes of inhaled poisons are:

   A. Carbon monoxide gas

   B. Toxic fumes from chemicals

C. Refrigeration gases

D. Chlorine and fumes from various sprays

E. Chemicals that give out fumes.

3) **Injected Poisons**

These are poisons injected into human body by poisonous animals. These have been discussed in Unit 5 of this Block.

4) **Absorbed Poisons**

These are poisons like caustic chemicals and poisonous plants with which one comes in contact and chemicals from these are rapidly absorbed causing burning, skin irritation, allergic reaction or severe systemic reactions as discussed in chemical burns in Unit 4 of Theory Block 3.

Recognition

The signs and symptoms which are seen in case of a poisoning are:

- Severe Nausea, Vomiting and Diarrhoea
- Seizures, fainting, confusion
- Pale, flushed, red skin
- Swelling, tenderness, cramps
- Breathing difficulty, Wheezing
- Skin irritation, change in color and other changes like burns etc.
- Shock and cardiac arrest.

General First Aid for Poisoning

The general first aid steps include the following:

- Call an ambulance.
- Also inform the police.
- Attend to the victim according to the type of poisoning (discussed in Subsection 1.10.2).
- Induce vomiting if the suspected poison is not corrosive or acidic.
- If any chemical, falls on the skin, ask the victim to remove the contaminated clothing and wash the areas exposed with water at room temperature (Fig. 1.24).

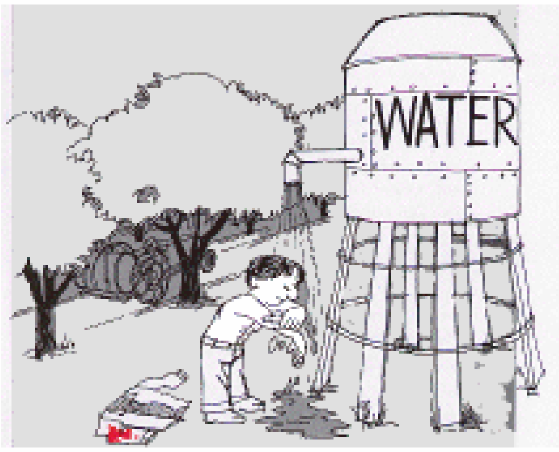

Fig. 1.24: Washing affected areas under water

- If any splash of chemical occurs in eye, ask the victim to rinse the eyes under gentle water stream for 10-15 minutes. Allow the stream to flow from the inner corner across the eye to the outer corner (Fig. 1.25). Also discussed in Unit 4 of previous Theory Block.

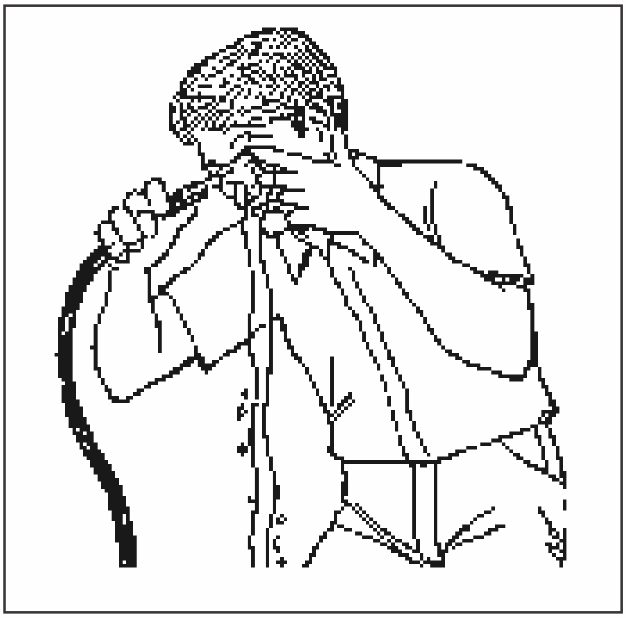

Fig. 1.25: Washing the eye with water

- Preserve any suspected bottle and vomited material .
- Collect history from bystanders/family/friends.
- Be prepared to give CPR, treat injuries, shock or fits. Resuscitate as per the status of the victim.
- Shift to hospital immediately.

1.10.2 First Aid in Various Types of Poisoning

Let us discuss some main forms of poisoning, their recognition and their first aid management.

A) Swallowed Poisons

1) Acid Poisoning

The episode of Acid Poisoning can be suicidal or homicidal or accidental. The various common acids used are nitric, sulphuric, hydrochloric, carbolic, oxalic and acetic acids.

Recognition

- Burns on or around the lips.
- Burning in the mouth, throat and stomach often followed by heavy vomiting.
- Diarrhoea and extreme thirst.
- In severe cases, victim may have unconsciousness, shock or seizure.

First Aid in Acid Poisoning

- Call for ambulance.
- Give half litre of water or milk to which chalk/milk of magnesia, plaster or whitewash (50 grams) has been added (Fig. 1.26).

Fig. 1.26: Plaster/chalk/white wash powder

- If milk or water is not available then olive oil, butter, white of an egg and barley water can be given.
- Shift to hospital immediately.

Do's and Don'ts

Do's

- Be with the victim.
- Try to note the type of acid consumed.
- Be prepared to resuscitate if required.
- Treat Shock.

Don'ts

- Don't induce vomiting. But you can rinse the mouth.

2) Alkali Poisoning

It can also be suicidal or accidental. Alkalis commonly used are ammonia, potassium hydroxide and sodium hydroxide, bleachers, detergents, washing soda.

Recognition

- Membrane of the mouth may be white and swollen
- Mouth has a soapy appearance
- Abdominal pain
- Vomiting may contain blood/mucus.

First Aid in Alkali Poisoning

- Give plenty of fluids: vinegar and citric acid, lemon or orange juice or barley water.
- Shift immediately to hospital.

Do's and Don'ts

Do's

- Be with the victim.
- Try to note the type of alkali consumed.
- Be prepared to resuscitate if required.
- Treat for Shock.

Don'ts

- Don't induce vomiting. But you can rinse the mouth.

3) Insecticide/Pesticide/Disinfectant like Phenyl/Petrol/Kerosense Poisoning

Various insecticides and pesticides like DDT etc. cause poisoning. Organophosphorus Compounds are also common components of pesticides and insecticides in agriculture and homes which are very lethal. Various disinfectants especially phenyl used for house hold cleaning are misused as poisons (Fig. 1.27).

Fig. 1.27: Insecticide/Pesticide/Disinfectants

Recognition

- Nausea, vomiting, fainting, fits
- Pain in abdomen , diarrhoea
- Unconsciousness
- Respiratory failure.

First Aid in Insecticide, Pesticide, Disinfectant, Kerosene or Petrol Poisoning

- Call ambulance
- Give plenty of water.
- You can add 2 tablespoons of Epsom salt in water (Fig. 1.28).

Fig. 1.28 : Epson Salt (Seendha Namak in Hindi)

- Shift to hospital.

Do's and Don'ts

Do's

- Be with the victim.
- Collect required details from the person or look for empty bottle present if in liquid or solid form to find out type of poison.

Don'ts

- Don't induce vomiting.

4) Cyanide Poisoning

Cyanide is the most lethal poisoning. It can also be inhaled or ingested. (Fig. 1.29).

Fig. 1.29: Cyanide

Recognition

- Headache, dizziness
- Nausea, decrease in BP, difficulty breathing
- Drowsiness, Convulsions
- Unconsciousness
- Foam in the mouth
- Respiratory failure
- Smell of bitter almonds

First Aid in Poisoning

- Call for an ambulance
- Start resuscitation immediately
- Shift to hospital.

Do's and Don'ts

Do's

- Be with the victim.
- Collect required details about what is swallowed from the person or look for empty bottles present to identify the posion.
- Time is crucial so you have to immediately shift the victim to hospital.

Don'ts

- Don't give anything or try to induce vomiting. Just shift the victim to nearest medical facility.

5) Metal Poisoning

Various metals especially lead and mercury lead to poisoning. Lead is used in paints and dyes and in old pipes of building. Mercury is commonly present in thermometer and if broken in mouth can cause mercury poisoning. Other metals like arsenic and phosphorous which are active components of rat poison also cause problems.

Recognition

- Metallic taste in mouth
- Nausea and abdominal pain
- Vomiting
- Stools may be dark red in colour
- Headache, drowsiness, fits
- Mouth is coated with colour as per the metal e.g. in lead poisoning, a blue line is seen on gums and in case of mercury, it is grey colored.

First Aid in Metal Poisoning

- Collect history of contact with metals like mercury or lead.
- Give plenty of warm water.
- Milk, white of egg, barley water can be given, if available.
- Induce vomiting if victim is conscious. Try to induce vomiting by putting fingers in the mouth and tickling the back part of the mouth. Try to keep the vomitus for investigation purposes (Fig. 1.30).

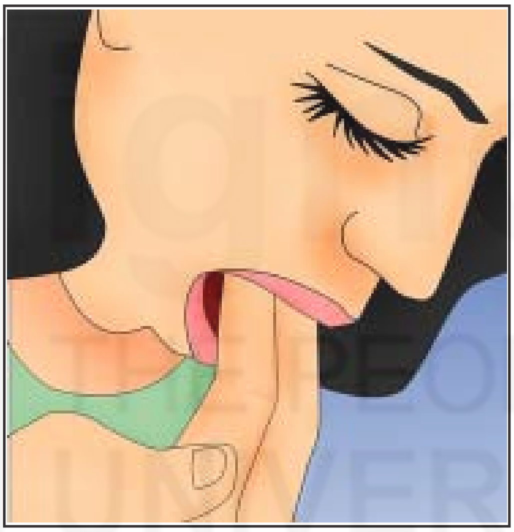

Fig. 1.30: Inducing vomitting

- Shift to hospital.

Do's and Don'ts

Do's

- Be with the victim.
- Collect required details/history about what is swallowed from the person or look for empty bottles present.

Don'ts

- Don't give oils.

6) Drug Poisoning

Various drugs lead to poisoning. Most commonly used medicines like Sleeping tablets and aspirin are misused for suicidal poisoning.

Recognition

The common sign and symptoms which helps in recognition of these poisons are:

1) Sleeping Tablets
   - Euphoria
   - Talkativeness
   - Headache
   - Dizziness
   - Feeling Sleepy.

2) Aspirin
   - Abdominal pain
   - Vomiting
   - Drowsiness
   - Ringing in the ears
   - Difficulty in breathing
   - Lot of sweating
   - Fast pulse.

First Aid in Drug Poisoning

- Call for ambulance right away. Collect required details about swallowed tablets etc.
- Give plenty of fluids orally. For sleeping tablets add spoonful of Epsom salt in water and for aspirin add soda bicarbonate in water.
- Induce vomiting if victim is conscious.
- Shift to hospital .

Do's and Don'ts

Do's

- Be with the victim.
- Collect required details about what is swallowed from the person or look for empty bottles present.

7) Poisons from plant sources

The various plant sources and their poisons include Castor oil Plant (common in children), mushroom , berries, jamal gota, bhang (an intoxicant), datura (its dried leaves and dried seeds are used as poisoning , mitha zahar, coca plant (cocaine), tobacco and opium.

Recognition

- Pain in throat and abdomen.
- Nausea, Vomiting, Diarrhoea
- Sensations and different tastes in mouth

- Excitement
- Visual hallucination
- Laughter
- Increase in appetite
- Dizziness, tingling and numbness
- Dry hot skin, rise in temperature
- Lot of sweating
- Breathing difficulty and unconsciousness.

First Aid in Poisoning from plant sources

- Call the ambulance.
- Give plenty of water. In case of mushroom castor oil can be given.
- Induce vomiting.
- Shift to hospital.

Do's and Don'ts

Do's

- Be with the victim.
- Immediately transport to the hospital.
- Collect required details about what is the cause of this poison.

8) Food Poisoning

It occurs by consumption of food, which is contaminated by bacteria and is stored or cooked incorrectly. Someone may feel the effects of food poisoning within a few hours, and will often be sick or have diarrhoea. However, in some cases it can take up to three days.

The effects of food poisoning can make someone feel very ill. The most important thing is for you to keep encouraging the person to drink water so they don't get dehydrated. Most people will get better without needing treatment.

Recognition

- Nausea and vomiting
- Headache
- Abdominal pain
- Diarrhoea
- Symptoms and signs of shock.

First Aid in Food Poisoning

- Call an ambulance.
- Keep the victim at rest
- Give plenty of fluids to drink

- Induce vomiting
- Shift the victim to hospital.

Do's and Don'ts

Do's
• Be with the victim.
• Immediate transport to the hospital.

9) Alcohol Poisoning

Alcohol poisoning is what happens to someone when they have drunk a large and a dangerous amount of alcohol within a short period of time.

Drinking too much alcohol stops the brain from working properly which can reduce the mental and physical body functions like sight, speech, coordination, balance and memory. It can also cause unconsciousness and even death.

Recognition

- Smell of alcohol
- Vomiting, fits
- Slurred speech
- Confusion, Lack of co-ordination
- Vision problems
- Flushing of face
- Rapid pulse
- Shallow breathing.

First Aid in Alcohol Poising

- Call an ambulance
- Give water, milk or white of egg
- Induce vomiting
- Shift the victim to hospital.

Do's and Don'ts

Do's
• Be with the victim.
• Immediate transport to the hospital.

B) Inhaled Poisons

If volumes of smoke or gases like carbon-monoxide are inhaled or breathed in, it can be deadly. If a victim has inhaled fumes they need immediate medical attention as they are likely to have low levels of oxygen in their blood and tissues.

Carbon monoxide is a poisonous gas. When inhaled, the gas does not allow the red blood cells to carry oxygen to the body's tissues and organs which can create problem and prove fatal. Moreover, this gas is difficult to detect as it has no taste or smell.

Unfortunately, lots of people who have carbon monoxide poisoning may not be aware that they are affected and may wrongly blame their symptoms on other reasons. The young and the old are at increased risk.

Recognition

If the casuality has had exposure to low levels of carbon monoxide for prolonged periods of time, they may complain of:

- Headaches
- Confusion
- Feeling aggressive
- Nausea and vomiting
- Diarrhoca

Severe symptoms may include:

- Grey-blue skin
- Rapid, difficult breathing
- Impaired level of response, leading to unconsciousness and even death.

First Aid in Inhaled Poisons

- Call the ambulance and fire services if you suspect formation and inhalation of fumes.
- Maintain scene safety :
  - Rescue the Victim and get him/her away from the source of the fumes into an environment with fresh air.
  - Do not enter the fume-filled area yourself.
  - If the casuality is in a closed space filled with exhaust fumes, open the doors letting the fumes escape before you enter.
- Support the victim in a comfortable position and encourage to breathe normally.
- Treat any burns or other injuries or shock.
- Be prepared to resuscitate.
- Stay with them until the emergency services arrive.
- Transport to hospital immediately.

Do's and Don'ts

Do's

- Call emergency services immediately.
- Be with the victim.
- Immediate transport to the hospital.
- Be alert for scene safety.
- Be prepared to resuscitate the victim.
- Loosen constrictive clothing of the victim.

Don'ts

- Don't enter if the scene is unsafe.
- Don't become a victim yourself.

C) Absorbed Poisons

These usually come in contact with the skin and eyes causing irritation, itching, redness in affected areas. The best first aid is to remove the clothes and wash the area or the eyes as already discussed in the General First Aid in Poisoning in Subsection 1.10.1. Use a mild soap to wash the area and rise as required. Use calamine lotion on the area. You can also study this in chemical Burns in Unit 4 of Theory Block 3.

D) Injected Poisons

These enter the body through bites and stings and have been discussed in Unit 3 of this Block.

Check Your Progress 2

1) How to use inhaler in case of asthma victim?

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2) List the steps of first aid in case of High Blood Sugar.

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3) Enumerate the first aid in following:

a) Mercury Poisoning

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b) Phenyl Poisoning

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1.11 LET US SUM UP

In this Unit, we studied regarding various common emergencies occurring in daily life and their recognition along with first aid management. The unit deals with definition, recognition and first aid management of victims with various common emergencies i.e. fits, fainting, chest pain, asthma, high and low blood sugar, drowning and poisoning. We studied a detailed overview of these conditions. The practical units 5, 6, 7, 8 and 9 in Practical Block 2 of this programme deals with the practical aspects of these common emergencies. For your information, a national poisons information centre with toll free number **1800 116 117** has been set up at AIIMS, New Delhi for gathering information on first aid in various poisons. In the next unit we will study regarding extreme heat and cold conditions.

1.12 KEYWORDS

Witnessed	:	Seen
Vigorous	:	Very strong and full of energy/powerful
Full-scale	:	Full, big, comprehensive
Violent	:	Using or involving physical force/extreme
Jerking	:	Move or cause to move suddenly/twitching
Foaming	:	Form or produce a mass of small bubble
Clenched	:	Pressed tightly
Uncontrollable	:	Unmanageable, Full of disorder
Excess	:	Surplus or abundance of something
Episode	:	Event , occurrence
Sponging	:	Wipe or clean with a wet sponge or cloth
Swallow	:	Pass through mouth down the throat into stomach
Temporary	:	For a short time
Reduction	:	Decrease
Over exhaustion	:	Extreme tiredness/exertion
Massive	:	Large
Dizziness	:	Losing consciousness and balance
Affected	:	Impact, Influence
Disturbance	:	Interruption of a settled and peaceful condition
Paralysis	:	Loss of the ability to move
Muscle tone	:	Firmness or strength of a muscle
Dribbling	:	Fall slowly in drops or a thin stream/trickle down
Bladder	:	Representing urine passing
Bowel	:	Represents stool passing
Deteriorating	:	Decreasing

Alarming	:	Cause to get alarmed or alert
Twitching	:	Jerking movement
Mishandle	:	Not handle or care for properly
Radiates	:	Spreads
Subsides	:	Drops/falls/collapse
Droop	:	Bend or hang downwards
Shortness	:	Being small or shallow
Cramping	:	Muscular contractions/Spasms
Outdoors	:	Not at home/outside
Crushing	:	Something that squeezes or presses strongly
Complication	:	Problem arising as a result of something else
Kneel	:	Position in which the body is supported by a knee
Cough	:	Expel air from the lungs with a sudden sharp sound
Pale	:	Light /white
Pressure	:	Some weight, force
Half-sitting	:	Sitting with upper part of body at 45 deg to the legs, in between sitting an lying down
Narrowed	:	Decrease in size and width
Wheeze	:	Whistling sound in the chest
Production	:	Make something
Hampers	:	Hinder or hold back or create problems
Gradually	:	Slowly
Immersed	:	Completely put in the water
Tilted	:	Lean or slope down
Distention	:	Swelling /enlargement
Fumes	:	Gas or vapour that smells strongly or is dangerous to inhale
Flushed	:	Red and hot
Rinse	:	Wash/clean
Suicidal	:	Deeply unhappy or depressed and likely to commit suicide
Homicidal	:	Hurt other with possibility of murder
Accidental	:	As a result of accident, happening by chance
Mucus	:	A substance secreted by the mucous membranes and glands
Soapy	:	Just like soap (with foam)
Appearance	:	Way that someone or something looks
Components	:	Parts
Tickling	:	Muscular twitching/jerking movement

Euphoria	:	Extreme Happiness
Intoxicant	:	Drugs which reduce pain and cause decrease in consciousness
Hallucination	:	Seeing things which actually don't exist
Tingling	:	Experience a slight prickling or stinging sensation
Numbness	:	Lack of feeling or sensation
Appetite	:	Hunger or Craving for something
Coordination	:	Working in cooperation or with one another
Slurred	:	Speak/stammer, not speaking clear words
Corrosive	:	Caustic, which burns something or is a harsh material.

1.13 ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress 1

1) Common emergency is a term used to denote the emergencies which occur commonly. These emergencies are associated with medical conditions, illness or sickness and are experienced many of the times. These emergencies occur everywhere and many persons/people report these problems on day to day basis. That's why they are called Common Emergencies
2) Refer Subsection 1.3.1
3) Refer Subsection 1.5.1
4) Refer Subsection 1.6.2

Check Your Progress 2

1) Refer Subsection 1.7.2
2) Refer Subsection 1.8.2
3) Refer Subsection 1.9.2

1.14 REFERENCES AND FURTHER READINGS

1. https://www.cdc.gov/epilepsy/about/first-aid.htm
2. https://hubpages.com/health/What-to-do-when-people-faint-First-aid-for-fainting
3. https://www.ucirvinehealth.org/blog/2016/05/stroke-dos-and-donts
4. https://www.boldsky.com/img/2016/06/ayurvedic-remedies-for-treating-fits-epilepsy-29-1467179051.jpg
5. http://www.epilepsiemuseum.de/alt/introen.html
6. http://4.bp.blogspot.com/-HXZre6YvdFY/Uf4EKWQg4VI/AAAAAAAAAm4/avP-E3ABOwA/s1600/8.jpg
7. http://drugline.org/medic/term/febrile-seizure/
8. http://www.sja.org.uk/sja/first-aid-advice/first-aid-for-parents/seizures-in-children.aspx
9. https://www.youtube.com/watch?v=owXhSD7XwUk

10. https://www.wikihow.com/Stop-Your-Child%27s-Febrile-Seizure
11. http://www.pediacare.com/category/fever-relief/page/2/
12. https://www.wikihow.com/Stop-Your-Child%27s-Febrile-Seizure
13. http://queitee.blogspot.in/2011/01/first-aid-for-seizures-1.html
14. http://www.stjohn.org.nz/First-Aid/First-Aid-Library/Seizures-or-Convulsions/
15. https://www.emedicinehealth.com/fainting/article_em.htm#fainting_causes
16. http://neurologist-ahmedabad.com/2017/10/04/fainting-3-possible-causes/
17. https://www.medicinenet.com/fainting/article.htm#heart_rhythm_changes
18 https://www.healthline.com/symptom/fainting#modal-close
19. https://familydoctor.org/condition/fainting/
20. https://www.nhs.uk/conditions/fainting/causes/
21. http://www.heart.org/HEARTORG/Conditions/Arrhythmia/Symptoms DiagnosisMonitoringofArrhythmia/Syncope-Fainting_UCM_430006_Article.jsp#.WnK0tqiWbIU
22. https://www.webmd.com/brain/understanding-fainting-basics#2
23. https://www.medicinenet.com/fainting/article.htm#what_causes_fainting_syncope
24. https://www.emedicinehealth.com/fainting/article_em.htm#fainting_diagnosis
25. https://www.youtube.com/watch?v=yro2VGKHlR4
26. https://www.medicinenet.com/stroke_symptoms_and_treatment/article.htm
27. http://www.wales.nhs.uk/sites3/images/840banner.jpg
28. https://www.mayoclinic.org/first-aid/first-aid-stroke/basics/art-20056602
29. http://www.wales.nhs.uk/sites3/home.cfm?orgid=840
30. http://www.mydr.com.au/heart-stroke/first-aid-for-stroke
31. https://www.mayoclinic.org/first-aid/first-aid-chest-pain/basics/art-20056705
32. https://www.medicinenet.com/causes_of_chest_pain_signs_and_symptoms/article.htm#chest_pain_in_women
33. http://www.stjohn.org.nz/First-Aid/First-Aid-Library/Chest-Pain/
34. https://www.cardiachealth.org/your-heart/your-coronary-arteries/angina/prinzmetals-angina/
35. https://www.webmd.com/heart-disease/most-common-areas-of-pain-caused-by-angina
36. https://www.ausmed.com/articles/chest-pain-assessment/
37. https://udaipurtimes.com/dos-and-donts-in-case-of-chest-pain/
38. http://www.india.columbiaasia.com/health-articles/be-emergency-ready-dos-and-donts
39. https://www.lung.ca/asthma

40. http://www.sja.org.uk/sja/first-aid-advice/illnesses-and-conditions/asthma-attack.aspx

41. https://www.webmd.com/first-aid/asthma-treatment

42. http://uniquetimes.org/natural-ways-to-fight-asthma/

43. https://www.pinnaclehealth.org/wellness-library/blog-and-staywell/health-resources/article/22365

44. https://chrisdudley.org/resources/first-people-diabetes/

45. http://firstaidwindsor.ca/first-aid-for-hyperglycemia/

46. https://www.stjohnqld.com.au/getmedia/f6098b8d-4640-404f-9335-3e34554a5966/diabetes.pdf.aspx?ext=.pdf

47. https://www.youtube.com/watch?v=gw7GpLvZHME

48. https://www.youtube.com/watch?v=L06DNMRcy98

49. https://www.reddit.com/r/LifeProTips/comments/4o6tbk/lpt_how_to_recognize_when_someone_is_drowning/

50. https://www.cprcertificationonlinehq.com/first-aid-for-drowned-victims/

51. http://www.onlinebangalore.com/heal/tips/poison.html

52. http://healthywa.wa.gov.au/Articles/N_R/Poisoning-first-aid

UNIT 2 EXTREME HEAT AND COLD CONDITIONS

Structure

2.0 INTRODUCTION

Extreme heat and cold weather conditions, have a bad effect on the health. Extreme heat and cold conditions effect the body of those persons, who spend lot of time outside the house making them exposed to severe heat and cold, environmental and climatic conditions. In this unit, we shall be discussing regarding the various adverse heat and cold conditions which are important to be understood by a first aid provider since proper knowledge is essential when dealing with such situations and also for handling the complications that may arise as a result of such problems.

The extreme heat and cold conditions affect human body and can cause the development of various problems that should be understood and cared for immediately. Hence, in this unit we will discuss about these conditions in detail.

2.1 OBJECTIVES

After completion of this unit, you shall be able to:

- define extreme heat and cold conditions;

- list the causes of extreme heat and cold conditions;
- recognize the types of extreme heat and cold conditions;
- explain the assessment of victim affected by extreme heat and cold conditions;
- describe the first aid to be provided in extreme heat and cold conditions; and
- enumerate the Do’s and Don’ts in extreme heat and cold conditions.

2.2 EXTREME HEAT AND COLD CONDITIONS

In this section we shall have an overview regarding extreme heat and cold conditions along with their assessment and general first aid.

2.2.1 Concept, Risk Factors and Recognition of Extreme Heat and Cold Conditions

Concept

The body’s temperature is controlled by a number of processes. You have learnt about the concept of Body temperature in Unit 2 of Practical Block 1. It is hence known to you that the body temperature in humans is regulated by the brain and is affected by a number of factors. Our normal temperature is about 37°C. When the body maintains normal temperature, it is said to be in the state of balance. However, when the temperature of the body increases or decreases as per the environment because the body's heat regulating mechanism is not able to regulate the body temperature, it is not normal and can cause life threatening emergencies that require immediate action and medical care. Let us discuss these conditions here.

Extreme Heat Conditions

Prolonged exposure to heat or high temperatures effects the body temperature of a person and people of all ages creating signs of illness due to heat (Fig. 2.1). Those at high risk are the very young, elderly people with some active or chronic health problems, and people taking certain medicines. Also people exposed directly to heat when doing certain activities may also be badly effected by the extreme heat, i.e. Athletes, military personnel, farmers, construction workers, labourers, besides this the individuals with problems of cardiovascular disease, obesity, diabetes mellitus, malnutrition and alcoholics are prone to get effected by extreme heat conditions.

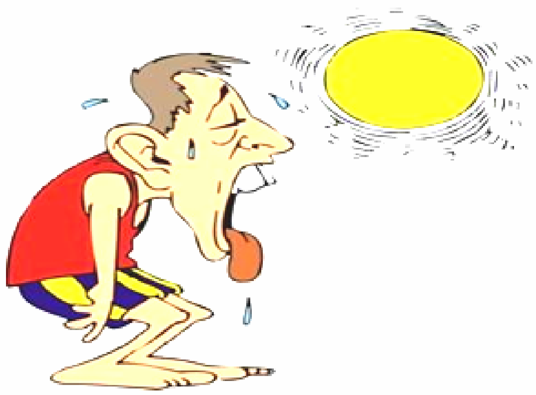


Fig. 2.1: Extreme Heat Conditions

Extreme Cold Conditions

Extreme cold effects the body of the victim. It usually occurs when the body is exposed to very low temperatures. As a result the body temperature also falls below normal level and injuries to exposed body parts can occur. These are common among those who have spent lot of days in cold weather or those who have no information on the danger of wearing wet, tight fitting socks and foot wear in extremely cold conditions. Injuries during extreme cold conditions are painful, debilitating and may take weeks to heal. These can effect upto such a great level that amputation of the affected part may be required (Fig. 2.2).

Fig. 2.2: Extreme Cold Conditions

Risk factors

- Exposure to extreme heat or cold environment
- Trauma/friction
- Dehydration
- Age-related physical problems
- Certain medical problems related to heart and lungs
- Drugs and alcohol
- Mental impairment that causes someone to be unaware of cold.

Recognition of Extreme Heat Conditions

The initial symptoms are mild but if the exposure of heat is not reduced, the situation can become life-threatening.

a) Mild symptoms are :
   - Nausea
   - Headache or dizziness
   - Sweating
   - Feeling uneasy, fatigue and feeling tired
   - Heating of the skin.

b) In case the exposure lasts a longer time, the symptoms are:
   - Fast pulse and fast breathing
   - Hot skin, redness may be present, sweating may be present
   - Fainting , Fits and Collapse.

Recognition of Extreme Cold Conditions

A victim experiencing extreme cold condition may show these signs and symptoms:

- Shivering
- Numbness of the exposed part
- Fainted or white or blue coloured appearance of the exposed part, hard skin with no pain, blisters may be present
- Confusion
- Weak pulse, slow respiration.

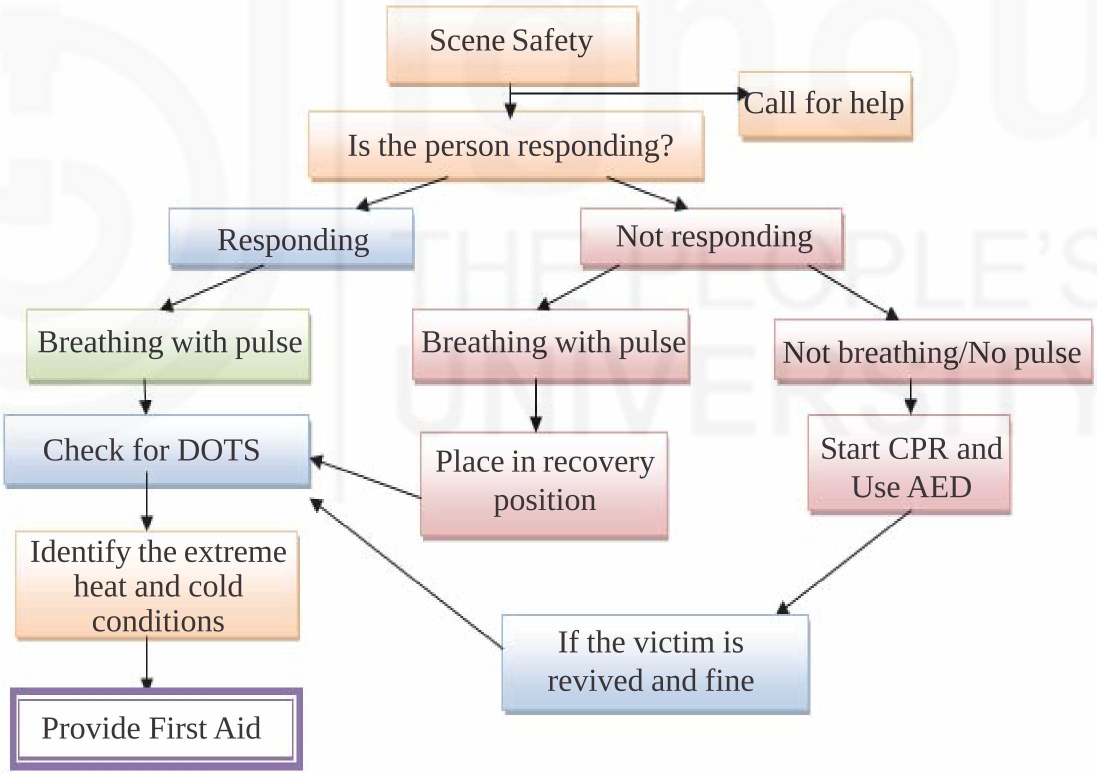
We shall learn about specific heat and cold conditions and their recognition in this Unit in Sections 2.3 and 2.4 respectively.

2.2.2 Assessment of the Victim and General First Aid

The assessment of victim and general first aid is as follows:

General Assessment

You have already learnt the assessment to be done while providing first aid in Unit 1 of Block 2 of the Theory Course. The general assessment is done as per following flowchart.



Specific Assessment

After general assessment, perform specific assessment as:

1) Check for DOTS and look for deformity, swelling, tenderness and any other sign and symptoms.
2) Check for the sign and symptoms of extreme heat and cold conditions. Check how severe is the condition is (as given in section on Recognizing extreme heat and cold conditions in 2.2.1).

3) Check for specific problems due to extreme heat (if there is heat stroke or heat exhaustion or sun stroke) or if there is extreme cold situations (frost bite, immersion foot etc). These have been discussed in upcoming sections 2.3 and 2.4 of in this unit.

4) Ask History from the victim or onlookers present.

General First Aid

After assessment , provide first aid as follows:

1) Extreme Heat Conditions

- Stop the activity causing the heat build-up.
- Move the casuality to a cooler place.
- Remove as much of their clothing as possible.
- Make the person lie down with their legs and feet raised (shock position) discussed in Unit 5 of this block and Unit 13 of Practical block 2 or if unresponsive and breathing, place the casuality in the recovery position (discussed in Unit 7 of Practical Block 1).
- If conscious give small sips of water if victim can drink. Cool the victim's body with water, and try to provide air with fan to increase the cooling effect.
- Rest and reassure.
- Perform Secondary assessment and Transport to medical facility if required.

Do's and Don'ts

Do's

- Monitor the temperature and predict the kind of heat emergency.
- Cool the victim slowly and carefully.
- Be alert for changes of responsiveness and other complications.
- Be ready to resuscitate as and when required.

Don'ts

- Don't give the person anything to eat or drink if they are vomiting or is unconscious.
- Never give liquids that contain alcohol or caffeine, as they interfere with the body's mechanism of controlling internal temperature.
- Never underestimate the seriousness of heat illness, particularly in children, the elderly or injured persons.
- Don't give medications like paracetamol or aspirin as they may be harmful.

2) Extreme Cold Conditions

- Remove the victim from the cold environment.
- Remove any wet clothing.
- Attempt to warm the casuality with blankets and clothing.
- If unresponsive but breathing, place the casuality in the recovery position (discussed in Unit 7 of Practical Block 1).

- Perform Secondary assessment and Transport to medical facility as required.

Do's and Don'ts

Do's

- Monitor the temperature.
- Warm the victim slowly and carefully.
- Be alert for changes of responsiveness and other complications.
- Be ready to resuscitate as and when required.

Don'ts

- Don't give the person anything to eat or drink if they are vomiting or unconscious.
- Don't rub or massage the affected area with your hands or with snow or ice.
- Don't break any blisters.
- Don't put pressure on affected areas and avoid walking on cold affected areas.
- Don't smoke or drink alcohol .
- Don't apply any medication on the effected part.
- Don't use hot water bottles, heating pads etc directly on the skin for warming purpose.

Note:

Be alert to deal with Cardiac arrest or any other complications that may arise. In case the situation worsens or if the situation is worse when approached, immediately transport to a medical facility. Remember that these emergencies can lead to death.

Thus, in this section we studied about the concept of heat and cold related emergencies. We also learnt about recognition and assessment of these situations and your role in these situations. In the next section, we will take up specific conditions and discuss them in detail.

2.3 EXTREME HEAT CONDITIONS

The various extreme heat conditions also called as heat related emergencies which need to be dealt by you as a first aid provider include heat stroke , heat exhaustion, heat cramps, sub burn, high fever and dehydration. In this section, we will discuss these in detail. Hence, lets begin.

2.3.1 Heat Cramps

Definition

These are cramps, contractions or spasms which occur in muscles due to hot environmental conditions. These are painful and usually experienced by persons who sweat a lot when working or playing outdoors. They usually tell that the body is getting exhausted due to sun's heat or in hot environmental condition (Fig. 2.3). It is a mild form of heat related emergency.

Fig. 2.3: Heat Cramps

Causes

In hot weather people who work hard and sweat a lot may sometimes get painful cramps in legs, stomach and arms. This occurs when body loses water and salt due to sweating.

Heat cramps are especially seen in older people, small children, overweight people, and people who have been drinking alcohol. They are also seen in people who work outdoors in sun or do strenuous exercise.

Recognizing Heat Cramps

The following symptoms help recognize heat cramps:

- Muscle pain
- Tightness in muscles
- Red skin
- Heavy sweating and feeling very thirsty
- Fatigue, tiredness, Dizziness, fainting

First Aid in Heat Cramps

- Move victim to a cool place. Make him/her rest.
- Give fluids. You can also give energy drinks or add one teaspoon of salt in a litre of water and drink it. One can also add little sugar and lime juice to the drink (Fig. 2.4).
- Massage affected muscles and gently stretch them.
- Apply moist towels or covered ice packs to forehead and cramped muscles, if available.
- If problem persists, seek medical help.

Fig. 2.4: Rest in shade and drink water with salt/sugar and lime juice

Do's and Don'ts

Do's

- Monitor properly to find if heat exhaustion or heat stroke is present (as discussed in later sections of this unit).
- Go to rest in shady and cool area.
- Stop the strenuous activity, immediately.

Don'ts

- Don't give anything to drink if nausea or vomiting is present.
- Don't wet the victim and don't pour water on the cramped muscles.

2.3.2 Heat Exhaustion

Definition

Heat Exhaustion is a heat related illness that can occur after one has been exposed to high temperatures. It is a more severe type of heat related emergency. It occurs when body is not directing blood supply to skin for cooling itself. It is a condition of overheating of the body with lot of sweating and a fast pulse. Heat exhaustion occurs when there is a mild rise in body temperature (to less than 40º C). (Fig. 2.5).

Fig. 2.5: Heat Exhaustion

The casuality can become dehydrated due to water and salt loss from body. It is important to recognize and care promptly, since time lapse can cause development of a severe condition of heat stroke.

Causes

It is related to loss of water and salt from the body due to excess sweating. It occurs when an exposure to high temperature like highly humid weather or environmental conditions occurs where there is no circulating air to cool the body. Heat Exhaustion can happen suddenly or overtime, when you exercise for longer periods in hot and humid weather. If not treated , it can lead to heat stroke which has been discussed in Sub section 2.3.3.

Recognition of Heat Exhaustion

The following signs & symptoms help to recognize Heat Exhaustion:

- Body temperature between 98.6ºF (37ºC) and below 104ºF (40ºC).
- Cool, moist, pale skin
- Lot of sweating

- Faintness/dizziness (also called as heat syncope)
- Fast pulse
- Muscle cramp (Heat Cramps)
- Headache
- Tiredness
- Nausea, Vomiting , diarrhoea.

First Aid in Heat Exhaustion

The first aid in Heat Exhaustion includes:

- Reassure the victim .
- Move to a cool, shady place and make him/her lie and lift the legs upto 20-30 cm.
- Loosen his/her tight clothing.
- Cool the victim by fanning him/her or by using wet towels on face, neck, chest, hands and feet. If possible bath can also be taken (Fig. 2.6).

Fig. 2.6: Applying wet towel on chest

- Provide plenty of water with a pinch of salt in it (as done in case of heat cramps).
- Transfer to nearby hospital if problem persists.

Do's and Don'ts

Do's

- Monitor properly to find if heat exhaustion or heat stroke is present (as discussed in later sections of this unit) as heat exhaustion can easily develop into stroke.
- Go to shady and cool area.
- Stop the strenuous activity, immediately.
- Give nothing by mouth if the victim is unconscious.
- Massage the cramped muscles, as needed.
- While giving fluid to the victim, make him sit and see that he is fully responsive.
- Tell the victim to drink the water slowly. Don't rush when drinking the water.

Don'ts

- Don't give anything to drink if nausea or vomiting is present.
- Don't wet the victim and don't pour water on the cramped muscles.
- Don't give coffee or tea.
- Don't give fatty or oily foods to eat.

2.3.3 Heat Stroke

Definition

Also called as sun stroke, it is an acute and serious emergency. It is a condition in which body's temperature rises suddenly upto 104ºF (40ºC) and body can no longer control its temperature by sweating. Thus, it occurs when heat regulating mechanism of body fails.

It is not very common but it develops suddenly and is very fatal if not immediately treated (Fig. 2.7).

Fig. 2.7: Heat Stroke

Heat stroke is a more severe condition than Heat Exhaustion as clear from Fig. 2.8. Heat Exhaustion if not cared for can complicate to Heat Stroke.

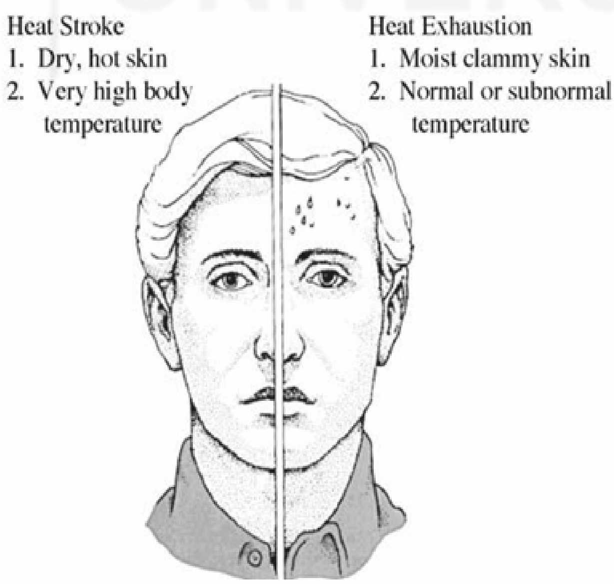


Fig. 2.8: Difference of Heat Stroke and heat Exhaustion

Causes

In this condition, when the environmental temperature particularly in summers is high the body's heat control system is not able to cope with it. The body overheats and the body's temperature regulating centre is not able to maintain the body's normal temperature. The victim is dehydrated, the body's water level goes down and thus, no sweating occurs. Hence, as the body heats up, severe damage to internal organs can occur which can complicate to death of the person. It occurs especially in older people and alcoholics in hot weather. It also effects people who are physically active and work in sun especially gardeners , labourers and other workers.

Associated risk factors are:

- Exposure to heat for long time
- Prolonged hot atmosphere
- Ill person , old age, babies, alcoholics and physically active
- Taking some medications

Most heat stroke related deaths occur in the elderly because their circulatory system is not able to compensate the stress caused by heat and they have decreased ability to sweat.

Recognition of Heat Stroke

The Sign and symptoms of Heat Stroke are as follows:

- High Body Temperature
- Skin is red, hot and dry
- Confusion, fainting
- No sweating
- Head ache, nausea, vomiting
- Fast heart rate
- Body cramps
- Dizziness, fainting (also called as heat syncope)
- Person may become unconscious.

First Aid in Heat Stroke

- Move the person in shade, cool place.
- Loosen constrictive clothes and jewellery. Take them off as required.
- If a large source of water like pond or bath tub is available, put the victim up till neck in it, to immediately cool him (Fig. 2.9).

Fig. 2.9: Putting victim in large water source

If large source of water is not available:

- Wrap the victim in cold wet sheet and pour water over him.
- Apply wrapped ice packs to neck, armpits and groin.
- Raise the feet and keep him/her cool with fan (Fig. 2.10).

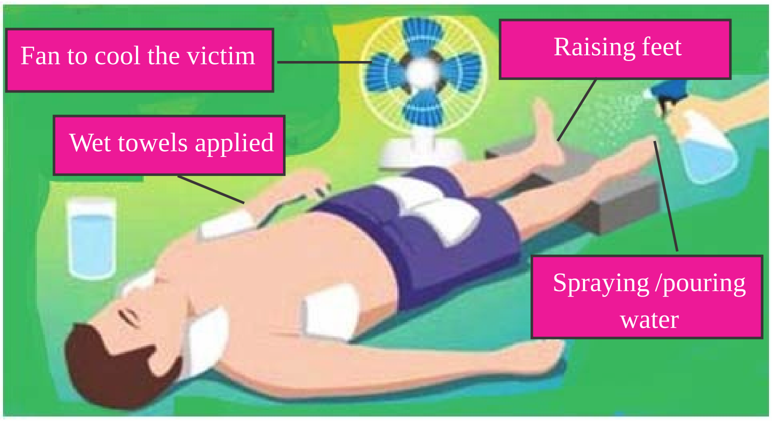


Fig. 2.10: Care in Sunstroke

- Take temperature and see if it has come down. If it comes down, dry the victim and monitor. If the temperature rises again, repeat the above steps.
- If conscious and temperature is lowered, give plenty of water to drink.
- Take medical help, if the condition does not improved.

Do's and Don'ts

Do's

- Monitor properly. Be alert for changes in consciousness and if resuscitation is required or not.
- If victim is not responsive but has pulse and breathing, put him in recovery position.
- Give nothing by mouth if the victim is unconscious.
- While giving fluid to the victim, make him sit and see that he is fully responsive.

Don'ts

- Don't give coffee or tea.
- Don't immerse the victim in ice, but only plain bathing water or shower or water from the tap with pipe.
- Don't give any medicine like paracetamol or aspirin.

2.3.4 Sun Burn

Definition

Sunburn occurs due to direct exposure of human body to the sun for a long time. It can be superficial or blisters may also form (Fig. 2.11). It is also called as sun poisoning.

Fig. 2.11: SunBurn

Causes

- Over exposure to sun rays particularly in very hot weather
- Windy climate with strong sun rays
- Winter on high mountains (ultraviolet rays are strong causing sunburn)
- Modern tanning beds
- Phototherapy lamps/arc lamps.

Recognition of Sun Burn

The following signs and symptoms help to recognize sunburn:

- Skin becomes red, swollen
- There will be itching at the site of burn
- Pain when rubbed or touched
- Affected skin will be very hot
- Superficial blisters.

First Aid in Sun Burn

- Shift the victim to cool and shady place.
- Loosen constrictive clothing.
- Apply cold tap water over the area or if large area is involved, shower can be taken.
- Apply soothing lotion like calamine or cold cream.
- Sips of water to be given at frequent intervals.
- If large area is involved, take medical help.

Do's and Don'ts

Do's

- Monitor the area affected properly.
- If victim is not responsive but has pulse and breathing, put him in recovery position.

Don'ts

- Don't give coffee or tea.
- Don't break blisters.

2.3.5 Hyperthermia/High Body Temperature

Definition

This is also called as Hyperthermia, where Hyper means abnormally increased and thermia stands for body temperature. It is different from fever. In fever, there is infection and body tries to raise the temperature in response to infection. But in Hyperthermia , the body's temperature increases because the body is not able to lose heat due to problem in the heat regulating system of the body.

Hyperthermia is elevation in temperature above 40ºC (104ºF) and can be life threatening even leading to death. It is due to hot environment and is a heat related emergency. It is hence, the overheating of the body and is a sign of some condition rather than a condition itself.

Causes

- Heat stroke, heat exhaustion
- Adverse Reaction to some drugs
- Exposure to heat and humidity
- Not drinking proper fluids
- Overdressing during hot climates
- Living in houses without proper ventilation and too much insulation
- Visiting highly populated and crowded places
- Walking or running under extreme heat
- Insufficient or impaired heat loss
- High Fever
- Alcohol consumption.

Recognition of Hyperthermia

The following signs & symptoms help recognize hyperthermia**:**

- Early stage : symptoms are similar to Heat Exhaustion (as studied in section 2.3.2)
- Later stages : symptoms are similar to heat stroke (as discussed in section 2.2.3)

First Aid and Do's and Don't in Hyperthermia

These are similar to Heat Exhaustion and Heat Stoke as discussed in previous section.

2.3.6 Dehydration

Definition

Dehydration happens when someone loses more fluid than they take in. It is a condition which occurs due to lack of fluid and salts which are necessary for the body (Fig. 2.12).

Fig. 2.12: Dehydration

Causes

- Extreme heat and over exposure to sun
- Exertion and over exhaustion
- Not drinking enough water/salts
- Lot of sweating due to exercise, weather conditions or fever
- Diarrhoea, vomiting.

Recognition of Dehydration

The following signs and symptoms help to recognize sunburn:

For Adults (Fig. 2.13)**:**

- Thirst/dry mouth
- Headache
- Dizziness/Fainting
- Dark urine or urinating less frequently (maybe one or two times a day)
- Sleepiness/Fatigue
- Constipation
- Dry skin
- Muscle cramps (especially during exercise).

Fig. 2.13: Signs of Dehydration in Adults

For Children (Fig. 2.14)**:**

- Lack of urine or wet diapers for 6-8 hours (lack of urine for 12 hours in older children)
- Little to no tears when crying
- Eyes look sunken into head

- Irritability, less active
- Fatigue or dizziness
- Loss of tone/firmness of the body tissue

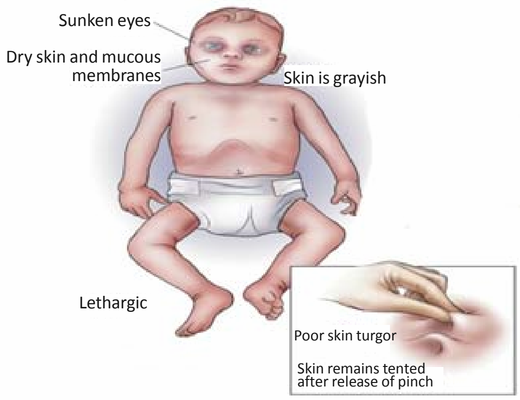


Fig. 2.14: Signs of Dehydration in Children

First Aid in Dehydration

- Shift the victim to cool and shady place.
- Loosen constrictive clothing.
- Tell to drink plenty of fluids.
- Ask to add ORS sachet and prepare the solution as per directions on the packet. Tell to consume sips of ORS solution at frequent intervals.
- Keep on monitoring the response and vital signs.
- If severe symptoms appear, take medical help.

Do's and Don'ts

Do's

- Check for associated conditions and take medical help.
- Start rehydration as early as possible.
- Tell the victim to rest and avoid conditions that can cause further loss of fluid from the body.
- Be alert for any changes in response and vital signs, especially in young children and babies.

Don'ts

- Don't give coffee or tea or alcohol.

Check Your Progress 1

1) Name three types of Extreme Heat conditions.

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2) List five important signs & symptoms of Heat exhaustion.

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3) Define Sun burn.

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2.4 EXTREME COLD CONDITIONS

In this section, we shall be learning about extreme cold conditions and their first aid management. It is important to understand these conditions as they can cause grave complications if not dealt with.

2.4.1 Frost Bite

Definition

It is the freezing of tissues of body parts like feet, nose, ear, finger or toe following exposure to prolonged and intense cold. There is formation of ice crystals within the blood causing the small blood vessels to become blocked cutting off the blood supply to the affected part. This causes cell dehydration, lack of nutrients and oxygen supply to the affected tissues/organs and this kills the living cells in affected part (Fig. 2.15). In some cases, the problems may be so severe that affected part may need to be cut or amputated.

Fig. 2.15: Frost Bite

It can be superficial or deep. In early stage, the affected area becomes red and is called **chilblain**. Later on, the affected area may become bluish or black in colour if no care is provided immediately (Fig. 2.16).

Fig. 2.16: Bluish/Blackish colour of affected part in Frost bite

The condition gets worse on wearing wet clothes, shoes and gloves and on exposure to strong cold winds. The areas if warmed, which should not be done, become swollen, discoloured and blisters get formed.

Causes

The cause is exposure to extreme cold environment. Climbing to high altitudes, exposure to ice cold water/snow blizzards or adventures to cold places where the temperature is low can lead to Frost Bite.

Recognition of Frost Bite

Frost Bite is associated with following sign and symptoms:

- Exposed parts at first are cold with prickling pain, stiffness and itching, then later on they become numb (without feeling)
- Colour of the parts effected changes from red to blue or black overtime
- Lack of movement of the part
- Swelling and blisters may be present.

First Aid in Frost Bite

When you come across someone with frost bite, the following first aid should be provided:

- Reassure the victim.
- If the victim is outside and warm place is far away:
  - Dry the affected area with a dry cloth.
  - Cover the area with dry cloth and find a warm house or warm environment for immediate rescue. (Fig. 2.17).

Fig. 2.17: Drying the affected area and covering it up

- If already in a protected environment or when you have reached a protected environment:
  - Cover the person with blanket and or loose layer of clothing (Fig. 2.18).

Fig. 2.18: Covering the victim

  - All wet or tight clothing and other constrictive jewellery e.g. wrist watch, bangles, rings must be removed from around the affected area.
  - Warm the affected part by covering the affected area and bringing it near to body to maintain the warmth. You can also re-warm the area by putting it in warm water, i.e. 38°C to 43°C (100.5°F to 110°F) but not hot water (Fig. 2.19).

Fig. 2.19: Soaking affected areas under warm water

  - If the ears or parts of the face are affected, apply towels wet in warm water and squeezed, to warm the areas (Fig. 2.20).

Fig. 2.20: Warming Ears/ face

- Stop the warming when the area becomes red again, the person feels pain since these are the signs that the affected part is fine.
- Change the clothes of the victim and put on warm dry clothing.
- Raise the effected part to relieve pain and swelling.

- Place clean cloth/dressings between affected toes or fingers if possible (Fig. 2.21) to avoid stiching together.

Fig. 2.21: Placing cloth/dressing between the fingers

- Seek medical help if the area affected is large or if no change is seen even after warming.

Do's and Don'ts

Do's

- If the person is conscious, you can provide him/her with hot drinks.
- If by any chance, you suspect that medical help is immediately required and the affected area will freeze again, don't warm the affected area. If you have warmed the area, cover it with blanket or warm cloth so that it does-not freeze again and then transport the victim.
- If you're outside and there is no availability of blankets or warm clothes, warm frostbitten hands by tucking them into your armpits after drying.
- Always check the temperature of water in the warming process. You can also check it by placing an uninjured hand or elbow in it and check if it is warm. If in any case the water becomes cold, fill in more warm water.
- Re-warm the affected area slowly.
- Re-warming the affected area should be done in 30 minutes.
- Stop the soaking of affected part in warm water when the skin gets its normal color or loses its numbness.
- If the person is wet, remove wet clothing.
- Tell the victim that burning pain and swelling will occur during warming.
- Remove any constrictive clothing during warming.
- Avoid rubbing the area if itching occurs.

Don'ts

- Don't try to separate fingers/toes if their skin sticks together.
- Don't rub or massage the affected area with your hands or snow or ice.
- Don't break any blisters.
- Don't move the frostbitten area unnecessarily.
- Don't disturb the blackened skin.
- Don't put pressure on affected areas and avoid walking on frostbitten feet or toes.
- Don't warm frostbitten skin with direct heat, such as a stove, hair dryer, heat lamp, fireplace or heating pad as this can cause burns.
- Don't smoke or drink alcohol .
- Don't apply any medication on the effected parts.
- Don't use hot water bottles on affected areas.

2.4.2 Immersion Foot

Definition

Also called as trench foot and occurs when there is long exposure (more than 12 hours) of the feet to wet and cold conditions between temperatures of 30 to 60º F (–1 to 15ºC). (Fig. 2.22).

Fig. 2.22: Immersion Foot

As a result of exposure to the wet and cold environment, the water is absorbed by the outer layer of skin of the feet and this causes the blood vessels of the feet to constrict especially in cold environment and lowers the blood supply to the area of foot. This lowers nutrition and oxygen supply to the affected part. If not taken care of immediately, the muscles and nerves of the feet can be permanently damaged. Also the cells of the foot can die over time leading to immense pain, disability and may be required to be cut off or amputated to save life. The recovery may take many weeks.

Causes

- Exposure to cooler climates with feet present in wet socks or wet shoes or boots
- Can occur even in deserts when socks are filled with sweat
- Plastic long boots which take time to dry after becoming wet
- Hiking in Wet cold environment and winter trips
- When one is susceptible to cold
- Poor nutrition, dehydration, inadequate clothing, tight clothing, socks and shoes are the predisposing factors.

Recognition of Immersion foot

The following signs and symptoms are seen in a victim of immersion foot:

- Cold swollen feet
- Blisters, redness of feet
- Skin is blotchy and peels easily
- Victim tells that feet feel hard with less or no sensation
- Itching, pain when exposed to heat, tingling
- Blue colour of nails
- Leg cramps.

First Aid in Immersion Foot

The following first aid steps should be performed:

- Remove wet shoes/boots and wet socks.
- Gently Clean and Dry feet using dry towel.
- Soak the affected feet in warm water 40ºC (102° to 110° F) for approximately 10 minutes (Fig. 2.23). You can also warm the affected area by exposure to warm air.

Fig. 2.23: Soak the feet in water

- Elevate the affected part in warm environment.
- After the colour of the affected area improves and the feet are dry, maintain warmth by wearing clean, dry socks and covering with warm blanket.
- Seek medical help if the situation doesn't improve.

Do's and Don'ts

Do's

- Always wear dry socks and shoes. But when sleeping or resting, do not wear socks.
- Dry the affected part properly and immediately.
- Remove wet clothing and replace with dry, warm clothing.
- Protect the victim from injury and infection.
- Look at your feet regularly.
- Use baby powder or talc to keep moisture away.

Don'ts

- Don't allow victim to walk on affected feet.
- Don't massage, rub, moisten or expose affected area to extreme heat or application of lotions.
- Don't break blisters.
- Do not apply heat or ice.

2.4.3 Snow Blindness

Definition and Causes

When eyes are exposed to glare produced by sun on snow for longer periods, injury to eyes can occur especially to the cornea of the eye. This is called snow blindness. It is also called as the sunburn of the eye in cold conditions (Fig. 2.24).

Fig. 2.24: Snow Blindness-effect on cornea

The major reason is exposure to sun's Ultraviolet Radiations. The condition is common at high altitudes on highly reflecting snow fields or exposure to solar eclipse. Artificial sources of Ultra Violet Radiations like flash from welding, carbon arc, photographic lamps, desk lamps etc. can also cause snow blindness.

Recognition of Snow Blindness

It is recognized by the following:

- Intense eye pain and Headache
- Red, watery swollen eyes
- Blurred vision and difficult to see light, changes in vision
- A feeling like a foreign body in the eye.

These symptoms may come only after 6 to 12 hours of exposure to the sunlight.

First Aid in Snow Blindness

- Immediately wash the eyes with cold water.
- Gently cover the eyes with eye pads or clean dark cloth (Fig. 2.25). You can also do eye bandaging as discussed in Unit 1 of Practical Block 2.
- Rest the victim in dark room to limit light.
- Seek medical help.

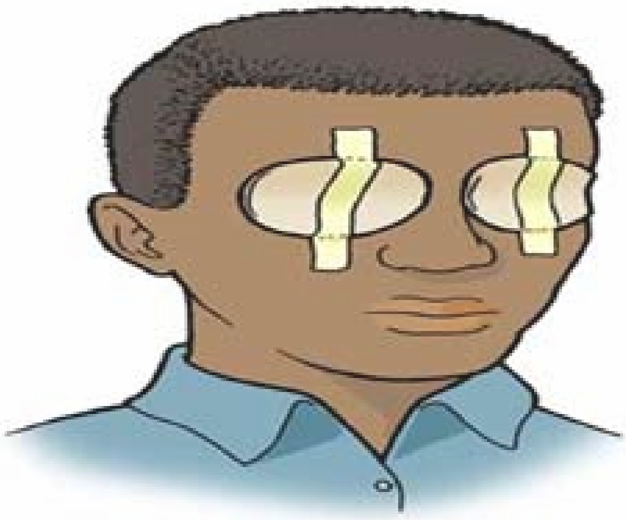

Fig. 2.25: Placing eye pads

Do's and Don'ts

Do's

- Remove contact lenses, if wearing.
- Dressing and bandage can be removed only on doctor's advice or when the victim can easily move and see from the eyes.
- Wearing dark goggles is advised.

Don'ts

- Don't rub /massage the eye.

2.4.4 Hypothermia/Low Body Temperature

Definition

Lower body temperature is also known as Hypothermia. It is the condition where in the body temperature is less than 35ºC or 95ºF due to exposure to extreme cold. It is especially problematic for children and elderly. It is a sign of many other problems also. It can complicate to cardiac arrest and even death. It occurs when the body loses heat more quickly than it produces.

Causes

- Prolonged and excessive chilling
- Associated with frost bite, immersion foot and other cold related emergencies
- Exposure to icy and chilly water source or other such environment e.g. snow and high altitudes.

Recognition of Hypothermia

The following sign and symptoms help recognize hypothermia:

- Feeling cold and shivering
- Confusion, Disorientation, Drowsiness
- Numb, pale, cold, dry skin
- Weak heart beat and slow respiration.

First Aid in Hypothermia

The first aid steps are as follows:

- Assess the victim.
- Move the victim away from the cold environment and keep him/her in warm room.
- Cover him/her with warm blanket.
- Remove wet clothes and replace with dry and warm clothing.
- Give warm drinks, if fully conscious.
- Continue to monitor vital signs.
- If the condition doesn't improve, immediately transfer to medical facility.
- If you can't move the victim to a warm place, make him/her rest on towel, blanket, warm coat so as to protect from the cold ground. Also cover the head and trunk of the body and call for ambulance to transfer him/her to hospital.

Do's and Don'ts

Do's

- You can use your own body heat to help someone with hypothermia by hugging the person gently but avoid anything else other than this.
- Warm the person's trunk first, not hands and feet by covering body with blanket.
- If using hot water bottles or chemical hot packs, wrap them in cloth; don't apply them directly to the skin.
- Be ready to assess and resuscitate if required.

Don'ts

- Don't give any alcohol or caffeinated drinks.
- Don't massage or rub someone with severe hypothermia, or handle them roughly especially during transport.
- Don't apply direct heat, such as a hot bath, heating pad or electric blanket.
- Don't immerse the person in warm water.

Check Your Progress 2

1) List the types of cold related conditions.

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2) Define Immersion foot.

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3) Write down signs and symptoms of hypothermia.

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2.5 LET US SUM UP

In this unit we have learnt regarding First Aid in extreme heat and cold related emergencies, their causes and recognition along-with the first aid management in these cases. The practical aspects of this Unit have been discussed in Unit 10 of Practical Block 2 which you must refer for getting acquainted with the practical procedures of these topics. In the next unit we shall be discussing regarding Bites and Stings and their First Aid management.

2.6 KEYWORDS

Climatic	:	Associated with climate or weather
Exposed	:	Uncover or open something
Processes	:	Series of actions or steps taken in order to achieve a particular goal
Malnutrition	:	Lack of proper nutrition
Obesity	:	Being overweight, very fat
Debilitating	:	To make someone or something physically weak
Uneasy	:	Uncomfortable
Tired	:	Exhausted/sleepy/fatigued
Sweating	:	Give out sweat
Numbness	:	Having no feeling or sensation
Appearance	:	Way in which something looks
Blisters	:	Small bubble on the skin filled with serum
Contractions/spasms	:	Sudden involuntary muscular twitching or jerky movement/cramps
Strenuous	:	Requiring effort or exertion
Worsens	:	Become bad or worse
Include	:	Involve or add something
Stretch	:	Make or be capable of being made longer or wider
Cramped	:	Suffering from cramps
Time lapse	:	Gap
Dehydrated	:	Having decrease of total water in the body
Promptly	:	Immediately/at once
Humid	:	Relatively high level of water vapour in the atmosphere
Syncope	:	Fainting
Acute	:	Severe/very critical
Fatal	:	Causing death
Constrictive	:	Tight
Immerse	:	Dip in water

Susceptible	:	Likely to be influenced or harmed
Impairement	:	State of being damaged
Superficial	:	On surface/ not deep
Tanning	:	Process by which skin becomes brown or browner or darker
Ventilation	:	Providing fresh air
Insulation	:	Protecting from cold
Grave	:	Very serious
Blizzards	:	Severe snowstorm
Adventures	:	Trips or experiences
Prickling	:	Having a tickling or stinging sensation
Tucking	:	Folding or putting in
Constrict	:	Make narrow
Hiking	:	Walk for a long distance, especially across country
Blotchy	:	Patchy or spotty
Moisten	:	Wet slightly
Glare	:	Shine with a strong or dazzling light
Intense	:	With lot of force
Confusion	:	Uncertainty/ Disorientation
Drowsiness	:	Feeling sleepy or lethargic
Prolonged	:	Continuing for a long time or longer than usual
Swelling	:	An abnormal enlargement of a part of the body

2.7 ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress 1

1) Heat stroke, Heat exhaustion, Heat cramps
2) Refer Subsection 2.3.2
3) Refer Subsection 2.3.4

Check Your Progress 2

1) Frost Bite, Immersion Foot, Snow blindness, Hypothermia
2) Refer Subsection 2.4.2
3) Refer Subsection 2.4.4

2.8 REFERENCES AND FURTHER READINGS

1. Mahopatra R, *First Aid for you & me*. Reprint 2012 Academic Publishers.
2. Clement I, *Text book on First Aid and Emergency Nursing* 1st edition, Jaypee Brothers.

3. William Swapna Naskar. R *Go swami mala Reprint 2016*. Kumar Publishing House.
4. WWW Collins dictionary.com/First aid Box. Retrieved through google.com Dec 2016.
5. http://www.garrardema.com/extreme-heat/
6. https://www.webmd.com/skin-problems-and-treatments/rm-quiz-frostbite-iq
7. http://www.nic-nagoya.or.jp/en/e/archives/394
8. http://slideplayer.com/slide/6444333/22/images/58/First+Aid+for+Heat-Related+Illnesses.jpg

   https://articles.mercola.com/sites/articles/archive/2016/07/06/heat-exhaustion.aspx
9. http://www.whmentors.org/saf/heat1.html
10. https://www.chihealth.com/st-elizabeth.html
11. http://www.emssafetyservices.com/2014/08/19/heat-related-emergencies/
12. https://www.healthline.com/health/heat-emergencies#prevention
13. https://www.hopkinsmedicine.org/healthlibrary/conditions/pediatrics/heat-related_illnesses_heat_cramps_heat_exhaustion_heat_stroke_90,P01611
14. https://ohioline.osu.edu/factsheet/AEX-981.4-10
15. http://www.natural-homeremedies.com/top-6-home-remedies-for-heatstroke/
16. http://standardfirstaidcourses.ca/hyperthermia-first-aid-tips/
17. http://firstaid.ygoy.com/2010/05/28/hyperthermia-first-aid/
18. https://www.mayoclinic.org/first-aid/first-aid-frostbite/basics/art-20056653
19. https://co.grand.co.us/DocumentCenter/View/555
20. http://www.everydaylearning.com.au/dl/Everyday%20Learning%20First%20Aid%20Sample.pdf
21. http://www.sterlingmedicaladvice.com/straight-no-chaser-the-dos-and-donts-of-frostbite/
22. https://www.worksafebc.com/en/resources/health-safety/books-guides/training-certification-standard-occupational-first-aid-training-agencies?lang=en
23. http://www.tripleonecare.co.nz/sites/default/files/images/TOCOnlineFABookJan016%20V4.5_0.pdf
24. https://www.mountnittany.org/articles/healthsheets/3584
25. https://i.kinja-img.com/gawker-media/image/upload/s
26. https://s.hswstatic.com/gif/frostbite-2.jpg
27. http://www.msdmanuals.com/home/injuries-and-poisoning/cold-injuries/nonfreezing-tissue-injuries
28. https://www.healthline.com/health/trench-foot#symptoms
29. https://www.army.mil/e2/c/downloads/369765.pdf

30. https://www.medicinenet.com/script/main/art.asp?articlekey=19378
31. http://magazine.rnli.org/Article/How-to-handle-hypothermia-78
32. http://newsprotein.com/symptoms-of-dehydration/
33. https://beprepared.com/blog/8724/first-aid-for-dehydration/
34. http://www.sja.org.uk/sja/first-aid-advice/hot-and-cold-conditions/dehydration.aspx
35. https://www.emedicinehealth.com/pdfguides/first_aid/dehydration-in-adults-treatment-61734.pdf
36. http://www.ugandanetwork.org.uk/images/rehydr.gif
37. http://urpullzone.smartdecision.netdna-cdn.com/wp-content/uploads/2016/05/signs-of-dehydration-crop.png
38. https://www.swchildrens.org/Health%20Library/Images/195ADF43-D24B-43F8-848C-479EC71B1581.jpg

UNIT 3 BITES AND STINGS

Structure

3.0 INTRODUCTION

Bites and stings are commonly seen in the rural and remote areas. Nowadays, however, they can occur in urban areas also. Lakhs of people every year are bitten or stung by someone or something. These emergencies include bites and stings due to various reasons. These bites or stings need to be identified and treated early as they affect some part or the whole of the body which can cause mild, moderate or severe reaction and can even be life-threatening. Most are not medical emergencies but however, treatment is usually required if there is bleeding, wounds or infection.

All bites and stings are not same. Different First Aid treatment and care is needed depending on the type of insect or animal that has caused the bite. Some species are more dangerous and cause more harm compared to others. Hence, in this unit we shall discuss the different types of bites and stings, causes, recognition and first aid in these situations. So, lets begin.

3.1 OBJECTIVES

After completion of this unit, you shall be able to:

- define Bites and Stings;
- list the causes of Bites and Stings;
- recognize the types of Bites and Stings;
- explain the assessment of victim affected by Bites and Stings;
- describe the first aid to be provided in Bites and Stings; and
- enumerate the Do's and Don'ts in Bites and Stings.

3.2 BITES AND STINGS

In this section we shall be discussing the concept, causes and types of Bites and Stings. We will also talk about assessment of the victim and general first aid in these situations.

3.2.1 Definition, Causes, Types and Recognition of Bites and Stings

Definition

Bites are the wounds caused by piercing or cutting off or stinging of the flesh of a person by an animal, insect or by another person. The bites happen when someone or something uses ones teeth to cut into or through flesh of the affected person. Bites may involve a local injection of poison into a wound. They may also involve infection. Bites should be managed like any other injury including cleaning, dressing/bandage and seeking medical help as necessary (Fig. 3.1).

Fig. 3.1: Bite

Stings occur when something touches skin or makes a very small hole in it so that one feels sharp pain. Sometimes part of animal remains in the body of affected person and is called **stinger**. Stings may be managed with first aid but severe allergic reactions due to venom may be present. In these cases there may be lot of danger to the victim and medical help is immediately required (Fig. 3.2).

Fig. 3.2: Sting

A sting is a defence mechanism of an insect. Stinging insects do not feed on blood but they inject a toxic venom through their stings in the human body. The animal and human bites however, don't inject venom or sting in the body.

Causes

The causes include :

- Variety of insects like – Wasps, Bees, Fine ants, Fleas , Lice, Leeches, Ticks, Bed Bugs.

- Scorpion and Snakes
- Animal and Humans.

The risk factors include:

- Pets as a common source of flea
- Crowded areas as they are unclean most of the time
- Birds nests are a source of many insects
- Old houses and furniture
- People who work outdoor (farmers, gardeners)
- Travelling from one country to another may raise the risk of insect bites.

Types

There are dozens of insects/animals whose bites or stings cause problems and therefore, to simplify they have been divided into two categories: venomous and non venomous.

Venomous are the ones which are poisonous in nature and include**:**

i) Wasps

ii) Bees

iii) Fire ants

iv) Scorpions

v) Snakes

Non venomous are the ones which are not poisonous in nature and include:

i) Fleas, Lice, Leeches, Ticks, Bed Bugs

ii) Dog/Monkey

iii) Bed bugs

iv) Human

We will discuss few common bites and stings in this unit in the next section 3.3.

Recognition of Bites and Stings

Most bites and stings show the following sign and symptoms:

- In case of mild problem, discomfort or pain, with itching, swelling and redness around the bite occurs which can heal in a few days.

Fig. 3.3: Some symptoms of bites

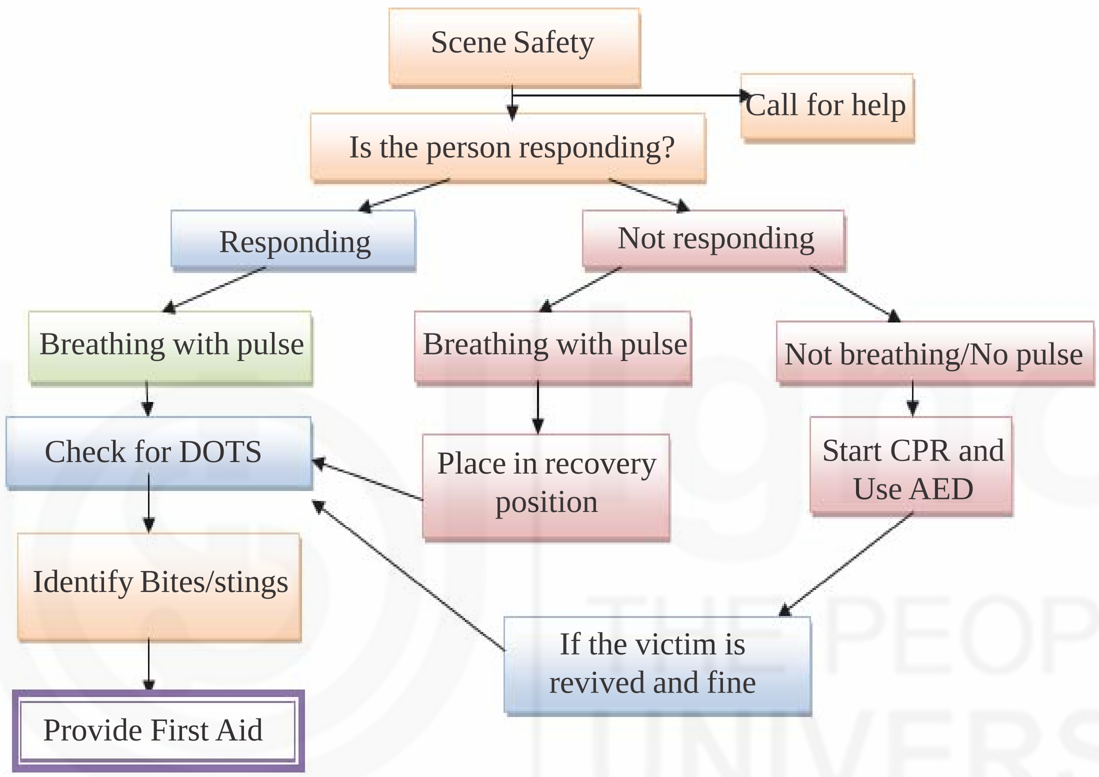
- In case of severe problem, blister or ulcer (break in the skin) may form. This takes longer to heal (Fig. 3.3).
- If bite or sting is present it may cause allergic reaction. This is seen as rash all over the body, itching all over the body and swelling of throat or tongue.

3.2.2 Assessment of the Victim and General First Aid

The assessment of the victim and first aid is as follows:

General Assessment

You have already learnt the assessment to be done while providing first aid in Unit 1 of Block 2 of the Theory Course. The general assessment is done as per following flowchart.



Specific Assessment

After general assessment, perform specific assessment as:

1) Check for DOTS and look for deformity, swelling, tenderness and any other sign and symptoms.
2) Check for the sign and symptoms of specific bite and sing (as given in section 3.3)
3) Ask History from the victim or onlookers present.
4) Assess what has caused the bite/sting and also how much poisonous the insect would be since the more poisonous the insect, the more life-threatening will be the complications.

General First Aid

The general first aid steps are:

- Assess the sign and symptoms.

- Maintain barrier technique when giving first aid. Hand wash and use PPE as discussed in Unit 3 of Theory Block 1.
- Remove stinger, if you are able to.
- Wash the area with mild soap and water and keep it clean and dry (Fig. 3.4). You can also clean the wound as discussed in Unit 1 of Theory Block 3 on Wounds and Bleeding. **This step is not done in snake bite.**

Fig. 3.4: Washing hands under running water

- Apply ice pack (wrapped in a thin cotton cloth or towel) or cool running water or wet cloth to reduce the swelling and relieve the pain (Fig. 3.5). **This step is not done for snake bite**.

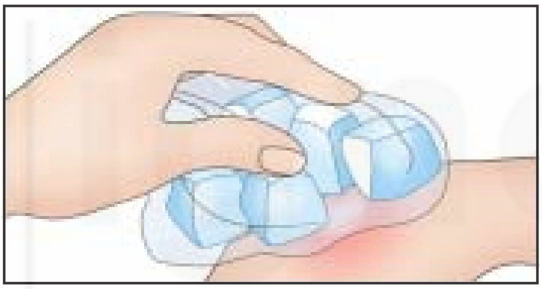


Fig. 3.5: Application of ice pack

- Provide first aid specific to the type of bites discussed later in this unit.
- Seek medical help.

The removal of stinger and first aid in specific type of bites have been discussed in upcoming sections.

3.3 VARIOUS BITES/STINGS

In this section, we shall discuss various types of bites and stings, their cause, first aid and do's and don'ts in the various bites and stings.

3.3.1 Scorpion/Spider Sting

Definition and Features

Scorpion Bite

Scorpions and spiders are common in many parts of India. Some scorpions are more poisonous than others. The Hindi name of scorpion is bicchoo.

Scorpions have a long tail with the presence of stinger . In the world, there are around 2000 species but only about 25-40 species are poisonous enough to cause serious damage to humans. In India, 86 different species of scorpions exist, of which 50 are deadly. Their stings are very dangerous and can be poisonous, as they cause severe pain and in children and signs of shock may occur. The most poisonous species of scorpion in the world is found in India and goes by the name of Indian Red Scorpion which has been said to be the most lethal in the world. When stung, victims have nausea, heart and lung problems. Besides this Yellow Fat-tailed Scorpion and Black Scorpion is also commonly found in India (Fig. 3.6). Together they cause 10-20 deaths every year.

Fig. 3.6a: Indian Red Scorpion b: Black Scorpion c: Yellow Scorpion

Spider Bite

Spiders are usually not poisonous. But some poisonous spiders exist, such as black widows and brown recluses which generally live in undisturbed areas, such as attics or sheds, and don't bite unless they are disturbed. In India, Tarantulas are the main poisonous variety of spiders. There are however, no reported incidents of fatalities from spider bites in recent years (Fig. 3.7). The Hindi name is Makdi or Makda.

Fig. 3.7a: Black Widow Spiders b: Brown Recluse Spiders c: Tarantulas

Recognition

The person with history of **scorpion bite** will have following symptoms:

a) Mild symptoms:

- A mark indicating the presence of bite (Fig. 3.8).

Fig. 3.8 a: Scorpion bite b: Sting/Bite Marks of a scorpion c. Severe bite mark

- Itching and swelling at the effected site
- Burning pain, increased sensation or numbness near the site of bite
- Ulcer at the affected site.

b) Severe symptoms:

- A mark indicating the presence of bite
- Severe pain
- Restlessness, excessive tears and excessive saliva production
- Nausea, vomiting
- Lot of sweating usually 4-6 hours after bite
- Victim may be excited or may cry a lot especially in children.

The person with history of **spider bite** will have the following symptoms:

- A mark indicating the presence of bite and a bump on the skin (Fig. 3.9).

Fig. 3.9 a: Bumps due to Spider bite **b: Bump on the skin due to spider bite**

- Itching and swelling at the effected site with ulcers and blisters (red and purple colour)
- Pain at the affected site
- Nausea, vomiting, Abdominal Pain, stiffness and cramping
- Rash can develop
- Headache, fever, chills
- Restlessness and fatigue.

First Aid in Scorpion/Spider bite

The first aid to be provided in case of scorpion/spider bite is as follows:

- Assess the site of bite or sting.
- Reassure the victim and make him/her calm.
- Try to identify the type of scorpion or spider that has caused the bite-you can snap it with camera in your mobile or try to see around the nearby area to find the scorpion or spider e.g. House spiders are not harmful but if the victim had been bitten outdoors, try to identify the type of spider or scorpion that has caused the bite and how much that insect is poisonous. Collect history from the victim and assess the sign and symptoms.
- Remove any constrictive clothing or jewellery around the bite area as swelling may occur after the bite and any kind of constriction around this area can decrease circulation.
- Wash the affected area with soap and water.

- Apply ice packs on the bite area after covering with towel or wet cloth over the bite to slow down the absorption of poison. Do this for 10 minutes, then remove it and again re-apply if required.
- Keep the affected area above the level of the heart (not below).
- If victim does not improve or shows symptoms of severe reaction or shock or if the identified scorpion/spider is of a dangerous variety. Rush the victim immediately to the hospital.

Do's and Don'ts

Do's

- Remove any constrictive clothing or jewellery.
- Ask victim to be calm and disallow any unnecessary movements of the affected area.
- Avoid taking food and water, if victim is having nausea and vomiting or abdominal pain.
- Keep victim still and calm to slow down the spread of the poison.
- If you are able to identify the scorpion/spider that has caused the bite and if it is the poisonous one, rush the victim immediately to hospital or call ambulance immediately as he/she requires urgent medical attention.
- Apply ice pack for 10 minutes and remove it. If the pain, itching persists, reapply again for 10 minutes and remove. Continue if required.
- Avoid the use of tourniquet.
- Continue assessing the victim.
- Use gloves when caring for the affected site.
- Keep the affected area above the level of the heart by raising it.

Don'ts

- Don't freeze the area by using ice pack directly on the bite or dipping the area in ice. Only wash the area and apply ice pack after covering in towel.
- Don't make a cut in the wound.
- Don't try to suck out any material from the site of the bite.
- Don't apply anything on the wound.

3.3.2 Snake Bite

Definition and Features

All snakes are not poisonous and all snake bites are not fatal. Most people die because of the fear and lack of information. The poison is also called as venom. There are nearly 3,000 species of snakes in the world and of these only 375 snake species are venomous. Thus, many of the species of snakes are harmless. In India, there are 270 species of snakes, out of which about 60 are highly venomous. Out of these, the Big 4 common varieties of dangerous snakes are:

1) The Indian Cobra: Also called as Nag, it is the commonest snake in India. It is also one of the most poisonous snakes of the world. A bite from a cobra can lead to paralyzing muscles, respiratory failure and cardiac arrest.

2) Common Krait : The Indian Krait or Common Krait is one of the most venomous snakes in India and the world. The venom of this snake affects the nervous system.

3) Russel's Viper : The venom of this snake affects the blood and stops its clotting. It is mostly dark brown or gray in colour.

4) Saw-scaled Viper: It is a rough scaled snake with large eyes, wider head than neck and stocky body. Refer Fig. 3.10 for figures of these species of snakes.

Fig. 3.10 a: Indian Cobra b: Common Krait c : Russel's Viper d: Saw-Scaled Viper

All the poisonous snakes are different from the non-poisonous ones. Refer Fig. 3.11 for difference in the snakes and Fig. 3.12 for difference in their bites.

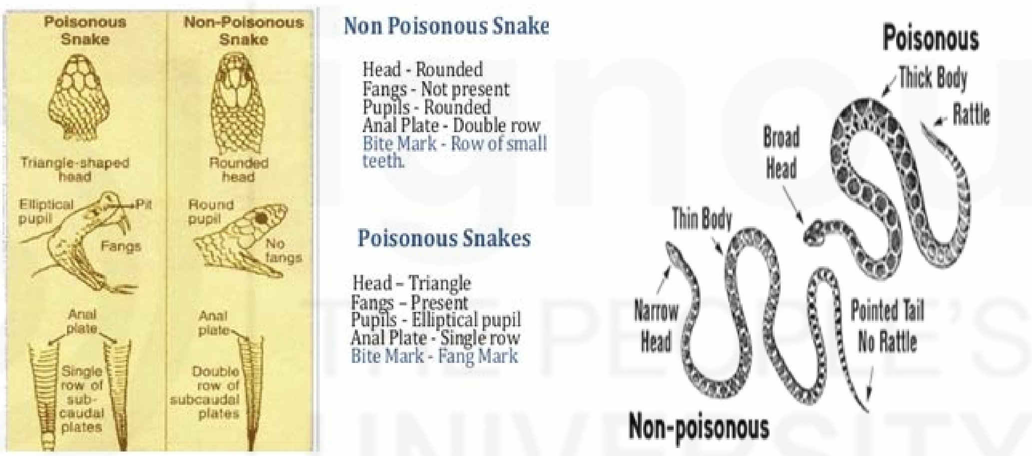


Fig. 3.11: Differences of Poisonous and Non-Poisonous snakes

Recognition of Snake Bite

The recognition of snake bite and if the bite is because of a poisonous snake can be done by the following signs and symptoms:

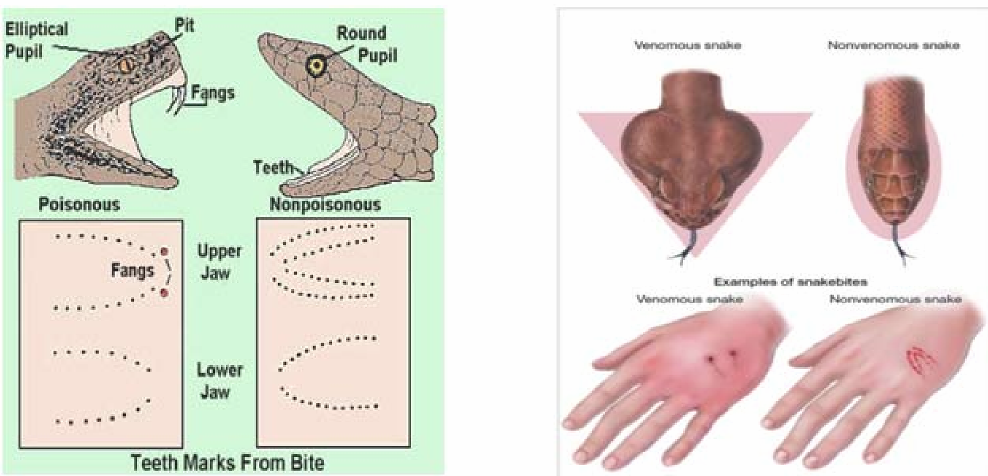


Fig. 3.12 : Differences of the bites of Poisonous and Non-Poisonous Snake Bites

1) Presence of Wound i.e. punctured wound due to the fangs of snake.

   One must be careful in observing this. If it is a poisonous snake, victim will have only 2 fang marks and the marks due to other teeth may or may not be present. But if the bite is of a non-poisonous snake, only 2 rows of teeth marks are seen and no fang marks (Fig. 3.12).

2) Other signs and symptoms:

   - Pain may be severe
   - Bleeding and Numbness at the site of bite
   - Swelling and change in colour of skin and blisters around the wound.

   If the Poisoning due to venom of snake bite has occured, the sign and symptoms are:

   - Weakness
   - Fainting, Drowsiness.
   - Nausea, Vomiting, Diarrhoea
   - Rapid pulse
   - Loss of muscle action especially around the eyes causing eyes to droop or double vision
   - Difficulty in breathing and speech
   - Area becomes bluish purple after bite in twelve hours
   - Foaming/Dribbling of saliva, paralysis
   - Convulsions, coma (Fig. 3.13).

Fig. 3.13: Some sign and symptoms of Snake Poisoning

In case of **Cobra and Krait**, which affect the nervous system, the sign and symptoms are:

- Drowsiness
- Weakness of muscles especially those around the eyes, causing eyes to droop or seeing things in double
- Paralysis and failure of respiration.

In case of **Viper**, which effects the clotting of blood, the sign and symptoms are:

- Headache
- Nausea and Vomiting
- Cough with blood
- Shock (Discussed in Unit 5 of this Block)

First aid in Snake Bite

- Tell the person to remain calm.
- Try to identify the pattern of bite or any snake nearby.
- Lay down the person quietly. Movement can cause the venom to spread through their body more quickly. Give him complete rest.
- Remove tight clothing or jewellery around the site of the bite since swelling may occur.
- Immobilize the limb in same way as done in case of fractured leg/arm by use of splint (discussed in Unit 2 of Theory Block 3). If the hospital is far away and it may take more than 30 minutes but less than 3 hours, pressure immobilization is done as given in Fig. 3.14.

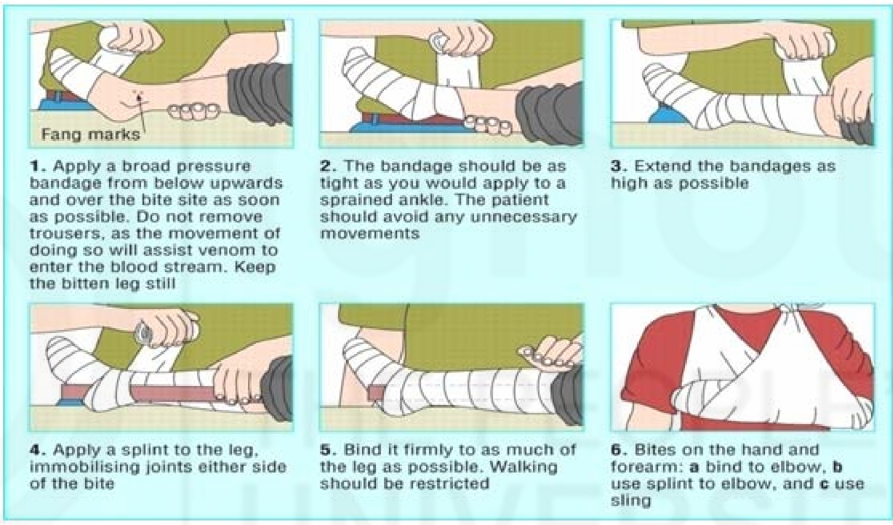


Fig. 3.14: Pressure immobilization with help of splints

- Rush immediately to the hospital.

Do's and Don'ts

Do's

- Remove any constrictive clothing or jewellery.
- Ask victim to be calm and disallow any unnecessary movements of the affected area.
- Remove shoes if the leg or foot was bitten.
- Avoid taking food, water or alcohol.
- Avoid the use of tourniquet.
- Continue assessing the victim.
- Avoid flushing the wound with water.

Don'ts

- Don't raise the affected area and don't allow any movements of this area.
- Don't tie a piece of cloth above the site of bite as was done traditionally, since this doesn't have any effect on controlling spread of poison/venom.
- Don't make a cut in the wound.
- Don't try to suck out any material from the site of the bite.
- Don't apply anything on the wound.
- Don't allow the victim to walk.
- Don't use ice pack on the bite.
- Don't allow victim to eat food or drink anything.
- Don't apply any kind of potentially harmful herbal or folk remedy.
- Don't freeze or apply extreme cold method like ice-pack or ice to the area of bite.
- Don't try to give anti-venom (medicine against snake bite) yourself. Only doctor knows that. So, Rush to the medical facility.
- Don't burn the wound or apply electric shock.
- Don't take the victim to snake charmer or any quack.

Check Your Progress 1

1) Define Bites and Stings.

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2) Fill in the Blanks.

a) A sting is a .... mechanism of an insect.

b) .... are a common source of flea.

c) In case of severe problem due to bites, ... may get formed.

d) The poison of insects/snake or scorpion is also called as ....

e) If a poisonous snake has caused the bite, there will be only .... marks on the affected area.

f) If the hospital is far-away and it may take more than 30 minutes but less than 3 hours, ... method is used in case of snake bite.

3) Tick if True or False:

a) Abdominal cramping usually exists in spider bites. (T/F)

b) Poisonous snake bites usually affect respiratory and nervous system. (T/F)

c) Avoid washing the affected area in snake bite. (T/F)

d) In case of snake bite , one must immediately rush to snake charmer. (T/F)

e) Raise the affected area above heart level in case of snake bite. (T/F)

3.3.3 Insect Bite

Definition and Features

There are various insects like wasps/hornet, bees, ants, fleas, lice, leeches, ticks, bed bugs , lady bug which cause bites and stings. These stings are actually very painful. Insect bites usually cause a mild reaction. The body reacts to sting or secretions which these insects inject into the person or transfer to the person's body through saliva. The features of various insect bites are:

1) Bee/Wasp Sting

When a bee (honey bee) stings, it loses the entire stinger and actually dies in the process. In this process, the sting often remains in the wound. The sting can cause pain, redness and swelling for a few hours (Fig. 3.15 a & c).

On other hand A wasp or hornet sting leads to sudden, sharp pain at first. A swollen red mark may appear later which is painful, itchy with swelling. The wasp however, does not lose its stinger and can hence, cause many stings to the victim. The Hindi name of bee is Madhu makhi or Makhi and of wasp is Tatiya (Fig. 3.15b).


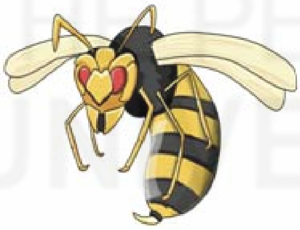
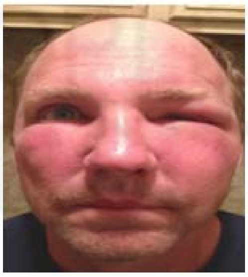

Fig. 3.15: Bee sting with stinger **b: Wasp/Hornet** **c. Redness and Swelling due to bee sting**

2) Ant Bite

Fire ants inject their venom by using biting parts of their jaw. They may inject venom many times. Indian Red ant also causes painful irritation at the point where it bites on the skin. Red bumps, itching appears in the area affected (Fig. 3.16).



Fig. 3.16 a: Red Ant **b: Fire Ant** **c: Ant Bite**

3) Fleas

Fleas are usually found on pets i.e. cats and dogs. They often bite below the knee or lower part of leg, commonly around the ankles. Fleas cause small, itchy red bumps that are in a line. Blisters can also develop (Fig. 3.17). The Hindi name for flea is pissu or pissoo.

Fig. 3.17 a: Flea **b : Flea Bites**

4) Lice

Mostly present in head, body and pubic area, these cause bites with itching and red bumps at the area affected (Fig. 3.18). The Hindi name is Joon (Ju).

Fig. 3.18 a: Lice **b : Lice Bites**

5) Leeches

Leeches exist in the damp/wet area and can suck blood. They cause wound and bleeding (Fig. 3.19). The Hindi name is Jonk.

Fig. 3.19 a: Leech **b: Leech Bite**

6) Tick

Ticks can burrow into the skin. There appears a small red bump on the skin with swelling, blisters and bruises. In severe reaction, they may cause Lyme disease which has a " bulls eye" shaped rash and fever. The Hindi name is Kilni (Fig. 3.20).



Fig. 3.20 a: Tick **b : Tick bite** **c: Bulls eye rash in Lyme Disease**

7) Mite

Mite bites cause very itchy red bumps Some mites burrow into the skin and cause a condition called scabies. The Hindi name of Mite is Ghun (Fig. 3.21).


Fig. 3.21 a: Mite **b. Mite Bites**

8) Bed Bug/Lady Bug

Bedbug/Lady bug bites occur on the face, neck, hands or arms. They're found as red bumps . They are not painful. The Hindi name of Bed Bug is Khatmal and of Lady bug is Gubrela Keet (Fig. 3.22).



Fig. 3.22 a: Bed Bug **b : Lady Bug** **c. Bed Bug Bites**

Recognition of Insect Bites

The insect bites are recognized by following sign and symptoms:

- Sharp pain at the site of sting
- Swelling, Redness and presence of Red Bumps/Lumps
- Sting may be present in the wound
- Itching
- Blister or Ulcer formation.

If it is a severe reaction, the following symptoms may be seen:

- Hives
- Abdominal cramps
- Nausea and vomiting
- Swelling of face, lips and throat
- Breathing problems
- Shock.

First Aid in Insect Bites

If the vicitm does not show any signs of allergic reaction, treat the site of the bite or sting for minor symptoms. The insects have sting which is left at the site of the puncture and has to be removed to prevent the person from danger. The treatment includes following steps:

- **Removal of sting**

  If the sting has been left embedded in the skin, hold tweezers as near to the skin as possible, grasp the sting and remove it (Fig. 3.23).

Fig. 3.23 : Removal of stinger with tweezers

  Do not squeeze the poison sac which is present on the top of sting because this will force the remaining poison into the skin. You can also use straight edge of credit card in sweeping motion. This is also discussed in Unit 11 and 13 of Practical Block 2.

- **Local treatment**
  - Wash the area of the bite with soap and water.
  - Place a cold compress or ice pack on the area wrapped in cloth or towel for about 10 minutes at a time to help reduce pain and swelling.
  - Use some home remedies like:

1) Bee venom is acid and can be neutralized by applying ammonia and soda.
2) Wasp venom is alkaline and can be neutralized by application of vinegar or lemon juice.
3) Apply calamine lotion or a paste of baking soda and water to the area several times a day to help relieve itching and pain (Fig. 3.24).

Fig. 3.24: Apply calamine lotion

- **If the Insect stings are inside the mouth or throat**
  - Give ice to suck to reduce swelling or Rinse the mouth with cold water.
  - If breathing becomes difficult, shift the victim to hospital.
- Refer the victim immediately to the hospital, if patient does not recover or shows symptoms of shock.

Do's and Don'ts

Do's

- Remove any constrictive clothing or jewellery.
- Be alert for severe allergic reaction or shock that may develop.
- Care for wound and bleeding as discussed in Unit 1 of Theory Block 3 on Wounds and Bleeding.

Don'ts

- Don't allow the victim to itch as it can worsen the problem.
- Don't mal-handle the stinger and remove it very carefully.

3.3.4 Animal Bites (Dog Bite/Monkey Bite)

Definition and Recognition

In India, dog bite is very common. In addition to this, monkeys may also bite and this should be taken seriously. Wounds after these bites are infected because dirt and germs are introduced in body from the teeth of these animals. It may cause break in skin and disease called rabies. Bite or scratch from a dog that breaks the skin poses the danger of infection, especially if the wound is deep (Fig. 3.25).

Fig. 3.25: Dog Bite

The recognition of Dog bite is done by observing the following sign and symptoms:

- Discomfort, pain at the site of bite
- Presence of puncture wound (Fig. 3.26)

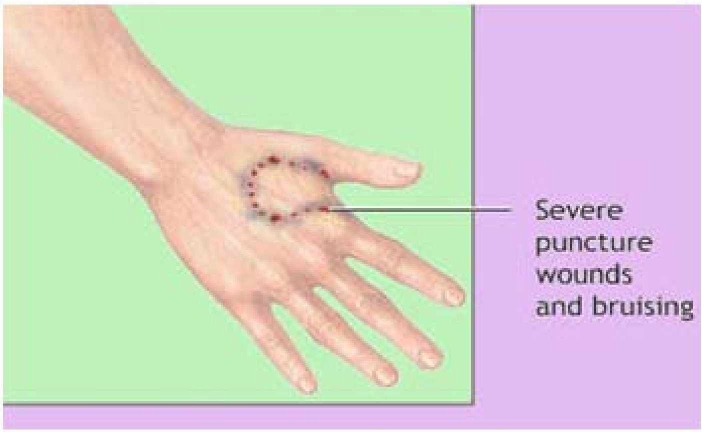

Fig. 3.26: Puncture wounds due to dog bite

- Headache, nausea and vomiting
- Agitation and confusion
- Difficulty in swallowing
- Foaming/Dribbling from mouth
- Respiratory failure
- Victim will have difficulty in drinking water.

First Aid in Dog Bite

Whenever there is an episode of dog bite, proceed with following steps:

- Assess the area when dog bite occurs.
- If bleeding is present, apply pressure over the bitten site to stop the bleeding. (as discussed in Unit 1 of Theory Block 3 on Wounds and Bleeding).
- Wash the wound well with mild soap and flush with running water or wash under running water. This is done in order to remove the saliva of the animal from the wound.
- Dry the wound.
- Wrap the bite area with dressing and bandage.
- Keep the injury elevated above the level of the heart to slow swelling and prevent infection.
- Refer the victim to the medical facility/doctor immediately.

Do's and Don'ts

Do's

- Remove any constrictive clothing or jewellery.
- Use gloves when caring for the affected site.
- Handle the wound carefully.
- Try to watch the dog for10 days for abnormal behaviour which are signs of rabies as follows:
  - No longer eating
  - No longer barking

- o If Shivers, becomes aggressive, barks at those it knows
- o Has convulsions and saliva dribbling from its mouth
- o If the dog has died or was killed, inform the respective veterinary facility.

Don'ts

- Don't chase any stray dogs and don't kill them.
- Don't make guesses about the dog whether the dog is healthy or is vaccinated. No matter what the case is, you must seek medical help right away.
- Don't apply icepack.
- Don't rub or apply force when cleaning the wound.
- Don't apply anything on the wound and avoid applying any traditional method like turmeric, soil, chillies, oil, herbs, chalk, betel etc.

3.3.5 Human Bite

Definition and Recognition

Very astonishing it may seem, but human bites are the third most common source of bites. These are common among toddlers and children who bite each other or an elder out of curiosity, aggression or otherwise when fighting, accidents, forceful collisions or giving closed fist blows causing injuries to the victim. These can be superficial or deep and can be either harmless or very serious. These are especially serious if the person who has done so has some infection and as a result the infection can spread (Fig. 3.27).

Fig. 3.27: Human Bite

The bites are recognized by following sign and symptoms:

- Cut in the skin which may be superficial or deep with breaking or tearing of the skin
- Sharp pain and Bruising
- Swelling , tenderness and redness
- Fever may occur or pus may be formed at the site if there is infection in the wound.

First Aid in Human Bite

- Stop the bleeding by applying pressure with a clean, dry cloth or dressing over the wound.
- Wash the wound thoroughly with soap and water.
- Remove any objects from the bite such as teeth, hair or dirt carefully.
- Dry the skin.
- Apply a clean dressing and bandage.
- Rush to hospital immediately.

Do's and Don'ts

Do's

- Remove any constrictive clothing or jewellery.
- Use gloves when caring for the affected site or wash your hands properly after giving first aid.
- Handle the wound carefully.
- Always seek medical help.

Don'ts

- Don't rub or apply force when cleaning the wound.

Check Your Progress 2

1) List the first aid steps when providing care to victim of Insect Bites.

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2) How will you recognize animal and human bite?

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3.4 LET US SUM UP

In this unit we have learnt regarding First Aid in Bites and Stings. We also discussed various Bites and Stings like Scorpion, Snakes, Spiders, Insects, Animals and Human, their causes and recognition alongwith the first aid management in these cases. The practical aspects of this Unit have been discussed in Unit 11 of Practical Block 2 which you must refer for getting acquainted with the practical procedures of these topics. In the next unit we shall be discussing regarding Altitude Illness and its First Aid management.

3.5 KEYWORDS

Reaction	:	Result of something that has been done
Piercing	:	Cutting through something/stinging
Injection	:	To put/inject /introduce something in the body
Venom	:	Poisonous substance
Defence	:	Resisting attack/ Protection
Mechanism	:	Method/Device
Toxic	:	Poisonous.
Redness	:	Reddish appearance
Swelling	:	Enlargement of body part
Blister	:	Fluid filled sac or sore on skin/ulcer
Rash	:	Area of redness or blisters or spots over the body
Complications	:	Problem arising due to a disease or situation
Lethal	:	Can cause death/fatal
Attic	:	Room inside the roof of the building
Fatalities	:	Occurrence of Death
Sensation	:	Feeling
Restlessness	:	Unable to relax or rest
Stiffness	:	Tightness in the area
Cramping	:	Spasms or contractions in the area affected
Chills	:	Feeling cold
Fatigue	:	Feeling of overall tiredness
Constriction	:	Tightening
Stocky	:	Broad built
Fang	:	Large sharp tooth
Foaming	:	Form or produce a mass of small bubbles
Dribbling	:	Falling down as a thin stream
Paralysis	:	Loss of the ability to move
Convulsions	:	Fits
Coma	:	Unconscious
Droop	:	Bend down
Flushing	:	Become red and hot
Secretions	:	Discharge or leakage from gland
Bumps	:	Projection or lump or bulge on skin

Burrow	:	Dig a hole or a tunnel
Hives	:	Rash on the whole body with swelling and itching
Embedded	:	Fixed
Squeeze	:	Press firmly
Neutralized	:	Make something ineffective by use of some other substance
Agitation	:	Anxious and excited (with lot of energy)
Confusion	:	Not being organized and oriented
Veterinary	:	Related to animals
Guess	:	Making idea about something without any evidence or knowledge
Vaccinated	:	Immunized or having injections to gain immunity against diseases
Astonishing	:	Surprising
Aggression	:	Feeling of violence or anger
Collisions	:	Crash or accident
Closed fist blows	:	Giving blows/punches with closed hand that can even break the teeth
Tearing	:	Pulling into pieces
Tenderness	:	Sensitivity to pain

3.6 ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress 1

1) Bites are the wounds caused by piercing or cutting off or stinging of the flesh of a person by an animal, insect or by another person. Stings occur when something touches skin or makes a very small hole in it so that one feels sharp pain. Sometimes part of animal remains in the body of affected person and is called stinger.

2) a) defence b) Pets c) blister/ulcer d) venom e) 2 fang f) pressure immobilization

3) a) T b) T c) T d) F e) F

Check Your Progress 2

1) Refer Sub section 3.3.3

2) Refer Sub section 3.3.4 and 3.3.5

3.7 REFERENCE AND FURTHER READINGS

1. http://www.jaico.be/insect-repellents/insects
2. http://www.health.vic.gov.au/edfactsheets/downloads/bite-and-stings.pdf
3. https://www.schn.health.nsw.gov.au/files/factsheets/bites_and_stings-en.pdf
4. https://poisoncontrol.utah.edu/publiced/pdfs/BiteStingTrifold.pdf

5. https://www.mayoclinic.org/first-aid/first-aid-insect-bites/basics/art-20056593
6. https://www.ehbo-koffer.nl/ehbo-kennisbank/ehbo-faq/hoe-stel-je-de-ernst-van-een-brandwond-vast/
7. https://brightside.me/inspiration-tips-and-tricks/5-bee-and-wasp-sting-emergency-hacks-that-can-save-your-life-359710/
8. https://www.wikihow.com/Treat-a-Cat-Bite
9. https://www.emedicinehealth.com/wilderness_scorpion_sting/article_em.htm
10. https://www.healthline.com/health/spider-bites
11. http://www.fortishealthcare.com/india/diseases/spider-bites-342
12. https://owlcation.com/stem/The-Most-Dangerous-Scorpions-in-the-World
13. http://www.ib-article.com/2012/12/7-scorpions-most-dangerous-in-world.html
14. http://www.walkthroughindia.com/wildlife/top-10-most-dangerous-animals-of-india/
15. http://www.vanguardsurvival.com/general-survival/insect-bites-stings-part-2/
16. https://www.epainassist.com/stings-and-bites/scorpion-bite
17. https://www.youtube.com/watch?v=5ynphH_fl2k
18. http://members.tripod.com/~c_kianwee/manage.htm
19. http://www.walkthroughindia.com/wildlife/the-6-most-venomous-snakes-in-india/
20. https://www.toppost.co/list-of-most-venomous-snakes-in-india/
21. https://www.slideshare.net/krishnavasudev75/snakebiteindian-scenario-and-treatment
22. https://secure2.convio.net/wc/images/content/pagebuilder/13742.jpg
23. http://www.emergencymedicinemims.com/cmsPage.php?case=20
24. http://thesurvivaldoctor.com/2016/05/24/how-to-identify-venomous-snake-bite/
25. https://www.slideshare.net/JoseLouies/snake-bite-firstaid-facts-prevention
26. https://en.wikipedia.org/wiki/Snakebite
27. https://www.healthline.com/health/snake-bites#first-aid
28. http://nhm.gov.in/images/pdf/guidelines/nrhm-guidelines/stg/Snakebite_QRG.pdf
29. https://www.poisonsinfo.nsw.gov.au/img.ashx?f=v&v=334583
30. http://www.bmj.com/content/331/7527/1244
31. http://www.moondragon.org/health/disorders/dogbite.html
32. https://health.clevelandclinic.org/2016/10/if-a-dog-bites-you-do-these-7-things-now/

33. http://www.wbpublibnet.gov.in/sites/default/files/first_aid/dog_bite_first_aid.pdf

34. http://www.ncdc.gov.in/Rabies_Guidelines.pdf

35. https://www.emedicinehealth.com/insect_bites/article_em.htm#causes_of_insect_bites

36. https://www.nhs.uk/conditions/insect-bites-and-stings/symptoms/

37. http://www.walkthroughindia.com/wildlife/the-9-most-painful-bites-of-indian-creatures/

38. https://www.terminix.com/pest-control/lady-bugs/bites/

39. http://trekkerpedia.com/2015/03/leech-bite-on-south-india-trek-treatment-infection-and-cure/

40. https://www.healthline.com/health/human-bites#risk-factors

41. https://www.emedicinehealth.com/human_bites/article_em.htm#what_are_human_bite_symptoms

42. https://www.bite-pro.com/content/human-bite-injuries.html

UNIT 4 ALTITUDE ILLNESS

Structure

4.0 INTRODUCTION

In the previous unit of this block you have learnt about the first aid in bites and stings. In this unit, we will be discussing about the first aid in altitude illness i.e. acute mountain sickness and motion sickness. These are common problems experienced by many people while they are climbing mountains and/or travelling by vehicles. Do you know that most of these problems can be prevented and first aid measures can be given in these situations. Let us now learn about acute mountain sickness and motion sickness and the first aid measures for managing these problems. So lets begin by studying this unit.

4.1 OBJECTIVES

After completion of this unit, you shall be able to:

- define altitude illness and motion sickness;
- list the causes of altitude illness and motion sickness;
- explain the assessment of victim with altitude illness;
- describe the first aid to be provided in altitude illness; and
- enumerate the Do's and Don'ts in management of altitude illness.

4.2 ALTITUDE ILLNESS

Altitude illness occurs when one travels to a high altitude too quickly. It is the effect on the body which occurs when someone goes or travels to a high altitude environment. Hikers, skiers, and adventurers who travel to high altitudes can sometimes develop altitude illness. Other name for this condition is acute mountain sickness.

It typically occurs at about 8,000 feet or 2,400 meters, above sea level at high altitudes or at high mountains. Most instances of altitude illness are mild and heal quickly. In rare cases, altitude illness can become severe and cause complications which may affect the lungs or brain.

4.2.1 Definition, Causes and Recognizing Altitude Illness

Definition

Altitude illness, also known as acute mountain sickness (AMS), is a group of symptoms which occur due to negative influence on health at high altitude, caused by low amounts of oxygen gas at these high altitudes (Fig. 4.1).

Fig. 4.1: Altitude illness

Altitude illnesses usually occurs when people travel to high altitudes where there is not enough oxygen in their blood because the oxygen level and pressure of air is very less. As altitude or height increases, air becomes thin and less oxygen is inhaled with each breath.

If persons, especially those who already have some medical disease like heart or lung problem or diabetes go to a high altitude quickly they can develop altitude illness.

Causes

The risk of experiencing acute mountain sickness is greater if the person lives by or near the sea and are not used to higher altitudes. So, when a person travels high up in the altitudes to a level of 8000 ft (2400 m) or above the sea level, the problem of altitude illness starts to appear (Fig. 4.2 and Fig. 4.3)

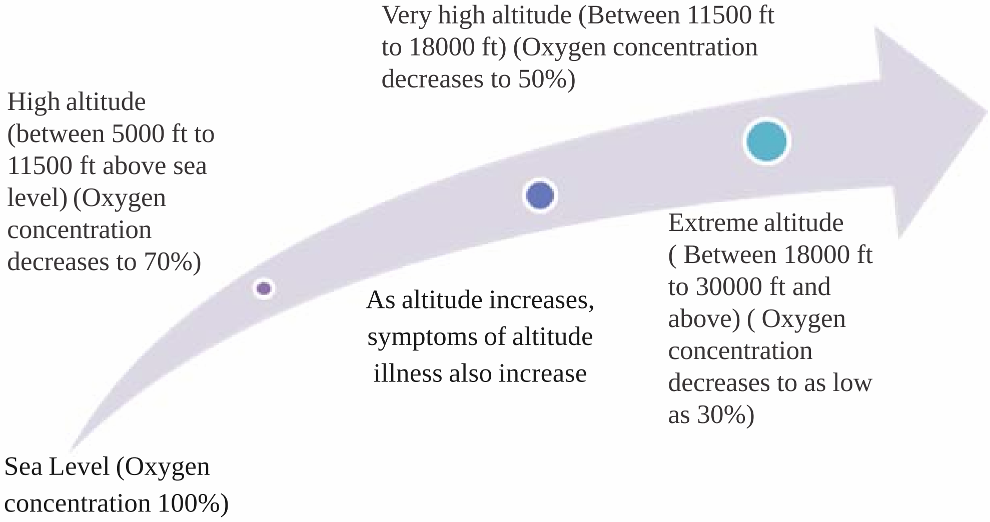


Fig. 4.2: Effect of altitude on altitude illness

Other risk factors include:

- quick movement to high altitudes
- physical exertion while travelling to a higher altitude
- travelling to extreme heights
- a low red blood cell count due to anemia
- heart or lung disease
- past history of acute mountain sickness.

Note:

- **Age, sex or physical fitness are not related to your chance of having altitude illess.**
- **Just because someone has never had an episode of Acute Mountain Sickness before, it doesn't mean he/she will not develop it ever.**

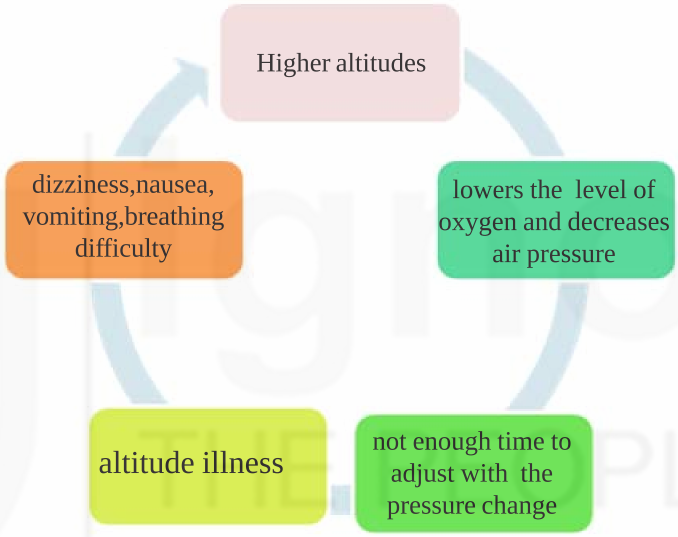


Fig. 4.3: Altitude illness –How it occurs

Recognizing Altitude Illness

The Altitude illness increases as the height/altitude increases. There are various signs and symptoms that help in recognition of altitude illness. These depend on the altitude on which the victim is present and are as under:

1) High altitude (5000 to 11,500 ft above sea level)

The sign and symptoms which normally appear at this level are usually mild (i.e. mild altitude illness) as follows:

- Increased breathing, difficulty to breathe (shortness of breath)
- Dizziness, fatigue, headache
- Nausea or vomiting
- Fast pulse, increased heart rate and increase in respiratory rate
- Feeling Tired or difficulty to get sleep

2) **Very High altitude (11500 to 18000 ft above sea level)**

The sign and symptoms which normally appear at this level are usually moderate to severe (i.e. moderate altitude illness) as follows:

- All the above symptoms
- Difficulty in breathing
- Chest congestion, Cough, Chest tightness
- Weakness and fatigue
- Fever may be present
- Swelling of face, hands, arms
- Nose bleeding may occur
- Dehydration
- Bluish skin due to lack of oxygen in the body.

3) **Extreme altitude (18000 ft to 30000 ft above sea level)**

The sign and symptoms which normally appear at this level are usually severe (i.e. severe altitude illness) as follows:

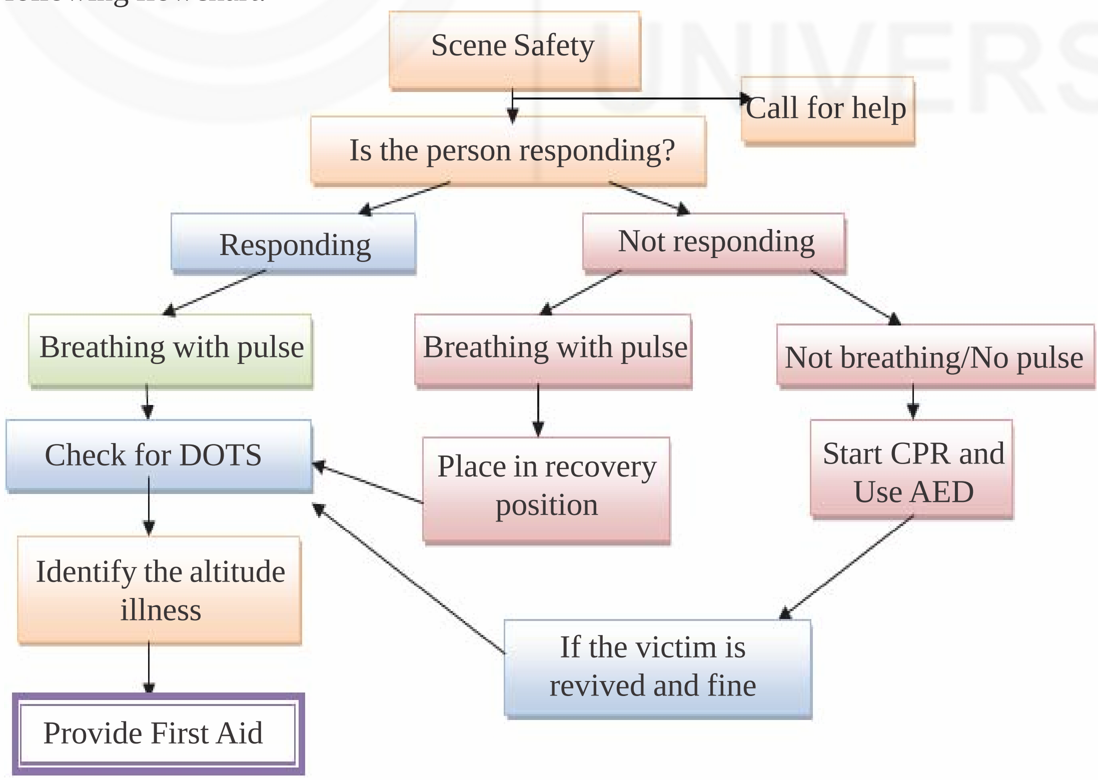
- All the above symptoms
- Failure of lungs to work (called respiratory failure)
- Swelling of Brain causing problems in standing, walking, posture, gait and consciousness
- Coma and Death may occur.

4.2.2 Assessment of the Victim and First Aid in Altitude Illness

The assessment and first aid in altitude illness is as follows:

General Assessment

You have already learnt the assessment to be done while providing first aid in Unit 1 of Block 2 of the Theory Course. The general assessment is done as per following flowchart.



Specific Assessment

After general assessment, perform specific assessment as:

1) Check for DOTS and look for deformity, swelling, tenderness and any other sign and symptoms.
2) Check for the sign and symptoms of altitude illness. Check how severe is the altitude illness (as given in section on Recognizing Altitude illness in 4.2.1).
3) Ask History from the victim regarding any existing disease or problem that he/she has. Also ask the victim if this is a first episode of altitude illness or if he has ever had this problem earlier especially when travelling to higher altitudes.

First aid in Altitude Illness

The first aid measures that need to be taken whenever an episode of altitude illness occurs are as follows:

1) For mild to moderate symptoms:
   - Tell the person to sit immediately and rest.
   - Encourage the victim to breathe deeply every few minutes to improve level of oxygen in their blood (Fig. 4.4).

Fig. 4.4: Breathing deeply

   - Provide him with water and tell him to drink enough water.
   - When he appears to be little fine, tell him/her to return to a lower altitude (descend).
   - If the person wishes to continue going up and his symptoms are mild, make him/her rest one day or more as per symptoms before moving to higher altitude (Fig. 4.5).

Fig. 4.5: Making the person rest

2) If there are severe symptoms, descend immediately around 500 m to 1,000 m (1500-3500 ft). Also, tell the victim to keep on breathing deeply as he/she descends. If he/she is dizzy or becomes unconscious, transport immediately to nearest medical facility or call an ambulance.

Do's and Don'ts

Do's

- Reduce the activity of the victim and make him/her rest immediately.
- Tell the victim to avoid alcohol and cigarettes.
- Descend as required which will improve the condition of the victim.
- Provide water to drink.
- Encourage them to breathe deeply every few minutes.
- Give any prescribed medicine to the victim if he/she is a known case.
- Avoid unnecessary physical exertion.
- Eat frequent, high-carbohydrate meals.
- Advice the person to climb gradually/slowly and return to lower level if any symptom arises.
- Advice to carry oxygen, when climbing above 9000 ft (2743.2 m).

Don'ts

- Don't keep on ascending if symptoms appear as it can cause life threatening problems.
- Don't allow a person with known history of altitude illness or anemia or other life threatening medical conditions to climb higher.
- Don't climb more than 300-500 m a day (about 1000 to 1600 ft).
- Don't allow to climb more than 600-900 m (about 2000-3000 ft which takes 3 to 4 days) without rest. Tell to rest for 1-2 days after 3 to 4 days of climbing.

Thus, in this section we have talked about Altitude illness (Acute mountain sickness) in detail. In the next section, we will discuss motion sickness in detail.

Check Your Progress 1

1) Fill up the blanks:

   a) .... is the effect on the body when going to high altitude.

   b) High altitude lowers the .... in the blood.

   c) .... and .... are the severe symptoms of acute mountain sickness.

   d) Shortness of breath occurs at .... altitudes.

   e) The Oxygen concentration at Extreme altitudes is usually ....

4.3 MOTION SICKNESS

In this section, we shall be discussing about motion sickness in detail along with the first aid in motion sickness.

4.3.1 Definition, Causes and Recognizing Motion Sickness

Definition

Motion sickness also called as travel sickness occurs when a person is travelling or is in motion. It is the feeling of the movement felt or sensed in the inner ear with motion of vehicle or any other source which disturbs the inner ear (Fig. 4.6).

It is a condition that occurs in some people who travel by car, train, airplane (air sickness) or boat (sea sickness). Many people get motion sickness when they are riding on roller coaster or other amusement park rides. It usually stops as the motion stops. Some people also get motion sickness while watching spinning or rotating objects.

Fig. 4.6: Motion Sickness

Causes

Motion sickness is caused due to imbalance between the sensory parts of our body (eyes, ears or sensory nerves) and the rest of the body. For example, when travelling, one's eyes sense that one is moving but the body is still or not in motion as a whole. This imbalance between the sensory organs of the body and rest parts of the body cause one to feel sick.

Motion sickness can be triggered by :

- Rocking boat
- Aeroplane travel
- Being on the back seat of car and reading or watching outside
- Travelling from train.

Risk factors for motion sickness

- pregnant women and children
- anyone who is travelling
- fear or anxiety about travelling
- being sensitive to a certain mode of travel
- poor air or ventilation in vehicle
- not being able to see outside to find where one is.

Note:

When the person who is travelling gets used to the movement, motion sickness stops. It also stops when motion is stopped. Sometimes just by thinking about movement one can have anxiety and symptoms of motion sickness. For example, a person with the problem of motion sickness might get symptoms when he/she is on an airplane before take off.

Recognizing of Motion Sickness

The Motion Sickness can be recognized by the following sign and symptoms:

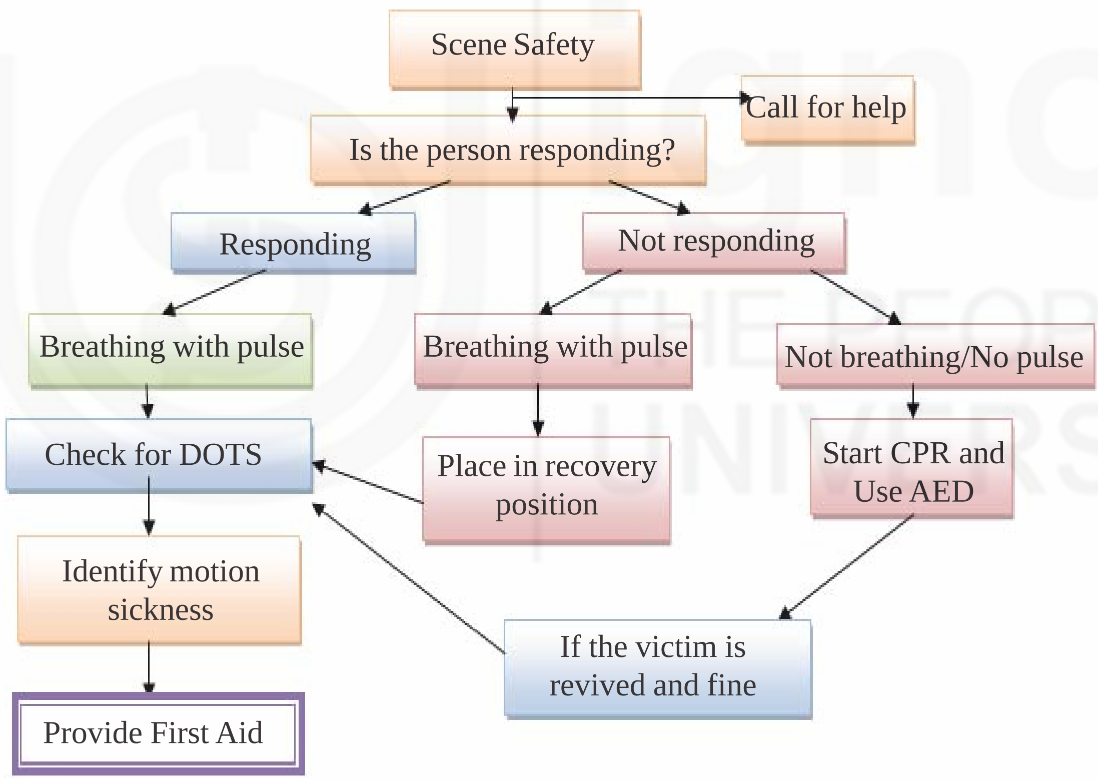
- feeling uneasy with cold sweats
- feeling dizzy and sleepy
- nausea and vomiting, may be severe with dehydration sometimes
- pale skin and increased saliva production
- headache and fatigue
- feeling of discomfort and being unwell
- shortness of breath.

4.3.2 Assessment of the Victim and First Aid in Motion Sickness

The assessment and first aid in motion sickness is as follows:

General Assessment

You have already learnt the assessment to be done while providing first aid in Unit 1 of Block 2 of the Theory Course. The general assessment is done as per following flowchart.



Specific Assessment

After general assessment, perform specific assessment as:

1) Check for DOTS and look for deformity, swelling, tenderness and any other sign and symptoms.
2) Check for the sign and symptoms of motion sickness.
3) Ask History from the victim about current situation and previous history of motion sickness. Ask onlookers o gather more information

First aid in Motion Sickness

The first aid measures that need to be taken whenever an episode of motion sickness occurs are as follows:

- Modify the environment (for example, change position or sit by the open window of the vehicle).
- Make the following changes for the person as per vehicle/mode of travel :
  - If in ship, request a cabin in the front or middle of the ship.
  - In plane, ask for a seat over the front edge of wing where movement is minimal. Once aboard, direct the air vent flow to the face (Fig. 4.7).

Fig. 4.7: Placing person over front edge of wing , b: direct airflow towards face

  - In train, take a seat near the front and next to a window. Face forward. Try to get a seat that faces the way the train is going.
  - In automobile, sit in the front of passenger's seat or by the side of the window and look at a distance.
- Recline the seat well back so that the victim's head is well back and supported to decrease effects of motion on inner ear. Try to arrange for lying down if possible and tell victim to shut the eyes, and keep the head still (Fig. 4.8).

Fig. 4.8: Position of head in motion sickness

- Over the time add distractions like looking far-away or at the horizon, listen to music, eating something like dry crackers or sucking on a candy.

Do's and Don'ts

Do's

- Get fresh air. Open the window to allow air to come in .
- Tell victim to recline the seat or lie down or keep head still.
- Allow for doing vomitting if it occurs.
- Ask victim to close eyes and breathe slowly. Focus on his/her breathing.
- Tell the victim to choose proper place to sit in the vehicle – usually where the motion is minimum.
- Ask the victim to always look straight ahead at a fixed point. Choose a far away point like horizon.
- If sitting or lying down doesn't help , tell the person to try standing.
- Tell victim to break up long journeys to get some fresh air, drink water or take a walk.

Ask victim to:

1) Eat light meals or snacks that are low in calories in the 24 hours before air travel.
2) Avoid salty foods and dairy products before travelling.
3) Avoid big, greasy meals and alcohol the night before air travel.
4) Avoid strong odors and spicy or greasy foods before and during travel.
5) Get a good night's rest before traveling.

Don'ts

Ask victim to follow these:

- Don't read, watch films or use electronic devices.
- Don't look at you phone or tablet.
- Don't look at moving objects, such as passing cars or rolling waves
- Don't overeat.
- Don't use rides if they make one unwell.
- Don't watch or talk to another traveler who is having motion sickness.
- Don't drink alcohol or eat a heavy meal before you travel.

Check Your Progress 2

1) Fill up the blanks:

   a) The feeling of the movements being sensed with the inner ear is different from the movements that is visualized is referred as ....

   b) The common symptoms of acute motion sickness are uneasiness with ....

   c) In plane, ask for a seat over the front edge of a wing where movement is ....

   d) .... is the name for motion sickness at sea.

4.4 LET US SUM UP

Thus, in this unit we have discussed about the concept, definition, causes, recognition, assessment and first aid in altitude illness i.e. acute mountain sickness and motion sickness. We have also discussed the various Do's and Don't that need to be taken care in case of altitude illness and motion sickness. It is hence clear that these conditions can arise especially while travelling, during adventure expeditions and outdoor sports and so on. As a first aid provider you must be equipped with knowledge and skill for the purpose of attending victims of such conditions to provide them with immediate care and help them with referral to medical centres. Thus, you must be alert to identify these problems as they arise and also provide first aid as and when required. You can refer Practical Block 2, unit 12 for practical aspects of this topic.

4.5 KEYWORDS

Experienced : Face, feel or encounter

Altitude : Great Height

Hikers : Persons who walk long distances, especially across country

Skiers : Persons who travel over snow or do skiing (Fig. 4.9)

Fig. 4.9: Skier doing skiing

Adventurers : Persons who undertake adventure which is a unusual or daring experience e.g. camping, mountain climbing etc.

Complications : Problem arising from some situation

Negative : Bad or not positive

Influence : Have an effect or impact

Inhaled : Breathe in through nose

Not used to : Not having an experience of something

Physical : Related to body

Exertion : Strain or doing heavy work

Extreme : To a great level

Recognition : Identify something

Shortness : Quality of being Small and difficult

Dizziness : Feeling faint

Congestion : Feeling heaviness or tightness in chest due to lot of secretions

Fatigue : Being very tired with no energy

Posture : The way of holding body or pose

Gait : Person's manner of walking

Gradually : Slowly and gently

Life threatening : Very fatal

Amusement : Entertainment

Rides : A roller coaster, roundabout in an amusement park (Fig. 4.10)

Fig. 4.10: Rides in an amusement park

Sense : Feel

Triggered : Cause to happen

Rocking : Gently moving to and fro from side to side

Sensitive : Delicate, easily affected by changes

Ventilation : Provision of fresh air

Uneasy : Uncomfortable

Dehydration : Excessive loss of water from the body

Saliva : Watery liquid present in the mouth

Aboard : Over or inside the aeroplane when its in air after taking off

Air vent : An opening that allows air to pass

Distractions : Anything that diverts attention

Expeditions : Trips

4.6 ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress 1

1) a. Altitude illness b. Oxygen c. Coma, death d. High e. 30%

Check Your Progress 2

2) a. Motion Sickness b. cold sweats c. minimal d. Sea sickness

4.7 REFERENCES AND FURTHER READINGS

1. Brunner and suddarth's(2013) " *Medical Surgical Nursing*" 13th edition, Philadelphia, wolters kluwelr health Lippincott Williams & Wilkins.

2. Lewis,S.L.Dirksen " (2014)*Medical Surgical Nursing*" 9th edition, Canada, Elsiever Publishers.

3. N.K. Anand, Shika Goel (2009). First Aid, Delhi , A.I.T.B.S Publishers, 156-161.
4. T.K Indrani (2008), *First aid for Nurses*, New Delhi, Jaypee Brothers Medical Publishers(P) Ltd,169-171.
5. N.N.Yalayyaswamy,(2006), *First Aid & Emergency Nursing*, Bangalore, Prithvi Books Agency 166-177.
6. www.ifrc.org
7. https://www.nhs.uk/conditions/altitude-sickness/
8. https://www.emedicinehealth.com/mountain_sickness/article_em.htm
9. https://en.wikipedia.org/wiki/Altitude_sickness
10. https://www.webmd.com/a-to-z-guides/altitude-sickness#1
11. https://www.medicinenet.com/motion_sickness_sea_sickness_car_sickness/article.htm
12. https://www.webmd.com/cold-and-flu/ear-infection/tc/motion-sickness-topic-overview#1
13. https://www.medicalnewstoday.com/articles/176198.php
14. https://www.nhs.uk/conditions/motion-sickness/
15. https://www.umm.edu/health/medical/altmed/condition/motion-sickness
16. https://familydoctor.org/condition/motion-sickness/
17. https://www.wikihow.com/Prevent-Air-Sickness-on-a-Plane
18. http://www.holidayswithkids.com.au/travel-tips-solutions/motion_sickness

UNIT 5 ALLERGIES AND SHOCK

Structure

5.0 INTRODUCTION

In the previous unit of this block, we discussed about the conditions related to altitude illnesses and its management. In this unit, we will be discussing about two conditions. The first condition is very common but very important to understand i.e. Allergy and the second is a most critical problem i.e. Shock.

The allergy or allergic reaction as some may call it, begins in the immune system which you have already studied in the Unit 2 of Block 1 of this Theory course. Allergy occurs when some substances called allergens cause an abnormal reaction in a person who is sensitive to them. These allergens can be dust, pets hair, pollen, insects, microorganisms, certain foods (Fish, eggs) and some medicines. The allergens are normally present in the environment and harmless but create problem only for those who are allergic to these substances. Some may be allergic to pollens, some to pets and so on. It is very important to learn about allergy as almost all of us may have experienced some kind of allergic reaction in our life time. Here, appropriate first aid given on time can prevent further deterioration of victim's condition.

The other important condition which we shall discuss is 'SHOCK' which is a true medical emergency. Shock is a serious condition which occurs after severe injury or illness leading to lack of blood flow in the body due to improper amount of blood or lack of proper mechanism of circulation. Hence, it can lead to heart attack or organ damage and death. So, recognizing symptoms of shock and its immediate first aid is essential to save the victim's life. Thus, in this unit we will discuss both these conditions.

5.1 OBJECTIVES

After completion of this unit, you shall be able to:

- define allergy and shock;
- enumerate the causes of allergy and shock;
- recognize the types of allergy and shock;
- explain the assessment of victim with allergy and shock;
- describe the first aid to be provided in allergies and shock; and
- list the Do's and Don'ts in management of allergy and shock.

5.2 ALLERGIC DISORDERS

In this section we will discuss regarding allergy, causes, recognition, assessment and general first aid in allergies.

5.2.1 Definition, Causes, Types and People at Risk for Allergies

Definition

An allergy is a hyper or increased or abnormal reaction of immune system when it is exposed to allergens. Allergens are the substances which cause allergy. A person can be allergic to pollens, dust, nuts, eggs, bites from some insects, poisons and some medicines and so on.

Causes

As such, there is no confirmed cause for allergy. Anyone can have allergy from any substance but commonly, it occurs due to disturbance in immune system. The main causes are as follows:

1) Dust
2) Pollen
3) Foods such as peanuts, cow's milk, soy, seafood and eggs
4) Pets Hair (dogs, horses, cats, rabbits)
5) Insect stings
6) Certain Medicines (Pencillin).

Types of Common allergies

A Depending upon the cause, the most common types of allergy are as follows (Fig. 5.1):

1) Skin allergy (which occurs by skin contact with cosmetics, hair dyes, pets, certain plants or jewellery)
2) Food and Drug Allergy (intake of eggs, nuts, shellfish, Soya, milk and certain medications like antibiotics)
3) Insect Bite/sting allergy (bee sting, spider, scorpion)
4) Allergy by Inhalation i.e. allergens enter in the body through nose (inhaling pollens, dust in breathing)

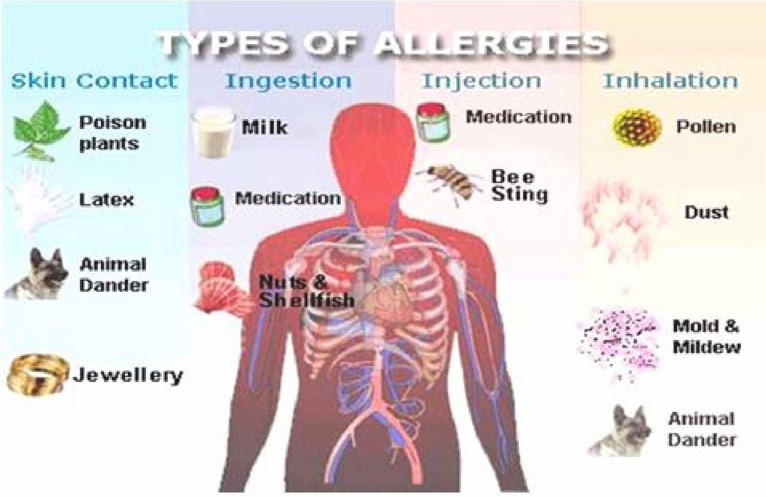


Fig. 5.1: Causes of allergy (source: http://www.medindia.net/patients/patientinfo/allergy.htm)

B) Depending on the types of sign and symptoms, allergies can be:

1) Mild /moderate allergy reaction: These are the allergies with mild to moderate symptoms.

2) Severe allergic reaction: Also called as anaphylactic shock, this type of allergy is the one in which the symptoms are more severe.

People who are at risk of allergies

The risk of allergy is increased in people who have any one of the following:

i) Strong family history of the allergy

ii) Very young or very old age

iii) Males are more likely to be more allergic than females

iv) Babies who got delivered by operation

v) People who do lot of cigarette smoking or tobacco use

vi) Air pollution.

5.2.2 Recognizing Allergies, Assessment of the Victim and General First Aid

Recognizing allergies

Allergies are recognized by a variety of signs and symptoms that appear when allergens affect the body. The general symptoms are :

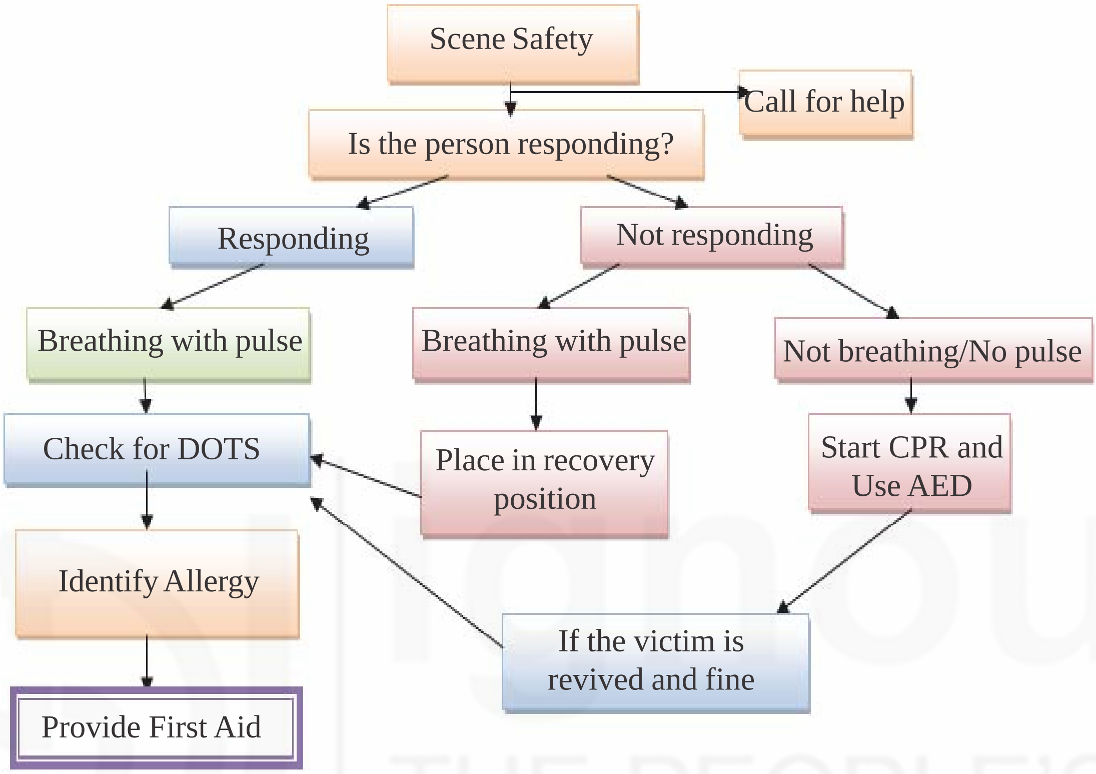
- Red area on the skin with itching and rash.
- Red eye with lot of tears and itching.
- Difficulty in breathing, lot of sneezing and discharge from nose.
- Swelling of hands/feet.
- Nausea, vomiting.
- Pain in abdominal area, diarrhoea.
- Rapid pulse and breathing.
- If severe allergic reaction occurs these symptoms appear to be even more worse and the person may become unconscious.

Specific allergies cause specific symptoms which has been discussed in the next section. (Section 5.3)

Assessment of the victim

General Assessment

You have already learnt the assessment to be done while providing first aid in Unit 1 of Block 2 of the Theory Course. The general assessment is done as per following flowchart.



Specific Assessment

After general assessment, perform specific assessment as:

1) Check for DOTS and look for deformity, swelling, tenderness and any other sign and symptoms.

2) Check for the sign and symptoms of allergy. Look for areas affected and recognize specific allergy based on sign and symptoms discussed in next section of this Unit.

3) Ask History for identifying the cause for allergy from the victim or onlookers/bystanders etc.

General First Aid

Once the assessment of the victim is complete, the first aid must be provided to the victim. The general first aid to be provided is as follows:

- Assess the victim and collect the history.
- Always in any kind of allergic reaction, Remove the source of allergen or move the casuality away from the allergen.
- Treat any symptoms. Allow the victim who has history of allergy to take medicine for allergy as prescribed by his/her physician or doctor.
- If there is severe reaction, follow these steps:

1) Lay person flat and don't allow to stand or walk.

2) If unconscious, follow the steps as given in the flowchart above in assessment of the victim.

3) Transfer the victim to hospital or seek medical advice.

Check Your Progress 1

1) Fill up the blanks

   a) An allergy is an exaggerated reaction of the .... system in response to exposure to foreign substances.

   b) The substances causing allergy are called ....

   c) The types of allergy due to specific medication which can put the person's life at risk is .... allergy.

   d) Severe allergic reaction is also called as ....

Thus, in this section we have discussed the causes, types, assessment and general first aid in allergies.

5.3 TYPES OF ALLERGY

The various types of allergies have been indicated in the Section 5.2. In this section we will study about them in detail.

5.3.1 Skin Allergy

Definition and Causes

The allergy of skin is caused by direct contact of allergen to skin causing allergic reaction on the skin. Most commonly rash and redness occurs on the skin (Fig. 5.2).

Fig. 5.2: Skin Allergy

The most common causes of Skin allergy are as follows:

- Poisonous plants (Oak)
- Hair dyes, Jewellery of gold and nickel
- Perfumes or chemicals in cosmetics and skincare products
- Rubber or latex products like gloves, balloons, rubber bands, condoms, bandages etc .
- Animal dander (tiny particles of skin that have been shed from animals with fur or feathers).

Recognizing Skin Allergy:

A person with skin allergies will have following symptoms (Fig. 5.3):

- Skin rash with redness
- Burning
- Itching and swelling on the skin
- Blisters on the skin (small pocket of body fluid present on the upper layer of skin) (Fig. 5.4)

Fig. 5.3: Symptoms of Skin Allergy

Fig. 5.4: Blister on the skin

First Aid in Skin Allergy

- Avoid the contact of the affected part with the allergen. Remove the allergen or your skin away from the allergen. If the sting or insect is the reason, remove it. (As discussed in Unit 3 of this Block).
- Care for the affected area by washing with soap and water. Use a mild soap like glycerine soap.
- If blisters are present apply an ice pack wrapped in a towel for atleast 30 minutes.
- Apply calamine lotion / moisturizer / anti-itching cream / prescribed cream over the affected area.
- If the symptoms don't improve or even worsen, refer to medical facility immediately.

Do's and Don'ts

Do's

- Wash the skin with lukewarm water thoroughly.
- Apply soothing lotion or Vaseline on the affected area.
- Take warm baths or showers and use mild soaps.
- Drink lots of water.

- Avoid further skin contact with the irritants/allergens.
- Apply ice pack, several times a day as required.

Don't's

- Don't use Nickel plated jewellery especially if there is history of allergy.
- Don't use rubber or latex products (Gloves, balloons, catheters) in case of rubber product allergy.
- Don't wear tight clothes.
- Don't scrub the skin with force but gently pat it while washing or taking bath.
- Don't apply any cosmetic lotion especially directly on the skin rash. Only use Vaseline and other soothing lotions.
- Don't cover the rash but keep it open, cool and moist.
- Don't apply ice pack directly on the skin but always cover it with towel and then apply on the skin.

5.3.2 Food/Drug Allergy

Definition and Causes

The food and drug allergy is an abnormal reaction which occurs by taking specific food items or medicine which acts as allergen. The chances of developing a food/drug allergy are more when the food is taken again and again or when drug is rubbed on the skin or given by injection, rather than taken by mouth. The various causes for food and drug allergy have been given in Fig. 5.5.

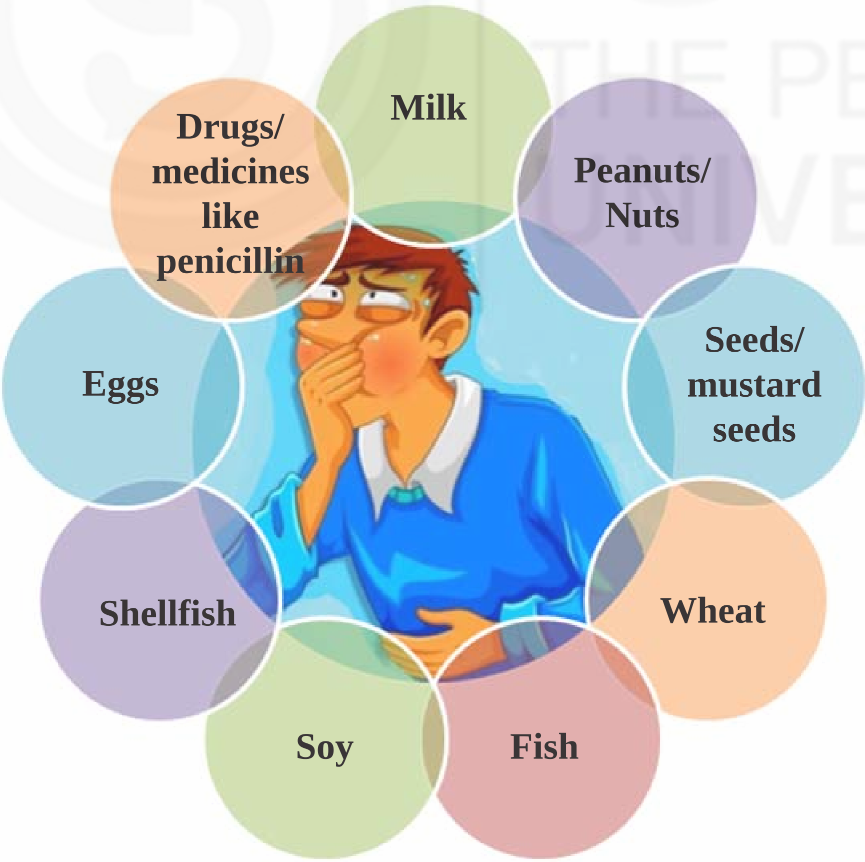


Fig. 5.5: Causes of Food and Drug Allergy

Recognizing Food/Drug Allergy

The common symptoms in food and drug allergy are as follows:

- Hives (reddish, swollen, itchy areas on the skin) (Fig. 5.6).

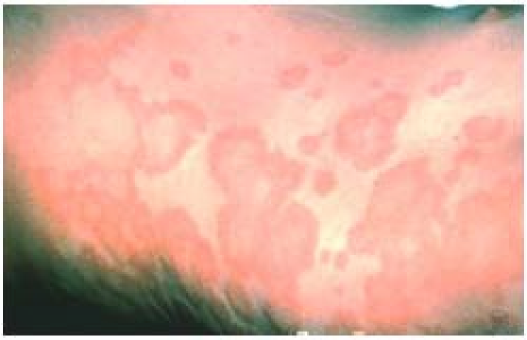


Fig. 5.6: Hives

- Dry and itchy rash
- Redness of the skin around the eyes
- Itchy mouth or ear canal
- Nausea, vomiting, Diarrhoea
- Stomach pain or cramps
- Nasal congestion or a runny nose or cough
- Trouble in swallowing
- Dizziness, weak pulse, fast respiration.

First Aid in Food/Drug Allergy

- Prevent further contact with allergen.
- Stay with the person and make him rest by relaxing him/her.
- Call for an ambulance if having difficulty breathing or if unwell.
- Observe for any change in condition while waiting for the ambulance to arrive.
- If the victim becomes unconscious, assess and resuscitate as per situation as discussed in flowchart on general assessment of the victim.

Do's and Don'ts

Do's

- Carefully check label for ingredient of food products to avoid those food items to which you/the victim is allergic. This will also help you as a first aid provider to find out the food that may have caused the allergy.
- Avoid the exposure to allergens, if already known.
- Care for the skin in case of rash and redness as discussed in skin allergy.

Don'ts

- Don't take the food or drug that causes allergic reactions.
- Don't take the drug which has not been prescribed by doctor.
- Avoid places where there is a chance that different types of food could come into contact with each other, such as buffets or bakeries and lead to allergies.

5.3.3 Insect Bites/Sting Allergy

Definition and Causes

The stings/bites of insects can cause allergic reactions especially during the warmer months. The various causes for this kind of allergy have been depicted in Fig. 5.7.

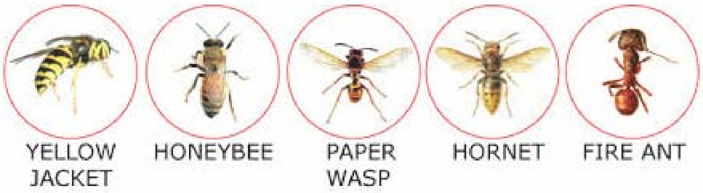


Fig. 5.7: Common Insect causing stings allergy

Recognizing Insect Bites/Sting Allergy:

A person with insect bites/stings can be recognized by following symptoms.

- Sharp pain or burning at the sting site.
- Redness, swelling, rashes and itching at local site (Fig. 5.8).
- Severe allergic reaction can occur in case of extreme stings or poisons from insects/bugs.

Fig. 5.8: Swelling and rashes of the affected area with insect sting allergy

First Aid in Insect Bites/Sting Allergy

- Assess the victim and maintain calm.
- If the insect has left its stinger in skin, remove the stinger with tweezers. You can also improvize by using straight edged object, such as a credit card, and using a brushing motion (pulling or squeezing a stinger may release more venom into your body). (As already discussed in Unit 3 of this Block).
- Gently clean affected area with soap and water to prevent infections.
- Raise the affected part and apply ice pack to reduce swelling and pain.
- Refer to medical facility, if swelling progresses or if the sting site seems infected.
- In case of severe allergic reaction, resuscitate as required and immediately shift to the medical facility.

Do's and Don'ts

Do's

- Follow all do's and don'ts as discussed in Unit 3 of this Block in section 3.3.3 on insect bites.
- Use calamine lotion on the skin.
- Teach to prevent this condition by keeping feet covered, wearing long pants, long-sleeved shirts, socks, shoes when going into heavy bushes and applying insect repellent creams.
- Tell to use nets over outdoor eating or sleeping areas.
- Tell to avoid wearing strong perfumes or fragrances, as these can attract insects.

Don'ts

- Don't break blisters.
- Don't walk barefoot in the grass.

5.3.4 Allergy by Inhalation

Definition and Causes

It is the allergy which occurs when allergens are inhaled or breathed inside the body by nose or mouth (Fig. 5.9). The causes are:

- **Inhaling** airborne spore, pollens from grass, trees and weeds during season change (April and September).
- **Inhaling** dust mites, dust particles, pet hair or moulds

Fig. 5.9: Allergy by Inhalation

Recognizing Allergy by Inhalation

The person with allergies due to inhalation has the following symptoms as given in Fig. 5.10.

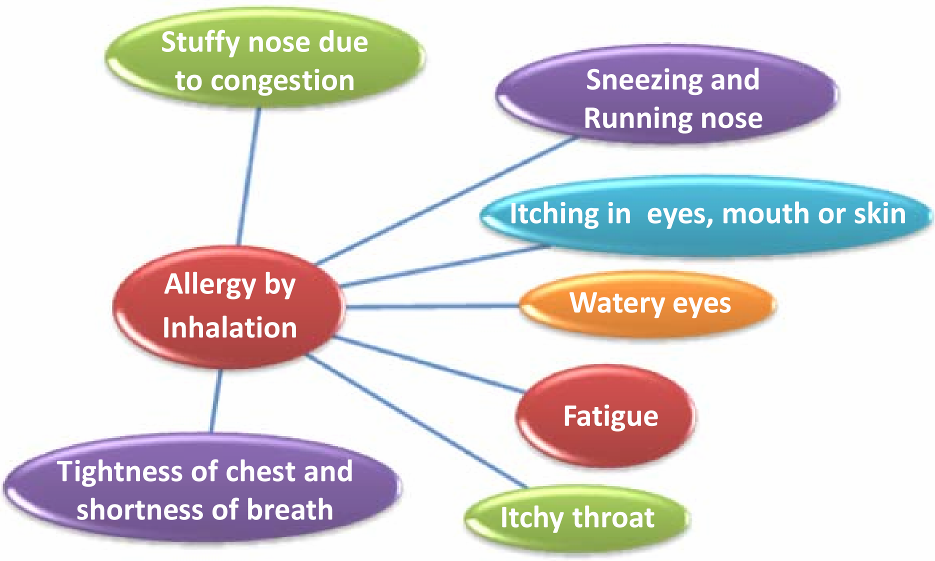


Fig. 5.10: Symptoms of Allergy by Inhalation

First Aid in Allergy by Inhalation:

- Avoid the triggers/causes of allergy.
- Clear the stuffy nose.
- Tell the victim to take prescribed drugs, if any.
- Consult doctor for prescribing drugs in allergies.
- Refer to the medical facility if the symptoms worsen.

Do's and Don'ts

Do's

Tell the victim to:

- Limit outdoor exposure and keep the windows and doors of the house and car closed as much as possible during the pollen season.
- Keep the home dry and well ventilated.
- For indoor allergens (dust mites), clean the mattress and pillows regularly.
- Wash bed linens every 2 weeks in hot (at least 130°F) water to kill any mites.
- Show sunlight to bed mattresses to get rid of bed bugs and dust .
- Vacuum cleaning of carpets.
- Take shower and change clothes after being outside.
- Clean cushions, soft toys, curtains and furniture regularly, either by washing (at a high temperature) or vacuum cleaner.

Don'ts

- Don't walk in grassy areas, such as parks and fields – particularly in the early morning, evening or night, when the pollen count is highest
- Don't do dry dusting.

Thus, in this section, we discussed in detail regarding various types of allergies, their definition, causes, recognition and their first aid. In the next section, we will discuss shock and its first aid management. So, lets continue.

Check Your Progress 2

1) List down the types of allergy.

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2) Discuss the management of patient with skin allergy.

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3) Enlist the common symptoms of drug allergy.

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4) Describe the first aid care of insect sting allergy.

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5.4 SHOCK

Till now we have discussed the allergic reactions, their types and management. Let us now discuss about another very important topic, SHOCK, its causes, recognition of shock and management.

5.4.1 Definition, Causes, Types and Recognizing Shock

Definition

Shock is a condition which should be treated as top priority. Shock is a very serious medical condition which can cause death because of decrease in blood flow to important organs like heart, brain, kidney. Shock usually occurs after severe injury or illness.

In simple words, shock occurs when there is lack of blood in the body (due to any cause) or failure of circulatory system to circulate blood in the body. Hence, there is decrease in supply of blood and oxygen to the various parts of body which can lead a person to collapse.

Causes of Shock

The most common causes of shock are listed in following Table 5.1

Table 5.1: Causes of Shock

Sl.No.	Causes
1	Major external bleeding after accidents, fall and trauma causing loss of blood
2	Major internal bleeding inside the body especially in Gastrointestinal bleeding and head injury
3	Severe Trauma, injuries-head, spinal,skull, wounds, fractures or Major Burns

4	When body cells looses the water /blood
5	Severe diarrhoea, excessive vomiting
6	Very severe allergic reactions
7	Conditions like heart attack or failure of heart to beat properly
8	Severe infection/infected wounds
9	Electric Shock, extreme heat and cold
10	Insects bites and stings and other emergencies

Types of Shock

There are basically 5 types of shock as listed below:

a) Hypovolemic Shock

It stands for Hypo meaning loss or lack of and volemic meaning water/ blood. Hence, it occurs when the body loses blood or fluids due to bleeding, severe vomiting and diarrhea. When there is decrease in fluids/ blood, small amount is only left in the body which is not sufficient for body's vital organs causing the person to collapse.

b) **Cardiogenic Shock**

Cardiogenic shock occurs when the heart has been damaged and can't pump and push enough blood to the organs of the body. The most common causes are heart attack and injury to heart.

c) Neurogenic Shock

Neurogenic shock occurs when there is an injury to brain and spinal cord which causes loss of function of nerves responsible for circulation of blood in the body. This decreases blood circulation to vital organs like heart, brain, kidney and liver.

d) **Anaphylactic Shock**

As already discussed in section 5.2, Anaphylactic shock is a severe allergic reaction. Person has a sudden decrease in blood pressure and difficulty in breathing. It should always be treated as a medical emergency, requiring immediate treatment.

e) Septic shock

Septic shock is a type of shock which occurs after a prolonged infection, infected wound or long fever. It results in sudden decrease in blood pressure which affects heart's ability to pump blood to vital organs such as the brain, heart, kidney and liver.

Recognizing Shock

Early identification of signs of shock is very important. The common signs and symptoms are as follows (Fig. 5.11):

- Skin is cold and clammy. Blue colour of lips and nail buds may be seen.

- Person may have restlessness, dizziness, confusion, lethargy, unconsciousness as there is decrease in blood flow to the brain.
- Nausea, vomiting, dryness of throat/lips.
- Decrease in BP, increase in Respiration, increase in pulse.
- Decrease in urine.

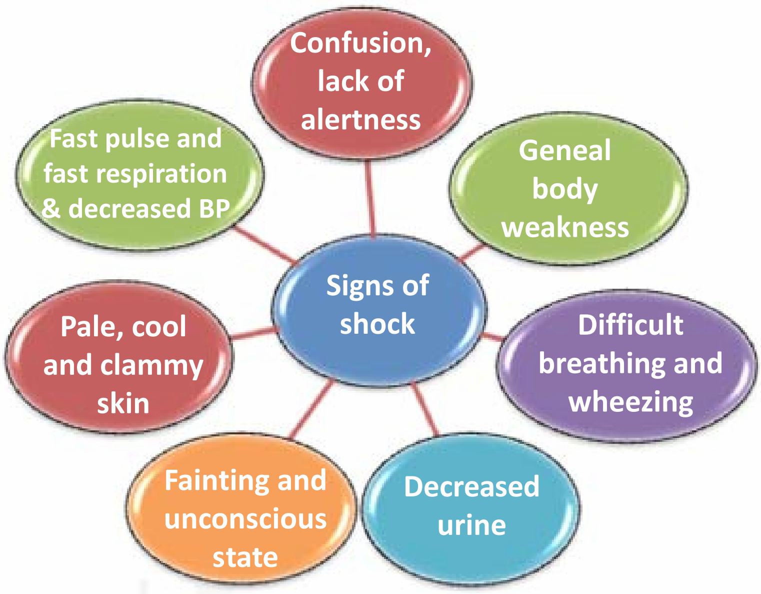


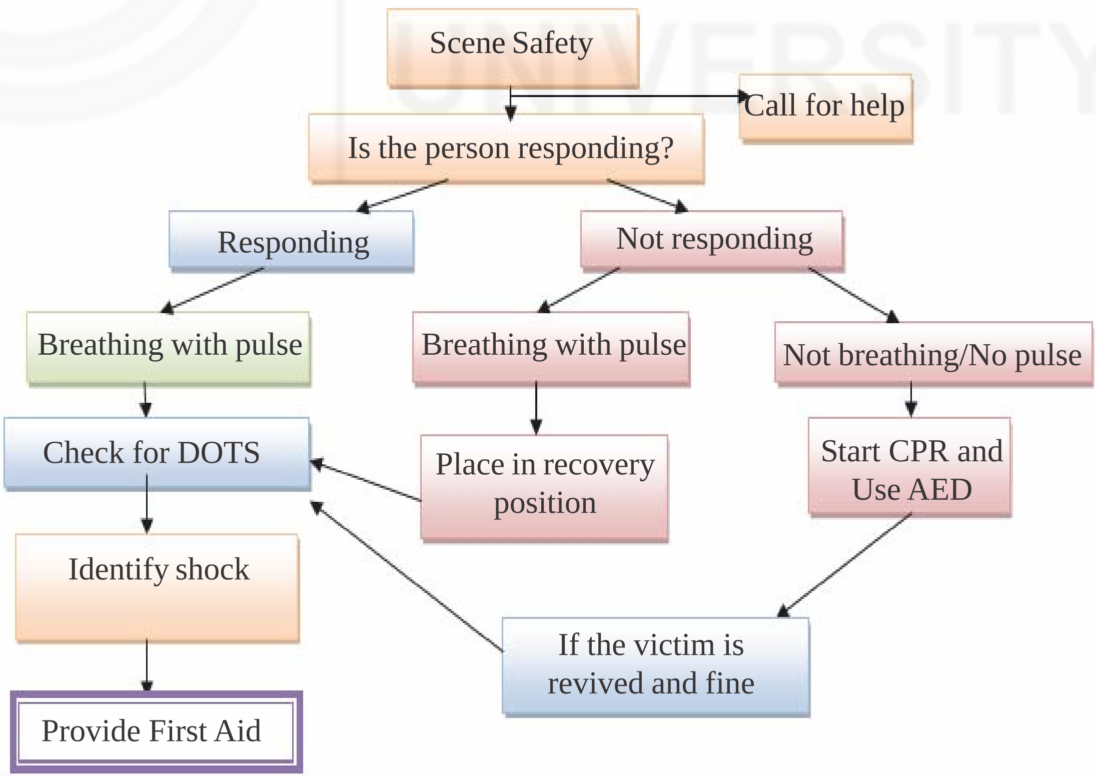
Fig. 5.11: Common Signs and Symptoms of shock

5.4.2 Assessment of the Victim

The assessment of the victim is as follows:

General Assessment

You have already learnt the assessment to be done while providing first aid in Unit 1 of Block 2 of the Theory Course. The general assessment is done as per following flowchart.



Specific Assessment

After general assessment, perform specific assessment as:

1) Check for DOTS and look for deformity, swelling, tenderness and any other sign and symptoms.
2) Check for the sign and symptoms of shock.
3) Ask History for identifying the cause if victim is conscious or onlookers/ bystanders are present etc.

5.4.3 First Aid in Shock

The first aid steps to be taken when the victim is in shock are as follows):

- Identify shock by assessing the signs and symptoms of shock.
- Loosen the clothing around neck, chest and abdomen.
- Raise the feet of the victim at least 20-30 cm above the ground. Improvise as required (Fig. 5.12). This is called as shock position.
- Keep the victim warm by covering with blankets or as per availability (Fig. 5.12).

Fig. 5.12: Raise the feet and cover the victim

- Turn the victim's head to one side, if required and only when there is no suspected neck/spinal cord injury.
- Transfer the victim to a health facility.

Refer Fig. 5.13 for flowchart on managing shock

Do's and Don'ts

Do's

- Identify shock.
- Keep the victim warm.
- Resuscitate as required.
- Immediately transfer the victim to health facility.
- Give hot tea/coffee with sugar, if conscious and speaking.
- Keep checking vital signs.

Don'ts

- Don't raise legs, if fracture to lower limb is suspected.
- Don't do unnecessary movements especially of head, back in suspected neck /spinal injuries.
- Don't give any thing by mouth, if the victim is unconscious.

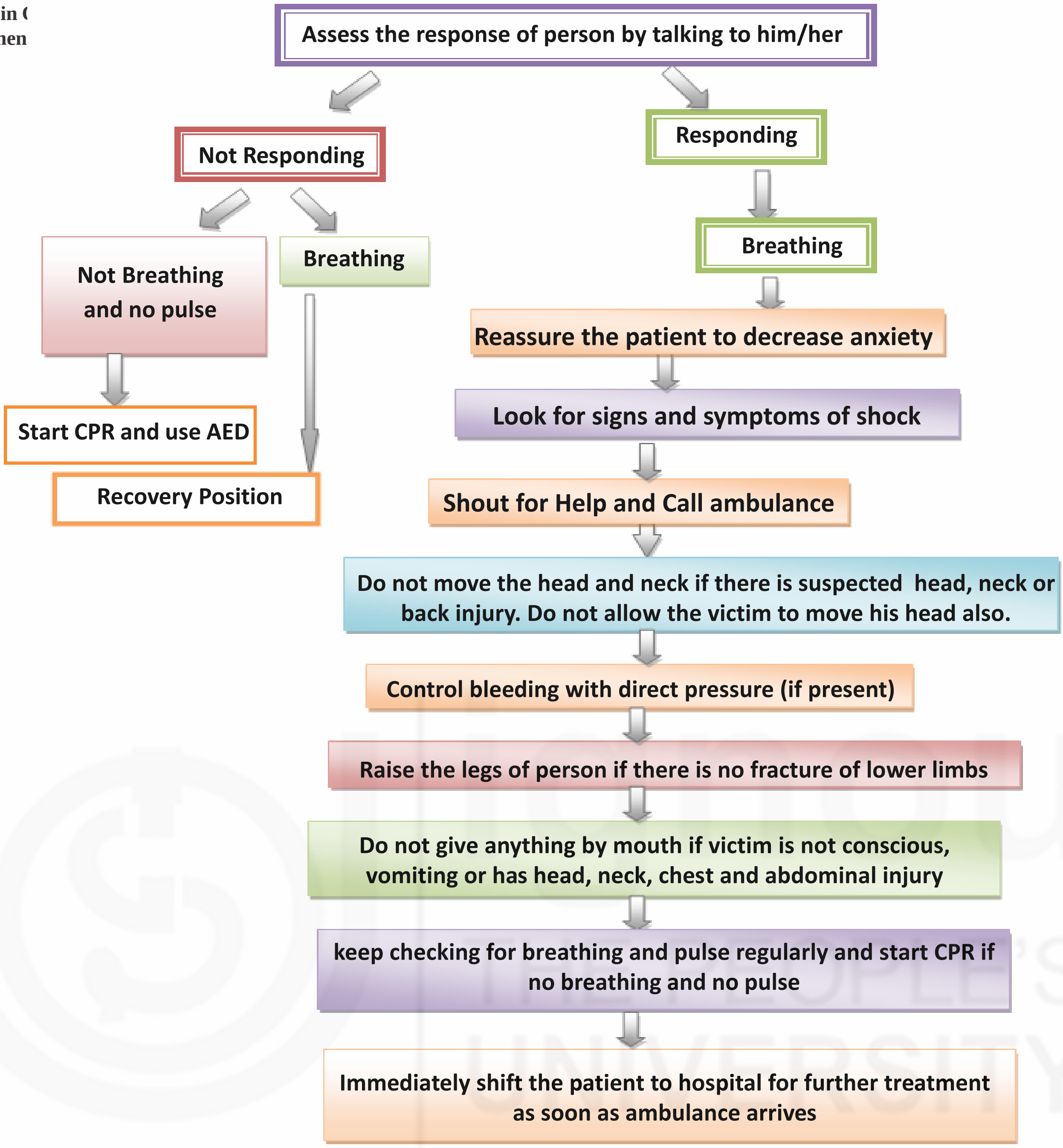


Fig. 5.13: Flow chart on managing Shock

Check Your Progress 3

1) Fill up the blanks

   a) The main cause of the neurogenic shock is injury to .... and ....

   b) The first and most important point in management of patient with shock is to .... sign and symptoms of shock.

   c) The first aid treatment for patient with no breathing and no pulse is ....

5.5 LET US SUM UP

Thus, in this unit, we have discussed the two very important topics, Allergy and Shock which if not identified and treated timely can lead to serious complications.

We have discussed the types of allergies, their symptoms and first aid management. In the same unit, we have also discussed Shock, the different types of shock, their symptoms and overall management.

With this we come to the end of our Theory block and the end of this Theory course. Hope , you have gained the essential knowledge about relevant topics in first aid and various emergency situations requiring first aid. It is expected from you now that you shall utilize this knowledge in providing first aid in various situations that you may encounter in day to day life.

5.6 KEYWORDS

Critical	:	Situation which may become dangerous overtime
Abnormal	:	Not common/not normal
Sensitive	:	Delicate
Deterioration/Worsen	:	Becoming worse, fall or drop to a lower level
Exposed	:	Become open to be unprotected from something
Disturbance	:	Create problem/affect someone's health
Severe/Extreme	:	A lot/Great/intense
Itching	:	Causing to itch
Rash	:	An area of redness
Sneezing	:	Sudden expulsion of air from the nose
Discharge	:	Flow-out of liquid
Swelling	:	Abnormal enlargement of a part of Body
Abdominal	:	Relating to the abdomen (i.e. stomach/intestines)
Diarrhea	:	Having loose stools many times in a day
Latex	:	A milky fluid found in plants and is a source of rubber
Lukewarm	:	Mildly warm
Soothing	:	Having a gently calming effect
Cramps	:	Painful contraction of a muscle
Nasal congestion	:	Nose blocked causing difficulty to breathe/ Stuffy nose
Swallowing	:	Pass down the throat from mouth
Label	:	Piece of paper attached to product giving information about it
Ingredient	:	Part or component of something
Buffets	:	Meal consisting of several dishes
Progresse	:	Advance or grow ahead
Attract	:	Creating interest about you in others
Barefoot	:	Without shoes

Inhaled	:	Breathing in while respiration
Weeds	:	Wild plant growing where it is not wanted
Triggers	:	Cause of something
Well ventilated	:	Airy
Damaged	:	Broken
Prolonged infection	:	Infection present for a long time

5.7 ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress I

1) a. Immune b. Allergens c. Drug d. Anaphylactic Shock

Check Your Progress 2

1) There are four types of allergy mentioned as below.
   i) Skin Allergy
   ii) Food and drug allergy
   iii) Insect bite allergy
   iv) Allergy by inhalation
2) Refer Section 5.3.1
3) Refer Section 5.3.2
4) Refer Section 5.3.3

Check your progress 3

1) a. Brain and Spinal Cord b. Identify/assess c. CPR & use of AED

5.8 REFERENCES AND FURTHER READINGS

1. Sampson, H.A. *Anaphylaxis and emergency treatment*. Pediatrics. 2003; 111:1601–1608.

2. Adkinson NF, et al. *Middleton's Allergy: Principles and Practice*. 8th ed. Philadelphia, Pa.: Saunders Elsevier; 2014.

3. Lieberman P, et al. *The diagnosis and management of anaphylaxis practice parameter*: 2010 update. Journal of Allergy and Clinical Immunology. 2010;126:477.

4. Blackham J.R. *The Indian manual of first aid.6^TH edition*. 2009. St. John Ambulance Assn., Indian Branch.

5. Allergic reactions. U.S. National Library of Medicine. Retrieved March 1, 2017, from http://www.nlm.nih.gov/medlineplus/ency/article/000006.htm.

6. What to do in a medical emergency: Shock. American College of Emergency Physicians. http://www.emergencycareforyou.org/EmergencyManual/WhatToDoInMedicalEmergency/Default.aspx?id=270. Accessed Feb. 17, 2017.

7. https://lifehacker.com/how-to-find-out-which-foods-are-making-you-sick-1557628052

8. https://www.firstaidtrainingbangkok.com/resources/treatments/illnesses/first-aid-for-allergies-anaphylaxis.html

9. http://www.first-rescue.co.uk/training/sports-emergency-first-aid-4-hours